Linear Elliptic and Parabolic PDE

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Part I

Elliptic Theory

The elliptic theory is the analytic core of the linear partial differential equations of this book. A second-order elliptic boundary-value problem asks for a function on a domain with prescribed values on the boundary and a prescribed image under an elliptic operator; the questions it raises are the standard three of analysis, existence, regularity, and qualitative structure. This part develops them in order: weak solutions produced by the variational method, the regularity theory that upgrades a weak solution to a classical one (Schauder and L^p), and the spectral decomposition of the operator. Throughout, the analysis foundations are cited rather than rebuilt, the Sobolev spaces with their trace and compactness theorems from the functional-analysis volume, the singular-integral bounds from harmonic analysis, and the spectral theorem for self-adjoint operators.

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Weak Formulations and the Lax-Milgram Theorem

The variational method turns a linear elliptic boundary-value problem into an equation for a bilinear form on a Hilbert space, where the existence and uniqueness of a solution follow from a single abstract theorem. This chapter sets up that reformulation. It rewrites the Dirichlet problem for a second-order operator as the search for a function in the Sobolev space H_0^1 satisfying an integrated identity against test functions, states and proves the Lax-Milgram theorem for a bounded coercive bilinear form, and applies it to produce weak solutions of a divergence-form Dirichlet problem. The Sobolev spaces H_0^1 that host the solutions, with their trace theorem and Poincaré inequality, are cited from the functional-analysis volume; the chapter develops the passage from a boundary-value problem to a solvable variational equation.

1.1 From Classical to Weak Solutions

The model problem of the elliptic theory is the Dirichlet problem for the Laplacian: given a bounded open set $U \subseteq \mathbb{R}^n$ and a function f on U, find u with

$$-\Delta u = f \quad \text{in } U, \qquad u = 0 \quad \text{on } \partial U \tag{1.1}$$

where $\Delta u = \sum_{i=1}^n \partial_i^2 u$ is the Laplacian and ∂U is the boundary of U. Read literally, (1.1) asks for a function twice continuously differentiable on U and continuous up to the boundary, so that both sides are defined pointwise and agree at every point. This is the *classical* formulation, and it is the naive reading of what it means to solve the equation. The task of this section is to loosen that reading into one that survives rough data and rough domains, and to identify the space of functions in which the loosened problem lives.

The classical formulation is fragile. A solution in the pointwise sense need not exist even when f is as regular as one could want, because the domain or the data can obstruct twice-differentiability at the boundary or in the interior.

Example 1.1.1: Failure of Classical Solutions
Two obstructions, one geometric and one from the data.

1. *A slit domain.* Let $U \subseteq \mathbb{R}^2$ be the slit disc, the unit disc with the positive x-axis removed, and write a point of U in polar coordinates (r, φ) with $0 < \varphi < 2\pi$. The removed segment is a genuine crack: along it the domain lies on both sides of the slit, with opening angle 2π at the origin, so near an interior point of the crack U cannot be written as the region above the graph of a function. Thus U is not even Lipschitz, a stronger obstruction than the corners the smooth-boundary results of this section require one to avoid. The function
$$u(r, \varphi) = r^{1/2} \sin \frac{\varphi}{2}$$
is harmonic ($\Delta u = 0$) on U; it vanishes on the two edges of the slit, where $\varphi = 0$ and $\varphi = 2\pi$ give $\sin 0 = \sin \pi = 0$, and on the outer arc $r = 1$ it equals $\sin(\varphi/2)$. Its full boundary data on ∂U is thus piecewise smooth: zero on the slit edges and $\sin(\varphi/2)$ on the arc. This is a Dirichlet problem with continuous, piecewise-smooth boundary data and $f = 0$, and u is its solution. Yet the gradient of u behaves like $r^{-1/2}$ as $r \to 0$, so u is not twice differentiable at the corner, and its second derivatives fail to be square-integrable there: against the area element $r\,dr\,d\varphi$ the integral $\int_{r<\varepsilon} |\nabla^2 u|^2\,dx$ scales like $\int_0^\varepsilon r^{-3} \cdot r\,dr = \int_0^\varepsilon r^{-2}\,dr = \infty$, so $u \notin H^2(U)$. No amount of smoothness in the data repairs this: the crack in ∂U forces the singularity, and u is a finite-energy H^1 solution with no classical, twice-differentiable representative. (Homogenizing the boundary condition changes nothing, since subtracting a fixed lift of the boundary data only shifts f to a smooth right-hand side while leaving the corner singularity of u intact.)

2. *Merely L^2 data.* Take U smooth, say the unit ball, and let $f \in L^2(U)$ be a function that is not continuous, for instance f equal to the indicator of a half-ball. A classical solution would have $\Delta u = f$ everywhere, forcing the continuous function Δu to equal the discontinuous f, which is impossible. Yet f is a perfectly reasonable right-hand side, the kind that arises whenever a source term is piecewise defined.

Both failures share a diagnosis: the classical problem requires two full derivatives of u at every point, and neither a boundary crack nor discontinuous data can give them. The remedy is to ask for less. Rather than testing the equation $-\Delta u = f$ at each point, one tests it in an averaged sense, against a family of smooth probe functions, and pushes the derivatives off u and onto the probes by integration by parts. What emerges is an identity requiring only *one* derivative of u, and only in an integrated L^2 sense.

Fix a probe $v \in C_c^\infty(U)$, a smooth function with compact support inside U. Multiply (1.1) by v and integrate over U:
$$\int_U (-\Delta u)\, v\,dx = \int_U f v\,dx$$
The divergence theorem, applied to the vector field $v\nabla u$, gives Green's first identity
$$\int_U (-\Delta u)\, v\,dx = \int_U \nabla u \cdot \nabla v\,dx - \int_{\partial U} v\,\frac{\partial u}{\partial \nu}\,dS$$

where ν is the outward unit normal and $\partial u/\partial\nu = \nabla u \cdot \nu$. The boundary integral vanishes because v has compact support in U, so $v = 0$ on ∂U. What remains is the identity

$$\int_U \nabla u \cdot \nabla v \, dx = \int_U f v \, dx \tag{1.2}$$

The left-hand side of (1.2) involves only first derivatives of u, paired against first derivatives of v; the second derivatives that the classical problem demanded have disappeared. For (1.2) to make sense one no longer needs u to be twice differentiable, only for ∇u to be a square-integrable vector field, so that the integral on the left is finite for every v whose gradient is likewise square-integrable. This is exactly the regularity encoded by the Sobolev space $H^1(U) = W^{1,2}(U)$, the functions in $L^2(U)$ whose weak first derivatives again lie in $L^2(U)$, developed in [2, Sobolev Space $W^{k,p}$].

The zero boundary condition persists too, but it must be re-read. A function in $H^1(U)$ is only an equivalence class up to sets of measure zero, and ∂U has measure zero, so "$u = 0$ on ∂U" cannot mean a pointwise restriction. The correct reading is given by the *trace theorem* of [2]: for U with Lipschitz boundary there is a bounded linear *trace* operator $T \colon H^1(U) \to L^2(\partial U)$ that agrees with pointwise restriction on functions continuous up to the boundary, and the closed subspace on which the trace vanishes is

$$H_0^1(U) = \{u \in H^1(U) \big| Tu = 0\}$$

equivalently the closure of $C_c^\infty(U)$ in the H^1 norm. This is the space of finite-energy functions with zero boundary values: the H^1 norm controls the Dirichlet energy $\int_U |\nabla u|^2 \, dx$, and membership in $H_0^1(U)$ is the weak sense in which u vanishes on ∂U. Requiring $u \in H_0^1(U)$ therefore imposes the boundary condition of (1.1) in the only form that a Sobolev function can carry it, and simultaneously gives the test functions: since $C_c^\infty(U)$ is dense in $H_0^1(U)$ and both sides of (1.2) are continuous in v for the H^1 norm, the identity holds for every $v \in C_c^\infty(U)$ if and only if it holds for every $v \in H_0^1(U)$.

These two moves, one derivative in place of two and a trace condition in place of a pointwise one, define the weak problem. It is stated most usefully for the general operator this chapter targets, so the definition is given for the divergence-form operator at once and specialized back to the Laplacian in the discussion that follows.

The data on the right is not restricted to L^2. The pairing $\langle f, v\rangle$ makes sense whenever f is a bounded linear functional on $H_0^1(U)$, that is, an element of the dual space $H^{-1}(U) = H_0^1(U)^*$; for $f \in L^2(U)$ this pairing is the L^2 inner product $\langle f, v\rangle = \int_U f v \, dx$, and $L^2(U) \subseteq H^{-1}(U)$ since $|\int_U f v \, dx| \le \|f\|_{L^2}\|v\|_{L^2} \le \|f\|_{L^2}\|v\|_{H_0^1}$. Writing the right-hand side as $\langle f, v\rangle$ thus covers both the L^2 data and the more singular data that arise, with no change to the definition.

Definition 1.1.2: Weak Solution of the Dirichlet Problem

Let $U \subseteq \mathbb{R}^n$ be bounded and open, let $Lu = -\operatorname{div}(A\nabla u) + c\,u$ be the divergence-form operator with $A = (a_{ij})$ bounded measurable and $c \in L^\infty(U)$, and let

$$a(u, v) = \int_U (A\nabla u) \cdot \nabla v + c\,uv\,dx$$

be its associated bilinear form. Given $f \in H^{-1}(U)$, a function $u \in H_0^1(U)$ is a *weak solution* of the Dirichlet problem $Lu = f$, $u|_{\partial U} = 0$, if

$$a(u, v) = \langle f, v\rangle \qquad \text{for all } v \in H_0^1(U) \tag{1.3}$$

For the Laplacian $L = -\Delta$ (the case $A = I$, $c = 0$) this reads $\int_U \nabla u \cdot \nabla v\,dx = \langle f, v\rangle$ for all $v \in H_0^1(U)$.

The definition relocates the entire problem into the Hilbert space $H_0^1(U)$. A weak solution is no longer a function satisfying a pointwise equation but a vector satisfying a linear identity against every other vector in the space, with the operator L replaced by the bilinear form a and the zero boundary condition absorbed into the ambient space. The membership $u \in H_0^1(U)$ carries the boundary condition, and the identity (1.3) carries the equation; together they say everything (1.1) said, in a form that asks only one derivative of u and tolerates f as rough as $H^{-1}(U)$. This is the reformulation the chapter opening promised, and every existence result to come is a statement about solving (1.3).

Example 1.1.3: The One-Dimensional Weak Problem

Take $n = 1$, $U = (0, 1)$, and $L = -\Delta = -\frac{d^2}{dx^2}$, so $a(u, v) = \int_0^1 u'v'\,dx$ and $H_0^1(0, 1)$ consists of the H^1 functions vanishing at both endpoints. For $f \equiv 1$ the weak problem asks for $u \in H_0^1(0, 1)$ with

$$\int_0^1 u'v'\,dx = \int_0^1 v\,dx \qquad \text{for all } v \in H_0^1(0, 1)$$

The classical candidate solves $-u'' = 1$ with $u(0) = u(1) = 0$, namely $u(x) = \frac{1}{2}x(1 - x)$. This u is smooth, lies in $H_0^1(0, 1)$, and $u'(x) = \frac{1}{2} - x$. Integration by parts against any $v \in C_c^\infty(0, 1)$ gives $\int_0^1 u'v'\,dx = -\int_0^1 u''v\,dx = \int_0^1 v\,dx$, and both sides extend by density to all $v \in H_0^1(0, 1)$, so u is the weak solution. The example shows the weak identity holding for a genuine classical solution; the general principle behind this coincidence is the content of the next result.

The weak formulation must be faithful to the classical one where the latter applies: a function smooth enough to solve (1.1) pointwise should be a weak solution, and conversely a weak solution smooth enough to differentiate twice should solve the equation pointwise. Both directions run on Green's identity, forward to discard the second derivatives and backward to recover them, and both extend verbatim from $-\Delta$ to the divergence-form operator L. The proposition records the equivalence and, in doing so, confirms that no solutions are lost in passing to the weak problem and none are gained beyond those the classical equation would sanction.

Proposition 1.1.4: Classical Solutions Are Weak, and Smooth Weak Solutions Are Classical

Let $U \subseteq \mathbb{R}^n$ be bounded and open with C^1 boundary, let $Lu = -\operatorname{div}(A\nabla u) + c\,u$ with $A = (a_{ij}) \in C^1(\overline{U})$ and $c \in C(\overline{U})$, and let $f \in C(\overline{U})$.

1. If $u \in C^2(\overline{U})$ solves $Lu = f$ in U and $u = 0$ on ∂U pointwise, then u is a weak solution in the sense of definition 1.1.2, with $\langle f, v\rangle = \int_U f v\,dx$.

2. Conversely, if $u \in C^2(\overline{U}) \cap H_0^1(U)$ is a weak solution, then $Lu = f$ at every point of U.

***Proof*:**
We prove the two directions in turn; both rest on Green's identity for the operator L, which for $u \in C^2(\overline{U})$ and $v \in C_c^\infty(U)$ reads

$$\int_U (Lu)\,v\,dx = \int_U (A\nabla u)\cdot\nabla v + c\,uv\,dx - \int_{\partial U} v\,(A\nabla u)\cdot\nu\,dS \tag{1.4}$$

and follows by applying the divergence theorem to the vector field $v\,A\nabla u$, using $\operatorname{div}(v\,A\nabla u) = \nabla v\cdot(A\nabla u) + v\operatorname{div}(A\nabla u)$ together with $\operatorname{div}(A\nabla u) = -Lu + cu$.

Part 1: a classical solution is weak. Let $u \in C^2(\overline{U})$ solve $Lu = f$ in U with $u = 0$ on ∂U. Fix $v \in C_c^\infty(U)$. Since v has compact support in U, the boundary integral in (1.4) vanishes, and substituting $Lu = f$ into the left-hand side gives

$$\int_U f v\,dx = \int_U (A\nabla u)\cdot\nabla v + c\,uv\,dx = a(u,v)$$

Because A and c are bounded on $\overline{U}$ and $u \in C^2(\overline{U})$, both sides are continuous in v for the H^1 norm; since $C_c^\infty(U)$ is dense in $H_0^1(U)$, the identity $a(u,v) = \int_U f v\,dx$ extends to every $v \in H_0^1(U)$. It remains to check $u \in H_0^1(U)$. As $u \in C^2(\overline{U})$ it lies in $H^1(U)$, and its trace is its pointwise boundary restriction $u|_{\partial U} = 0$; by the trace characterization of $H_0^1(U)$ from [2, Zero-Trace Characterization, Trace Theorem], $u \in H_0^1(U)$. Thus u satisfies definition 1.1.2 with $\langle f, v\rangle = \int_U f v\,dx$.

Part 2: a smooth weak solution is classical. Let $u \in C^2(\overline{U}) \cap H_0^1(U)$ satisfy $a(u,v) = \int_U f v\,dx$ for all $v \in H_0^1(U)$, in particular for all $v \in C_c^\infty(U)$. For such v the boundary term in (1.4) again vanishes, so

$$\int_U (Lu)\,v\,dx = a(u,v) = \int_U f v\,dx$$

which rearranges to

$$\int_U (Lu - f)\,v\,dx = 0 \qquad \text{for all } v \in C_c^\infty(U)$$

The function $Lu - f$ is continuous on U, since $u \in C^2$, $A \in C^1$, $c \in C$, and $f \in C$. A continuous function that integrates to zero against every $v \in C_c^\infty(U)$ vanishes identically: were $(Lu - f)(x_0) > 0$ at some $x_0 \in U$, continuity would give a ball $B \subseteq U$ on which $Lu - f > 0$, and choosing $v \in C_c^\infty(U)$ nonnegative with $\operatorname{supp} v \subseteq B$ and $v(x_0) > 0$ forces $\int_U (Lu - f)v\,dx > 0$, a contradiction; the case $(Lu - f)(x_0) < 0$ is symmetric. Hence $Lu - f = 0$ at every point of U, completing the proof.

The two directions of proposition 1.1.4 are exactly the forward and backward passes of Green's identity, and the density of $C_c^\infty(U)$ in $H_0^1(U)$ is what lets a statement checked against compactly supported probes stand as a statement against all of $H_0^1(U)$. The final step of Part 2, that a continuous function orthogonal to every test function vanishes, is the fundamental lemma of the calculus of variations; it is the mechanism by which an integrated identity delivers a pointwise equation, and it recurs throughout the regularity theory whenever a weak solution is shown to be strong. The proposition guarantees that the weak problem is a genuine extension of the classical one: it agrees with the classical problem on smooth data and smooth domains, and departs from it only by admitting solutions, such as the crack function of example 1.1.1, that the classical formulation could not hold.

The passage from $-\Delta$ to the general $Lu = -\operatorname{div}(A\nabla u) + c\,u$ changed nothing in the definition: the same integration by parts that produced (1.2) produces the bilinear form $a(u,v) = \int_U (A\nabla u)\cdot\nabla v + c\,uv\,dx$, with the leading coefficient A recording anisotropic or variable-coefficient diffusion and the zeroth-order coefficient c recording absorption or reaction. The divergence form is precisely the shape that lets one derivative move onto the test function, and it is the form the variational method requires.

1.2 Bilinear Forms and the Lax-Milgram Theorem

The weak formulation of the previous section presents the Dirichlet problem as an equation for a bilinear form: find u in a Hilbert space with $a(u,v) = F(v)$ for every test function v, where F records the data. Solving it means inverting the form. This section isolates the two properties of a that make the inversion possible, boundedness and coercivity, and proves the abstract theorem that delivers the solution. The theorem is stated for an arbitrary Hilbert space H and an arbitrary bounded functional $F \in H^*$; the elliptic form $a(u,v) = \int_U (A\nabla u)\cdot\nabla v + c\,uv\,dx$ on $H_0^1(U)$ is the case that motivates it and reappears as the running illustration, but the argument never uses anything beyond the two hypotheses. The next section verifies that the elliptic form satisfies them, at which point the abstract theorem produces the weak solution.

Throughout, H is a real Hilbert space with inner product $\langle\cdot,\cdot\rangle$ and norm $\|u\| = \langle u,u\rangle^{1/2}$, and $a\colon H\times H \to \mathbb{R}$ is a bilinear form, linear in each argument separately. Two quantitative conditions on a govern the theory.

Definition 1.2.1: Bounded and Coercive Bilinear Form

Let H be a Hilbert space and $a\colon H\times H \to \mathbb{R}$ a bilinear form. Then a is *bounded* (or *continuous*) if there is a constant $M > 0$ with

$$|a(u,v)| \le M\|u\|\,\|v\| \qquad \text{for all } u,v \in H$$

and a is *coercive* (or H*-elliptic*) if there is a constant $\alpha > 0$ with

$$a(u,u) \ge \alpha\|u\|^2 \qquad \text{for all } u \in H$$

Boundedness is a continuity requirement: it says the form does not amplify by more than a fixed factor, so a is continuous as a map $H\times H \to \mathbb{R}$. Coercivity is a nondegeneracy requirement from below, and it is the substantial hypothesis. It says the quadratic functional $u \mapsto a(u,u)$ controls the squared norm, which forces $a(u,u)$ to grow as u leaves the origin and, in particular, to vanish only at

$u = 0$. The constant α is the coercivity constant and it enters the a priori bound on the solution directly, so its size is quantitative information, not a mere existence certificate. A symmetric bounded coercive form is exactly an inner product on H equivalent to the original, a fact developed at the end of the section; coercivity is the abstraction of the positive lower bound that uniform ellipticity gives for the elliptic form.

The running example is the elliptic form itself. It is the reason the two hypotheses are the right ones to isolate.

Example 1.2.2: The Elliptic Form and the Dirichlet Inner Product

Take $H = H_0^1(U)$ with the inner product $\langle u, v\rangle_{H_0^1} = \int_U \nabla u \cdot \nabla v \, dx$, which is equivalent to the full H^1 inner product on $H_0^1(U)$ by the Poincaré inequality of [2, Poincaré Inequality]. Consider first the Dirichlet form

$$a_0(u, v) = \int_U \nabla u \cdot \nabla v \, dx$$

which is the weak form of $-\Delta$. It is symmetric, and it *is* the inner product just named, so it is trivially bounded with $M = 1$ and coercive with $\alpha = 1$. Coercivity here is the statement $a_0(u, u) = \|\nabla u\|_{L^2}^2 = \|u\|_{H_0^1}^2$: the energy of u is its squared norm.

The general divergence-form operator $Lu = -\operatorname{div}(A\nabla u) + cu$ produces

$$a(u, v) = \int_U (A\nabla u) \cdot \nabla v + c\, uv \, dx$$

Boundedness comes from Cauchy-Schwarz and the bounds $\|A\|_\infty, \|c\|_\infty < \infty$: since $|(A\nabla u) \cdot \nabla v| \le \|A\|_\infty |\nabla u||\nabla v|$ and $|c\, uv| \le \|c\|_\infty |u||v|$, integrating gives $|a(u, v)| \le M\|u\|_{H^1}\|v\|_{H^1}$ with $M = \|A\|_\infty + \|c\|_\infty$. Coercivity is the substantive condition, and it is where uniform ellipticity of A (the bound $\xi^\top A(x)\xi \ge \theta|\xi|^2$) and the Poincaré inequality enter; the sign condition $c \ge 0$ makes the low-order term help rather than hurt. The verification is carried out in full in section 1.3. For now the point is that the elliptic form is bounded outright and coercive under the natural hypotheses, so it is exactly the class of forms the abstract theorem is built to invert.

With boundedness and coercivity in hand, the abstract solvability theorem can be stated. The equation to solve is $a(u, v) = F(v)$ for all $v \in H$, where F is a bounded linear functional recording the data of the problem. The theorem asserts that this equation has one and only one solution, and that the solution depends boundedly on the data, with the coercivity constant α as the explicit constant.

Theorem 1.2.3: Lax-Milgram Theorem

Let H be a Hilbert space and $a\colon H \times H \to \mathbb{R}$ a bounded coercive bilinear form, with bound M and coercivity constant α as in definition 1.2.1. Then for every bounded linear functional $F \in H^*$ there is a unique $u \in H$ with

$$a(u,v) = F(v) \qquad \text{for all } v \in H$$

and the solution satisfies the a priori bound

$$\|u\| \le \frac{1}{\alpha} \|F\|_{H^*}$$

***Proof*:**
The strategy is to convert the two-variable form a into a single bounded operator $A\colon H \to H$, translate the equation $a(u,v) = F(v)$ into an operator equation $Au = w$, and then show that coercivity forces A to be invertible. The Riesz representation theorem of [2, Natural Isomorphism To $\mathcal{H}^*$ (Riesz Representation)] gives the conversions.

Step 1: Represent the form by a bounded operator. Fix $u \in H$. The map $v \mapsto a(u,v)$ is linear in v, and it is bounded, since $|a(u,v)| \le M\|u\|\,\|v\|$ shows it is a functional of norm at most $M\|u\|$. By the Riesz representation theorem [2, Natural Isomorphism To $\mathcal{H}^*$ (Riesz Representation)], there is a unique element of H, which we call Au, with

$$a(u,v) = \langle Au, v\rangle \qquad \text{for all } v \in H$$

The assignment $u \mapsto Au$ defines a map $A\colon H \to H$. It is linear: for $u_1, u_2 \in H$ and scalars, bilinearity of a gives $\langle A(u_1+u_2), v\rangle = a(u_1+u_2, v) = a(u_1,v) + a(u_2,v) = \langle Au_1 + Au_2, v\rangle$ for every v, and a Hilbert-space element is determined by its inner products against all v, so $A(u_1+u_2) = Au_1 + Au_2$; scalar multiplication is identical. The operator is bounded: taking $v = Au$ in the representation and applying boundedness of a,

$$\|Au\|^2 = \langle Au, Au\rangle = a(u, Au) \le M\|u\|\,\|Au\|$$

so $\|Au\| \le M\|u\|$ and $\|A\| \le M$.

Step 2: Translate the data. The functional F is itself an element of H^*, so the Riesz representation theorem [2, Natural Isomorphism To $\mathcal{H}^*$ (Riesz Representation)] produces a unique $w \in H$ with

$$F(v) = \langle w, v\rangle \qquad \text{for all } v \in H$$

and $\|w\| = \|F\|_{H^*}$. The equation to solve, $a(u,v) = F(v)$ for all v, now reads $\langle Au, v\rangle = \langle w, v\rangle$ for all v, which holds for all v if and only if

$$Au = w$$

The problem has become the single operator equation $Au = w$: solving the variational problem for every $F \in H^*$ is exactly inverting A.

Step 3: Coercivity bounds A below. Coercivity gives, for every $u \in H$,

$$\alpha\|u\|^2 \leq a(u,u) = \langle Au, u\rangle \leq \|Au\|\,\|u\|$$

using Cauchy-Schwarz at the last step. Dividing by $\|u\|$ for $u \neq 0$ (the inequality is trivial for $u = 0$) yields the lower bound

$$\|Au\| \geq \alpha\|u\| \qquad \text{for all } u \in H$$

This single inequality does most of the remaining work. It shows at once that A is injective (if $Au = 0$ then $\|u\| = 0$) and that any solution obeys the a priori bound: from $Au = w$ and $\|Au\| \geq \alpha\|u\|$,

$$\alpha\|u\| \leq \|Au\| = \|w\| = \|F\|_{H^*}$$

which is the claimed estimate $\|u\| \leq \|F\|_{H^*}/\alpha$. Uniqueness follows from injectivity: if $Au_1 = w = Au_2$ then $A(u_1 - u_2) = 0$, so $u_1 = u_2$.

Step 4: The range of A is closed. It remains to show A is surjective, so that $Au = w$ has a solution for the given w (indeed for every $w \in H$). First the range $A(H)$ is closed. Let (Au_n) be a sequence in the range converging to some $y \in H$; then (Au_n) is Cauchy, and the lower bound applied to $u_n - u_m$,

$$\|u_n - u_m\| \leq \frac{1}{\alpha}\|A(u_n - u_m)\| = \frac{1}{\alpha}\|Au_n - Au_m\|$$

shows (u_n) is Cauchy as well. By completeness of H, $u_n \to u$ for some $u \in H$, and boundedness of A (Step 1) gives $Au_n \to Au$; by uniqueness of limits $y = Au \in A(H)$. Hence $A(H)$ is closed.

Step 5: The range of A is dense, hence all of H. A closed subspace equals H once it is shown to be dense, and density follows from injectivity applied to the orthogonal complement. Suppose $z \in H$ is orthogonal to the range, so $\langle Au, z\rangle = 0$ for all $u \in H$. Taking $u = z$ and using the representation together with coercivity,

$$0 = \langle Az, z\rangle = a(z,z) \geq \alpha\|z\|^2$$

forces $\|z\| = 0$, that is $z = 0$. Thus the orthogonal complement of $A(H)$ is trivial, so $A(H)$ is dense; being also closed by Step 4, $A(H) = H$. Therefore A is a bijection, the equation $Au = w$ has the unique solution $u = A^{-1}w$, and this u solves $a(u,v) = F(v)$ for all v with the a priori bound of Step 3, completing the proof.

The Lax-Milgram theorem is what the variational method rests on, and the shape of its statement is worth reading off directly. It converts a question about a bilinear equation into the two hypotheses of definition 1.2.1: verify boundedness and coercivity of the form, and existence, uniqueness, and a quantitative stability estimate all follow at once, for every choice of data F. The a priori bound $\|u\| \leq \|F\|_{H^*}/\alpha$ is the well-posedness of the problem in the sense of Hadamard, that the solution operator $F \mapsto u$ is bounded, so small changes in the data produce small changes in the solution. In the elliptic application F is built from the source term f, and the bound becomes the energy estimate $\|u\|_{H_0^1} \leq C\|f\|$ that controls the weak solution by its data.

Two features of the proof deserve emphasis. First, coercivity did all of the heavy lifting through the single lower bound $\|Au\| \geq \alpha\|u\|$ of Step 3, which simultaneously gave injectivity, the a priori bound, and the closedness of the range; boundedness of a entered only to make A a well-defined bounded operator. Second, the argument is entirely constructive in structure but not in the solution: it locates $u = A^{-1}w$ without exhibiting it, so producing the solution in a concrete problem still requires the coercivity constant and the Riesz element w, which is where the elliptic estimates of the next section do their work. The theorem does not require a to be symmetric, and this is exactly what makes it more general than the Riesz representation theorem it is built from: a general divergence-form operator with a first-order or a nonsymmetric coefficient matrix produces a nonsymmetric form, and Lax-Milgram handles it with no extra hypothesis.

When the form *is* symmetric, the theorem specializes in an illuminating way. A symmetric bounded coercive form is an inner product, so Lax-Milgram collapses onto the Riesz representation theorem, and the solution acquires a variational characterization it does not have in general: it is the minimizer of an energy. This is the abstract form of Dirichlet's principle, and it is the reason the method is called variational.

Proposition 1.2.4: The Symmetric Case and Dirichlet's Principle

Let $a\colon H \times H \to \mathbb{R}$ be a bounded coercive bilinear form that is in addition *symmetric*, $a(u, v) = a(v, u)$ for all u, v. Then:

1. a is an inner product on H whose induced norm $\|u\|_a = a(u, u)^{1/2}$ is equivalent to the original norm, satisfying $\sqrt{\alpha}\,\|u\| \leq \|u\|_a \leq \sqrt{M}\,\|u\|$.
2. For $F \in H^*$, the unique $u \in H$ with $a(u, v) = F(v)$ for all v furnished by theorem 1.2.3 is exactly the Riesz representative of F with respect to the inner product a.
3. That same u is the unique minimizer over H of the energy functional

$$J(v) = \tfrac{1}{2}\, a(v, v) - F(v)$$

***Proof*:**
We prove the three statements in turn.

1. Bilinearity of a gives additivity and homogeneity in each argument; symmetry is assumed; and coercivity gives positive-definiteness, $a(u, u) \geq \alpha\|u\|^2 > 0$ for $u \neq 0$. These are the inner-product axioms, so a is an inner product with induced norm $\|u\|_a = a(u, u)^{1/2}$. The two-sided bound is immediate from the hypotheses: coercivity gives $\|u\|_a^2 = a(u, u) \geq \alpha\|u\|^2$, and boundedness gives $\|u\|_a^2 = a(u, u) \leq M\|u\|^2$. Taking square roots yields $\sqrt{\alpha}\,\|u\| \leq \|u\|_a \leq \sqrt{M}\,\|u\|$, so $\|\cdot\|_a$ and $\|\cdot\|$ are equivalent and (H, a) is again a Hilbert space (completeness transfers across equivalent norms).
2. Since (H, a) is a Hilbert space and F is bounded on it (boundedness of F in $\|\cdot\|$ transfers to $\|\cdot\|_a$ by the equivalence just proved), the Riesz representation theorem of [2, Natural Isomorphism To $\mathcal{H}^*$ (Riesz Representation)] applied in the inner product a produces a

unique $u \in H$ with $a(u, v) = F(v)$ for all v. This is precisely the equation of theorem 1.2.3, whose solution is unique; the two solutions therefore coincide, and u is the Riesz representative of F relative to a. In the symmetric case Lax-Milgram is Riesz representation in a change of inner product.

3. Let u be the solution from part (2) and let $v \in H$ be arbitrary. Write $v = u + h$ with $h = v - u \in H$. Expanding J and using bilinearity and symmetry of a,

$$J(u+h) = \tfrac{1}{2}\, a(u+h, u+h) - F(u+h) = \tfrac{1}{2}\, a(u,u) + a(u,h) + \tfrac{1}{2}\, a(h,h) - F(u) - F(h)$$

The defining equation $a(u, h) = F(h)$ cancels the two middle cross terms, leaving

$$J(u+h) = \big(\tfrac{1}{2}\, a(u,u) - F(u)\big) + \tfrac{1}{2}\, a(h,h) = J(u) + \tfrac{1}{2}\, a(h,h)$$

Coercivity gives $a(h, h) \geq \alpha \|h\|^2 \geq 0$, so $J(v) = J(u + h) \geq J(u)$, with equality if and only if $a(h, h) = 0$, that is $h = 0$ and $v = u$. Hence u is the unique minimizer of J, as we sought to show.

The equivalence in part (1) is the concrete content of "symmetric coercive form equals inner product": the energy $\|u\|_a^2 = a(u, u)$ measures the same notion of size as the original norm, up to the constants α and M, so nothing is lost by working in whichever inner product is convenient. Part (3) is Dirichlet's principle in the abstract, and it is the source of the method's name. Solving the boundary-value problem is the same as minimizing an energy: the weak equation $a(u, v) = F(v)$ is exactly the vanishing of the first variation of J, the Euler-Lagrange equation of the minimization problem. For the Dirichlet form $a_0(u, v) = \int_U \nabla u \cdot \nabla v \, dx$ of example 1.2.2 with $F(v) = \int_U f v \, dx$, the functional is

$$J(v) = \tfrac{1}{2} \int_U |\nabla v|^2 \, dx - \int_U f v \, dx$$

the classical Dirichlet energy minus the work of the source, and its minimizer over $H_0^1(U)$ is the weak solution of $-\Delta u = f$ with zero boundary values. The symmetry hypothesis is essential to this reformulation: a nonsymmetric form has no energy to minimize, the cross terms in the expansion of J do not cancel, and only the bare existence statement of theorem 1.2.3 survives. That the general theorem needs no symmetry is what lets it reach the nonsymmetric operators that the energy method alone cannot.

1.3 Weak Solvability of the Dirichlet Problem

The abstract machinery is now in place: Lax-Milgram (theorem 1.2.3) solves the weak identity $a(u, v) = \langle f, v \rangle$ (definition 1.1.2) once the form is bounded and coercive (definition 1.2.1). What remains is to check that the form attached to a genuine second-order elliptic operator meets those two hypotheses. For $Lu = -\operatorname{div}(A\nabla u) + cu$ with A bounded measurable and uniformly elliptic and $c \in L^\infty(U)$ nonnegative, the form

$$a(u, v) = \int_U (A\nabla u) \cdot \nabla v + c\, uv \, dx$$

is shown to be bounded and coercive on $H_0^1(U)$, whereupon Lax-Milgram delivers a unique weak solution of the Dirichlet problem together with an energy estimate. The two structural hypotheses on L enter at precisely one place each: boundedness of a uses the L^∞ bounds on the coefficients, and coercivity uses uniform ellipticity together with the Poincaré inequality, the point at which the boundedness of U and the vanishing boundary trace of functions in $H_0^1(U)$ become indispensable.

The operator L is the general second-order operator in divergence form. Its principal part $-\operatorname{div}(A\nabla u)$ is governed by the coefficient matrix $A = (a_{ij})$, and the requirement that L be *elliptic* is a positivity condition on A: the associated quadratic form $\xi \mapsto \xi^\top A(x)\xi$ must be bounded below by a fixed positive multiple of $|\xi|^2$, uniformly in x. This is the analytic substitute for the sign of the Laplacian, and it is what will make the Dirichlet form control the gradient. The zeroth-order coefficient c contributes the term $\int_U c\,uv$; its sign is what separates the coercive case treated here from the indefinite case treated later.

Definition 1.3.1: Uniform Ellipticity

Let $U \subseteq \mathbb{R}^n$ be a bounded open set. A measurable coefficient matrix $A\colon U \to \mathbb{R}^{n\times n}$, $A = (a_{ij})$, is *bounded* if there is a constant $\Lambda > 0$ with

$$|A(x)\xi| \le \Lambda\,|\xi| \qquad \text{for a.e. } x \in U \text{ and all } \xi \in \mathbb{R}^n$$

that is, the operator norm $\|A(x)\|$ is bounded by Λ for a.e. x, and *uniformly elliptic* if there is a constant $\theta > 0$, the *ellipticity constant*, with

$$\xi^\top A(x)\xi = \sum_{i,j=1}^{n} a_{ij}(x)\,\xi_i\xi_j \ge \theta\,|\xi|^2$$

for a.e. $x \in U$ and all $\xi \in \mathbb{R}^n$. The second-order operator $Lu = -\operatorname{div}(A\nabla u) + cu$ with such an A and with $c \in L^\infty(U)$ is called a *uniformly elliptic operator in divergence form.*

The two inequalities pull in opposite directions on the same matrix. Boundedness caps the size of A as a linear map from above, so that the integrals defining a converge and a is continuous; uniform ellipticity bounds the quadratic form $\xi^\top A\xi$ from below, so that a sees the full length of the gradient rather than allowing it to collapse in some direction. Neither inequality asks A to be symmetric or continuous, only measurable and bounded, which is the natural class of coefficients for the L^2 theory. A symmetric A makes a symmetric and connects the problem to energy minimization (proposition 1.2.4), but the existence theory below never needs symmetry.

Example 1.3.2: The Laplacian and a Perturbed Metric

Two instances anchor the definition.

1. For $A = I$ the identity matrix and $c = 0$, the operator is $Lu = -\operatorname{div}(\nabla u) = -\Delta u$, the negative Laplacian. Here $\xi^\top A\xi = |\xi|^2$, so both $\Lambda = 1$ and $\theta = 1$; the Laplacian is uniformly elliptic with ellipticity constant 1, and $a(u,v) = \int_U \nabla u \cdot \nabla v$ is the Dirichlet form.
2. Let $A(x) = (1 + b(x))\,I$ with $b \in L^\infty(U)$ satisfying $0 < \theta - 1 \le b(x) \le \Lambda - 1$ for a.e. x, so that $\theta \le 1 + b(x) \le \Lambda$. Then $\xi^\top A(x)\xi = (1 + b(x))|\xi|^2 \ge \theta|\xi|^2$, and since $A(x)$ is the scalar multiple $(1 + b(x))I$ its operator norm is $\|A(x)\| = 1 + b(x) \le \Lambda$, so A is bounded and uniformly elliptic with constants θ and Λ. This models an isotropic medium of spatially varying, but uniformly nondegenerate, conductivity; the coefficient b need not be continuous, only essentially bounded away from -1.

With the class of operators fixed, the task is to verify the two hypotheses of Lax-Milgram for the associated form. The norm on $H_0^1(U)$ is the H^1 norm, $\|u\|_{H^1}^2 = \|u\|_{L^2}^2 + \|\nabla u\|_{L^2}^2$; recall that $H_0^1(U)$ is the completion of $C_c^\infty(U)$ in this norm, the space of finite-energy functions with vanishing boundary trace, and that on it the Poincaré inequality of [2, Poincaré Inequality] bounds the L^2 norm of a function by the L^2 norm of its gradient. That inequality is what converts the lower bound on $\|\nabla u\|_{L^2}$ given by ellipticity into a lower bound on the full norm, and its proof depends on both the boundedness of U and the zero boundary condition encoded in $H_0^1(U)$.

Proposition 1.3.3: Boundedness and Coercivity of the Elliptic Form

Let $U \subseteq \mathbb{R}^n$ be bounded and open, let A be bounded and uniformly elliptic with constants Λ, θ as in definition 1.3.1, and let $c \in L^\infty(U)$ with $c \geq 0$ a.e. Then the bilinear form

$$a(u,v) = \int_U (A\nabla u) \cdot \nabla v + c\,uv\,dx$$

is bounded and coercive on $H_0^1(U)$: there are constants $M, \alpha > 0$ with

$$|a(u,v)| \leq M\,\|u\|_{H^1}\,\|v\|_{H^1} \qquad \text{and} \qquad a(u,u) \geq \alpha\,\|u\|_{H^1}^2$$

for all $u, v \in H_0^1(U)$. One may take $M = \Lambda + \|c\|_{L^\infty}$ and $\alpha = \theta/(1 + C_P^2)$, where C_P is the Poincaré constant of U.

***Proof*:**
Both estimates are established directly from the definition of a; boundedness uses the L^∞ bounds on the coefficients and the Cauchy-Schwarz inequality, coercivity uses uniform ellipticity and the Poincaré inequality.

Step 1: Boundedness. Fix $u, v \in H_0^1(U)$. The form splits as

$$a(u,v) = \int_U (A\nabla u) \cdot \nabla v\,dx + \int_U c\,uv\,dx$$

and each term is bounded separately. For the principal term, the bound on A gives $|A(x)\nabla u(x)| \leq \Lambda\,|\nabla u(x)|$ almost everywhere, so by the Cauchy-Schwarz inequality in $\mathbb{R}^n$ applied to the dot product, $|(A\nabla u) \cdot \nabla v| \leq |A\nabla u|\,|\nabla v| \leq \Lambda\,|\nabla u|\,|\nabla v|$ almost everywhere. Integrating and applying the Cauchy-Schwarz inequality in $L^2(U)$,

$$\left| \int_U (A\nabla u) \cdot \nabla v\,dx \right| \leq \Lambda \int_U |\nabla u|\,|\nabla v|\,dx \leq \Lambda\,\|\nabla u\|_{L^2}\,\|\nabla v\|_{L^2}$$

For the zeroth-order term, $|c(x)| \leq \|c\|_{L^\infty}$ almost everywhere gives, again by Cauchy-Schwarz,

$$\left| \int_U c\,uv\,dx \right| \leq \|c\|_{L^\infty} \int_U |u|\,|v|\,dx \leq \|c\|_{L^\infty}\,\|u\|_{L^2}\,\|v\|_{L^2}$$

Since $\|\nabla u\|_{L^2} \leq \|u\|_{H^1}$ and $\|u\|_{L^2} \leq \|u\|_{H^1}$, and likewise for v, both terms are bounded by a multiple of $\|u\|_{H^1}\|v\|_{H^1}$. Adding them,

$$|a(u,v)| \leq \big(\Lambda + \|c\|_{L^\infty}\big)\,\|u\|_{H^1}\,\|v\|_{H^1}$$

so a is bounded with $M = \Lambda + \|c\|_{L^\infty}$.

Step 2: A lower bound on the gradient. Set $u = v$. Uniform ellipticity, applied pointwise with $\xi = \nabla u(x)$, gives $(A\nabla u) \cdot \nabla u = \nabla u^\top A \, \nabla u \geq \theta \, |\nabla u|^2$ almost everywhere. The zeroth-order term is nonnegative because $c \geq 0$: $\int_U c \, u^2 \, dx \geq 0$. Hence

$$a(u, u) = \int_U (A\nabla u) \cdot \nabla u \, dx + \int_U c \, u^2 \, dx \geq \theta \int_U |\nabla u|^2 \, dx = \theta \, \|\nabla u\|_{L^2}^2$$

This is where the sign hypothesis on c is used: a nonnegative c can only help the lower bound, never subtract from it. The estimate $a(u, u) \geq \theta \, \|\nabla u\|_{L^2}^2$ already controls the gradient part of the H^1 norm, but not the L^2 part; the Poincaré inequality gives the rest.

Step 3: Upgrading to the full norm by Poincaré. Because U is bounded and $u \in H_0^1(U)$ has zero boundary trace, the Poincaré inequality of [2, Poincaré Inequality] provides a constant $C_P = C_P(U) > 0$, depending only on U, with

$$\|u\|_{L^2} \leq C_P \, \|\nabla u\|_{L^2} \qquad \text{for all } u \in H_0^1(U)$$

Squaring and adding $\|\nabla u\|_{L^2}^2$ to both sides,

$$\|u\|_{H^1}^2 = \|u\|_{L^2}^2 + \|\nabla u\|_{L^2}^2 \leq \left(C_P^2 + 1\right) \|\nabla u\|_{L^2}^2$$

Therefore $\|\nabla u\|_{L^2}^2 \geq \|u\|_{H^1}^2 / (1 + C_P^2)$, and combining with Step 2,

$$a(u, u) \geq \theta \, \|\nabla u\|_{L^2}^2 \geq \frac{\theta}{1 + C_P^2} \, \|u\|_{H^1}^2$$

so a is coercive with $\alpha = \theta/(1 + C_P^2) > 0$, completing the proof.

The proof isolates the role of each hypothesis. The L^∞ bounds on A and c make a continuous, which is a statement about size alone and needs no positivity. Coercivity is where the analytic content sits: uniform ellipticity turns the principal term into a genuine lower bound on $\|\nabla u\|_{L^2}$, the sign condition $c \geq 0$ keeps the zeroth-order term from eroding that bound, and the Poincaré inequality converts a bound on the gradient into a bound on the whole H^1 norm. The Poincaré step is the one that fails on an unbounded domain or without the boundary condition, which is why the theory is set on $H_0^1(U)$ over a bounded U and not on all of $H^1(\mathbb{R}^n)$. The coercivity constant $\alpha = \theta/(1 + C_P^2)$ records this dependence transparently: it degrades as the ellipticity constant θ shrinks (the operator approaches degeneracy) and as the Poincaré constant C_P grows (the domain becomes large or thin in some direction).

With boundedness and coercivity in hand, the existence theorem is a direct application of Lax-Milgram. The right-hand side $f \in L^2(U)$ enters through the linear functional $v \mapsto \langle f, v \rangle = \int_U f v \, dx$, which is bounded on $H_0^1(U)$: by Cauchy-Schwarz and $\|v\|_{L^2} \leq \|v\|_{H^1}$, $|\langle f, v \rangle| \leq \|f\|_{L^2} \|v\|_{L^2} \leq \|f\|_{L^2} \|v\|_{H^1}$, so $\langle f, \cdot \rangle$ lies in the dual $H_0^1(U)^*$ with $\|\langle f, \cdot \rangle\|_{H_0^1(U)^*} \leq \|f\|_{L^2}$. This is the last ingredient the abstract theorem requires.

Theorem 1.3.4: Weak Solvability of the Dirichlet Problem

Let $U \subseteq \mathbb{R}^n$ be bounded and open, let $Lu = -\operatorname{div}(A\nabla u) + cu$ be uniformly elliptic in divergence form with A bounded, ellipticity constant θ, and $c \in L^\infty(U)$, $c \geq 0$. Then for every $f \in L^2(U)$ the Dirichlet problem

$$Lu = f \text{ in } U, \qquad u = 0 \text{ on } \partial U$$

has a unique weak solution $u \in H_0^1(U)$ in the sense of definition 1.1.2, that is, a unique $u \in H_0^1(U)$ with

$$a(u, v) = \int_U fv\, dx \qquad \text{for all } v \in H_0^1(U)$$

Moreover the solution obeys the energy estimate

$$\|u\|_{H^1} \leq \frac{1 + C_P^2}{\theta} \|f\|_{L^2}$$

where C_P is the Poincaré constant of U; in particular u depends continuously on f.

***Proof*:**

Take $H = H_0^1(U)$ with its H^1 norm, which is a Hilbert space, and let a be the elliptic form. By proposition 1.3.3, a is bounded and coercive on H with coercivity constant $\alpha = \theta/(1 + C_P^2)$. Define $F\colon H \to \mathbb{R}$ by $F(v) = \int_U fv\, dx$. As computed above, F is a bounded linear functional on H with $\|F\|_{H^*} \leq \|f\|_{L^2}$.

The hypotheses of the Lax-Milgram theorem (theorem 1.2.3) are met: a is a bounded coercive bilinear form on the Hilbert space H, and $F \in H^*$. That theorem produces a unique $u \in H$ with $a(u, v) = F(v)$ for all $v \in H$, which is precisely the statement that u is a weak solution of the Dirichlet problem in the sense of definition 1.1.2. Existence and uniqueness follow at once.

For the energy estimate, the same theorem gives the a priori bound $\|u\|_H \leq \|F\|_{H^*}/\alpha$. Substituting $\|F\|_{H^*} \leq \|f\|_{L^2}$ and $\alpha = \theta/(1 + C_P^2)$,

$$\|u\|_{H^1} \leq \frac{\|f\|_{L^2}}{\alpha} = \frac{1 + C_P^2}{\theta} \|f\|_{L^2}$$

The map $f \mapsto u$ is linear, since the weak formulation is linear in f and the solution is unique, so this bound is exactly the continuity of the solution operator $L^2(U) \to H_0^1(U)$, completing the proof.

The theorem is the first existence result of the book. For any L^2 datum f, the divergence-form Dirichlet problem is well-posed in $H_0^1(U)$: a solution exists, it is unique, and it varies continuously with the data, the three components of Hadamard well-posedness delivered in one stroke by an abstract Hilbert-space theorem. The price is that the solution lives in $H_0^1(U)$ and satisfies the equation only weakly, against test functions; whether this weak solution is in fact smooth, and solves $Lu = f$ pointwise, is the separate question of regularity taken up in the later chapters on the Schauder and L^p theories. The energy estimate is the quantitative core of the statement: it is the a priori bound that will recur throughout the elliptic theory, and its constant $(1 + C_P^2)/\theta$ exhibits the same two sources of degradation as the coercivity constant, a small ellipticity constant θ or a large domain constant C_P.

Example 1.3.5: The Dirichlet Problem for the Laplacian

Take $A = I$ and $c = 0$, so that $L = -\Delta$ and the Dirichlet problem is the Poisson equation

$$-\Delta u = f \text{ in } U, \qquad u = 0 \text{ on } \partial U$$

on a bounded open $U \subseteq \mathbb{R}^n$. The form is $a(u, v) = \int_U \nabla u \cdot \nabla v \, dx$, with ellipticity constant $\theta = 1$ and coefficient bound $\Lambda = 1$; the sign condition $c \geq 0$ holds trivially since $c = 0$. Theorem 1.3.4 applies and yields, for every $f \in L^2(U)$, a unique $u \in H_0^1(U)$ with $\int_U \nabla u \cdot \nabla v \, dx = \int_U f v \, dx$ for all $v \in H_0^1(U)$, obeying

$$\|u\|_{H^1} \leq (1 + C_P^2) \, \|f\|_{L^2}$$

Here the whole force of the theorem reduces to the coercivity of the Dirichlet form, which is the Poincaré inequality itself: $a(u, u) = \|\nabla u\|_{L^2}^2 \geq \|u\|_{H^1}^2/(1 + C_P^2)$. On the unit ball $U = B_1(0) \subseteq \mathbb{R}^n$ the Poincaré constant admits the explicit bound $C_P \leq 1/\sqrt{\lambda_1}$, where λ_1 is the first Dirichlet eigenvalue of the Laplacian on the ball, so the energy estimate reads $\|u\|_{H^1} \leq (1 + \lambda_1^{-1}) \, \|f\|_{L^2}$; the eigenvalue λ_1 reappears as the sharp coercivity threshold in the spectral theory of the operator.

The example shows the mechanism in its simplest form, but it also marks the boundary of the present method. Coercivity was secured by the sign condition $c \geq 0$, which guaranteed $\int_U c\, u^2 \geq 0$ and so kept the zeroth-order term from working against ellipticity. This condition is sufficient but not necessary, and it is genuinely used: if c is allowed to be negative, and negative enough, the term $\int_U c\, u^2$ can overwhelm $\theta\|\nabla u\|_{L^2}^2$ and drive $a(u, u)$ below zero for some $u \in H_0^1(U)$, at which point a is no longer coercive and Lax-Milgram no longer applies. Concretely, coercivity survives as long as $\|c^-\|_{L^\infty}$ stays below θ/C_P^2, where $c^- = \max(-c, 0)$ is the negative part; past that threshold the argument breaks. The indefinite case, where existence and uniqueness are governed by the spectrum of L and recovered through the Fredholm alternative rather than by coercivity, is treated in the chapter on existence by variational methods. For the coercive operators of this section the theory is complete: every L^2 datum produces exactly one finite-energy weak solution, with a bound.

2

The Sobolev Setup for Boundary-Value Problems

The variational method of chapter 1 produced weak solutions in the Sobolev space H_0^1, and the chapters that follow, existence by the Fredholm alternative, the two regularity theories, and the spectral decomposition, are all carried out in the Sobolev scale. This chapter assembles the function-space framework they use. It is deliberately short and citational: the Sobolev spaces, their embeddings, the trace theorem, the Poincaré inequality, and the Rellich compactness theorem are developed in the functional-analysis and harmonic-analysis volumes and are imported here as tools. What the chapter develops is the packaging those tools take for boundary-value problems, the negative space H^{-1} dual to H_0^1 and the Gelfand triple it completes, the compact embedding that makes the solution operator compact, and the reduction of inhomogeneous boundary data to the homogeneous case by a trace lift.

2.1 The Sobolev Scale and the Gelfand Triple

The task of this section is to name the imported results and then build the one object the boundary-value theory needs but the analysis volumes do not give in this form: the negative Sobolev space $H^{-1}(U)$ dual to the Dirichlet space, together with the Gelfand triple it completes. The weak equation $a(u,v) = \langle f, v\rangle$ of chapter 1 was solved for $f \in L^2(U)$, but its natural right-hand side is any bounded functional on $H_0^1(U)$, and the space of those functionals is exactly $H^{-1}(U)$. Once it is in place, the operator $L = -\operatorname{div}(A\nabla\cdot) + c$ reads as a bounded map $H_0^1(U) \to H^{-1}(U)$, and the equation $Lu = f$ becomes an equality in a single Banach space rather than a family of identities tested against every v.

Throughout, $U \subseteq \mathbb{R}^n$ is a bounded open set, with Lipschitz boundary ∂U where the trace enters; $H_0^1(U)$ carries the H^1 norm $\|u\|_{H^1}^2 = \|u\|_{L^2}^2 + \|\nabla u\|_{L^2}^2$, and $(u,v) = \int_U uv\,dx$ is the L^2 inner product.

The Sobolev machinery is developed in full in the analysis volumes; here it is recorded as a list of tools, each with a one-line statement and a citation naming what it provides for boundary-value problems. None of these is re-derived.

- *The Sobolev spaces and the scale.* For a nonnegative integer k, the space $H^k(U) = W^{k,2}(U)$ consists of the L^2 functions whose weak derivatives up to order k lie in $L^2(U)$, a Hilbert space under $\|u\|_{H^k}^2 = \sum_{|\alpha|\le k} \|\partial^\alpha u\|_{L^2}^2$; the L^p theory underlying weak derivatives is [3], and the $W^{k,p}$ spaces and their completeness are [2, Sobolev Space $W^{k,p}$, $W^{k,p}$ is a Banach Space]. The fractional and negative-order refinement, the scale H^s for real s defined on $\mathbb{R}^n$ by the Fourier weight $(1+|\xi|^2)^{s/2}$, is [4, Sobolev Spaces (H^s)]; it is the ambient scale in which H^1 and the H^{-1} built below both sit.
- *The Sobolev embedding theorem.* On a bounded Lipschitz domain, $H^k(U)$ embeds continuously into $L^q(U)$ for the appropriate q, and, once k exceeds $n/2$, into a Hölder space $C^{m,\gamma}(\overline{U})$ with m the largest integer strictly below $k - n/2$ [2]; this is what converts Sobolev regularity of a weak solution into pointwise regularity in the later chapters.
- *The trace theorem.* For a bounded Lipschitz domain there is a bounded linear trace operator $\operatorname{trace}\colon H^1(U) \to L^2(\partial U)$, restricting a smooth function to its boundary values, with image the fractional space $H^{1/2}(\partial U)$ and kernel exactly $H_0^1(U)$ [2]; this is what gives "$u = 0$ on ∂U" its meaning for a function that is only defined almost everywhere, and $H_0^1(U)$ is the space of finite-energy functions with vanishing trace.
- *The Poincaré inequality.* On a bounded U there is a constant $C_P = C_P(U) > 0$ with $\|u\|_{L^2} \le C_P\|\nabla u\|_{L^2}$ for all $u \in H_0^1(U)$ [2, Poincaré Inequality]; on $H_0^1(U)$ it makes $\|\nabla u\|_{L^2}$ an equivalent norm, and it is the inequality that produced coercivity in theorem 1.3.4.
- *The Rellich-Kondrachov theorem.* For a bounded Lipschitz domain the inclusion $H^1(U) \hookrightarrow L^2(U)$ is compact [2]; this is the compactness that will make the solution operator compact and open the door to the Fredholm alternative and the spectral theory.

These five tools are all that the elliptic theory draws from the Sobolev framework. The remainder of the chapter is the book's own: it packages them into the dual space and triple that the boundary-value problems are stated in.

Theorem 1.3.4 solved $Lu = f$ for $f \in L^2(U)$, but the argument only ever used f through the functional $v \mapsto \int_U fv\,dx$ on $H_0^1(U)$. Nothing in Lax-Milgram requires that functional to arise from an L^2 function: any bounded linear functional on $H_0^1(U)$ is an admissible right-hand side. Collecting all of them gives the correct space of data for the Dirichlet problem, and it is strictly larger than $L^2(U)$, large enough to contain distributions such as a divergence of an L^2 vector field or, in dimension one, a point evaluation. This dual space is what the negative index in H^{-1} records.

Definition 2.1.1: The Negative Sobolev Space

Let $U \subseteq \mathbb{R}^n$ be bounded and open. The *negative Sobolev space* $H^{-1}(U)$ is the continuous dual of the Dirichlet space,

$$H^{-1}(U) := \left(H_0^1(U)\right)^*$$

the Banach space of bounded linear functionals $f\colon H_0^1(U) \to \mathbb{R}$, equipped with the dual norm

$$\|f\|_{H^{-1}} = \sup\left\{\langle f, v\rangle \;\middle|\; v \in H_0^1(U),\ \|v\|_{H^1} \le 1\right\}$$

where $\langle f, v\rangle$ denotes the value of f at v, the H^{-1}-H_0^1 *duality pairing.*

The space is defined purely by duality, so it inherits completeness from the general fact that the dual of a normed space is a Banach space; no separate Sobolev construction is needed. Its elements are not functions in any pointwise sense but continuous functionals, and the pairing $\langle f, v\rangle$ is the only way they act. The dual norm measures how large f can make the pairing against a unit-norm test function, so a functional is small in H^{-1} precisely when it pairs weakly with every finite-energy function of controlled size. The notation $\langle \cdot, \cdot\rangle$ is chosen to extend the L^2 inner product: the next proposition shows that when f happens to lie in $L^2(U)$, the pairing $\langle f, v\rangle$ agrees with $\int_U fv\,dx$, so the two brackets never conflict.

Example 2.1.2: An L^2 Datum and a Point Source

Two functionals illustrate the range of $H^{-1}(U)$, one arising from a function and one that cannot.

1. Any $f \in L^2(U)$ defines an element of $H^{-1}(U)$ by $\langle f, v\rangle = \int_U fv\,dx$. This pairing is bounded on $H_0^1(U)$: by the Cauchy-Schwarz inequality and $\|v\|_{L^2} \le \|v\|_{H^1}$,

$$|\langle f, v\rangle| \le \|f\|_{L^2}\|v\|_{L^2} \le \|f\|_{L^2}\|v\|_{H^1}$$

so $f \in H^{-1}(U)$ with $\|f\|_{H^{-1}} \le \|f\|_{L^2}$. Thus every square-integrable datum is already an element of the negative space, and the inequality shows the inclusion is norm-decreasing.

2. Take $n = 1$ and $U = (-1, 1)$, and fix $x_0 \in U$. The point evaluation $\delta_{x_0}\colon v \mapsto v(x_0)$ is a bounded functional on $H_0^1(U)$: by the one-dimensional Sobolev embedding $H_0^1(U) \hookrightarrow C(\overline{U})$ from [2, Morrey's Inequality], there is a constant C with $|v(x_0)| \le \|v\|_{C(\overline{U})} \le C\|v\|_{H^1}$ for all $v \in H_0^1(U)$, so $\delta_{x_0} \in H^{-1}(U)$. It is not represented by any L^2 function, since no $g \in L^2(U)$ satisfies $\int_U gv\,dx = v(x_0)$ for all test functions v; the point source is a genuine element of $H^{-1}(U)$ outside $L^2(U)$, and the Green's function studied in the later potential-theory chapter is precisely the response of $-\Delta$ to such a source. In dimensions $n \ge 2$ the point mass leaves H^{-1}: on the H^s scale one has $\delta \in H^{-s}$ exactly when $s > n/2$ [4, Sobolev Spaces (H^s), Fourier Transforms Of Important Distributions], so $\delta \notin H^{-1}$ for $n \ge 2$. A divergence $\operatorname{div} F$ of an L^2 vector field F does remain a typical element beyond L^2.

The first item is the map that embeds $L^2(U)$ into $H^{-1}(U)$, and it is the hinge of the triple built next; the second shows that the negative space is genuinely larger, which is the reason to introduce it rather than work in L^2 throughout. With $H^{-1}(U)$ defined and shown to receive $L^2(U)$, the three spaces $H_0^1(U)$, $L^2(U)$, and $H^{-1}(U)$ can be assembled into the nested chain that organizes the whole theory.

The Dirichlet space sits inside $L^2(U)$, which in turn sits inside its dual $H^{-1}(U)$ through the pairing of the previous example. The two inclusions are continuous and dense, and the pairing that realizes the second one extends the L^2 inner product, so the same bracket $\langle\cdot,\cdot\rangle$ describes both the duality H^{-1} against H^1_0 and, on the overlap, the L^2 product. A chain of a Hilbert space between a dense subspace and its dual, with the middle space identified with its own dual, is a *Gelfand triple*; the one here is the setting in which every elliptic boundary-value problem of the book is posed.

Proposition 2.1.3: The Gelfand Triple

Let $U \subseteq \mathbb{R}^n$ be bounded and open. The inclusions

$$H^1_0(U) \hookrightarrow L^2(U) \hookrightarrow H^{-1}(U)$$

hold, where the first is the set inclusion and the second sends $f \in L^2(U)$ to the functional $\langle f,\cdot\rangle = \int_U f\,\cdot\,dx$ on $H^1_0(U)$. Both maps are continuous, injective, and have dense image, and the second extends the L^2 inner product in the sense that $\langle f,v\rangle = (f,v)$ for $f \in L^2(U)$ and $v \in H^1_0(U)$. Moreover the uniformly elliptic operator $L = -\operatorname{div}(A\nabla\cdot) + c$ of definition 1.3.1 maps $H^1_0(U)$ boundedly into $H^{-1}(U)$, and for $u \in H^1_0(U)$ the weak equation $Lu = f$ of definition 1.1.2 is the equality $Lu = f$ in $H^{-1}(U)$.

***Proof*:**
The two inclusions are treated in turn, then the extension property, and finally the operator L.

Step 1: The first inclusion. A function in $H^1_0(U)$ is in particular in $L^2(U)$, and $\|u\|_{L^2} \le \|u\|_{H^1}$ by the definition of the H^1 norm, so $H^1_0(U) \hookrightarrow L^2(U)$ is a continuous injection. Its image is dense because $C^\infty_c(U)$ is contained in $H^1_0(U)$ and is dense in $L^2(U)$ [2, Smooth Functions Dense in L^p]. A subspace containing a dense set is itself dense.

Step 2: The second inclusion. For $f \in L^2(U)$, item (1) of example 2.1.2 shows that $\langle f,\cdot\rangle = \int_U f\,\cdot\,dx$ is a bounded functional on $H^1_0(U)$ with $\|\langle f,\cdot\rangle\|_{H^{-1}} \le \|f\|_{L^2}$, so the map $f \mapsto \langle f,\cdot\rangle$ carries $L^2(U)$ into $H^{-1}(U)$ and is continuous. It is injective: if $\langle f,\cdot\rangle = 0$ then $\int_U fv\,dx = 0$ for all $v \in H^1_0(U)$, hence for all $v \in C^\infty_c(U)$, and a function orthogonal to every test function vanishes almost everywhere [2, The du Bois-Reymond Lemma], so $f = 0$ in $L^2(U)$. What carries the step is the inclusion $C^\infty_c(U) \subseteq H^1_0(U)$ together with this fundamental lemma. Density of the image in $H^{-1}(U)$ holds because $L^2(U)$ already contains $C^\infty_c(U)$, which is dense in $H^{-1}(U)$ by duality with the reflexive Hilbert space $H^1_0(U)$ [2, Hahn-Banach Consequences, Natural Isomorphism To $\mathcal{H}^*$ (Riesz Representation)].

Step 3: The pairing extends the inner product. For $f \in L^2(U)$ and $v \in H^1_0(U)$, the definition of the second inclusion gives $\langle f,v\rangle = \int_U fv\,dx = (f,v)$ directly, which is the asserted agreement of the duality pairing with the L^2 inner product on their common domain. The single bracket $\langle\cdot,\cdot\rangle$ is therefore unambiguous.

Step 4: The operator L. For $u, v \in H^1_0(U)$ set

$$\langle Lu, v\rangle := a(u,v) = \int_U (A\nabla u)\cdot\nabla v + c\,uv\,dx$$

the elliptic form of chapter 1. For fixed u this is linear in v, and by the boundedness of a established there, $|\langle Lu,v\rangle| \le M\|u\|_{H^1}\|v\|_{H^1}$ with $M = \Lambda + \|c\|_{L^\infty}$, so $\langle Lu,\cdot\rangle$ is a bounded functional on $H^1_0(U)$, that is, $Lu \in H^{-1}(U)$ with $\|Lu\|_{H^{-1}} \le M\|u\|_{H^1}$. Since $u \mapsto Lu$ is linear, $L\colon H^1_0(U) \to H^{-1}(U)$ is a bounded linear operator. Finally, for $f \in H^{-1}(U)$ the statement that u is a weak solution (definition 1.1.2), namely $a(u,v) = \langle f, v\rangle$ for all $v \in H_0^1(U)$, reads $\langle Lu, v\rangle = \langle f, v\rangle$ for all v, which is the equality $Lu = f$ of elements of $H^{-1}(U)$, completing the proof.

The proposition recasts the weak formulation as a single equation in a single space. In chapter 1 the statement "u is a weak solution" was a family of scalar identities, one for each test function v; here it is the one equation $Lu = f$ between two elements of $H^{-1}(U)$, and the family of identities is just the unwinding of that equality through the pairing. The gain is conceptual economy. The operator L, which acts pointwise as a second-order differential operator on smooth functions, is now a bounded linear map between Banach spaces, and questions about solving $Lu = f$ become questions about the map $L\colon H_0^1(U) \to H^{-1}(U)$, whether it is injective, surjective, invertible. Theorem 1.3.4, read through the triple, is the assertion that for a coercive L this map is a bounded bijection onto $H^{-1}(U)$ with bounded inverse; the energy estimate is the bound on L^{-1}.

The chain also fixes an identification that is easy to misread. The middle space $L^2(U)$ is identified with its own dual by the Riesz representation theorem [2, Natural Isomorphism To $\mathcal{H}^*$ (Riesz Representation)], and it is that identification, not another, that makes $L^2(U)$ sit *inside* $H^{-1}(U)$ rather than only map to it: an L^2 function is its own Riesz functional, restricted to test functions. The two end spaces $H_0^1(U)$ and $H^{-1}(U)$ are dual to each other but are deliberately *not* identified with their duals, since identifying $H_0^1(U)$ with its dual as well would force $H_0^1(U) = L^2(U) = H^{-1}(U)$ and collapse the triple. Keeping the middle identification and only the middle one is what gives the scale its three distinct floors, the finite-energy functions, the square-integrable functions, and the functionals, with L climbing down two of them from H_0^1 to H^{-1}.

2.2 Compactness and Inhomogeneous Boundary Data

Two tasks remain. The first is *compactness*: on a bounded domain the inclusion $H_0^1(U) \hookrightarrow L^2(U)$ is not just bounded but compact, and this single fact promotes the solution operator L^{-1} from a bounded operator to a compact one. The second is the treatment of a nonzero boundary condition: the existence theory of chapter 1 solved only the homogeneous problem $u = 0$ on ∂U, and the trace theorem lets an inhomogeneous datum $u = g$ be absorbed into the right-hand side, reducing it to the case already solved.

The compactness of the inclusion is the Rellich-Kondrachov theorem, imported from the functional-analysis volume. On a bounded domain a set of functions bounded in $H_0^1(U)$, hence with uniformly controlled gradients, cannot oscillate or escape to infinity freely, and the theorem makes this precise: bounded sequences in $H_0^1(U)$ have L^2-convergent subsequences. Boundedness of U is essential here, exactly as it was for the Poincaré inequality; on $\mathbb{R}^n$ the inclusion is bounded but not compact, since a bump function may be translated to infinity without changing its norm.

Theorem 2.2.1: Compactness of the Solution Operator

Let $U \subseteq \mathbb{R}^n$ be bounded and open. Then the inclusion

$$H_0^1(U) \hookrightarrow L^2(U)$$

is a compact operator (Rellich-Kondrachov, [2]). Consequently, if $Lu = -\operatorname{div}(A\nabla u) + cu$ is uniformly elliptic in divergence form with A bounded and $c \in L^\infty(U)$, $c \geq 0$, so that the coercive existence theory of chapter 1 applies, then the solution operator

$$L^{-1}\colon L^2(U) \to L^2(U)$$

that sends a datum f to the unique weak solution of $Lu = f$, $u = 0$ on ∂U, is a compact operator.

***Proof*:**
The compactness of the inclusion is the Rellich-Kondrachov theorem for the bounded domain U, whose proof is given in [2]; it is imported here and not re-derived. The $H_0^1(U) \hookrightarrow L^2(U)$ form used here is the restriction of the Lipschitz-domain statement $H^1(U) \hookrightarrow L^2(U)$ recorded in section 2.1 to the closed subspace $H_0^1(U)$, and it needs no boundary regularity: for bounded U it follows on extending by zero to a large ball, so the theorem holds for any bounded open U. The claim to prove is that the solution operator, viewed as a map $L^2(U) \to L^2(U)$, inherits compactness from it.

The argument is a factorization. By theorem 1.3.4, for every $f \in L^2(U)$ the problem $Lu = f$, $u = 0$ on ∂U, has a unique weak solution $u \in H_0^1(U)$, and the energy estimate gives

$$\|u\|_{H^1} \leq C\,\|f\|_{L^2}$$

with C a constant depending only on the Poincaré constant C_P of U and the ellipticity constant θ. Writing $S\colon L^2(U) \to H_0^1(U)$ for the map $f \mapsto u$, this estimate says exactly that S is a bounded linear operator, with operator norm at most C. Let $\iota\colon H_0^1(U) \hookrightarrow L^2(U)$ denote the inclusion. The solution operator $L^{-1}\colon L^2(U) \to L^2(U)$ is the composite

$$L^{-1} = \iota \circ S$$

since applying S produces the solution $u \in H_0^1(U)$ and ι views it back in $L^2(U)$.

Now S is bounded and ι is compact by the first part. The composition of a compact operator with a bounded operator is compact: if (f_k) is a bounded sequence in $L^2(U)$, then (Sf_k) is a bounded sequence in $H_0^1(U)$ by boundedness of S, and compactness of ι extracts a subsequence along which $(\iota S f_k)$ converges in $L^2(U)$. Thus $L^{-1} = \iota \circ S$ maps bounded sequences to sequences with L^2-convergent subsequences, which is compactness, as we sought to show.

The theorem is the structural payoff of setting the problem on a bounded domain. Coercivity, in chapter 1, made L^{-1} exist and be bounded; boundedness of U, through Rellich, makes it compact. The mechanism is worth isolating, because it recurs whenever an elliptic problem is inverted: the inverse gains a full derivative of regularity, landing its output in $H_0^1(U)$ rather than $L^2(U)$, and on a bounded domain that gain in regularity is what the compact embedding converts into compactness of the operator. Symbolically, L^{-1} is compact because it factors through the smaller space $H_0^1(U)$, and the inclusion of that smaller space into $L^2(U)$ is compact.

Compactness of L^{-1} is the hinge on which the later chapters turn. A compact operator on a Hilbert space is a norm-limit of finite-rank operators, and its spectral and Fredholm theory is as rigid as that of a matrix. Two uses stand out. The Fredholm alternative for the indefinite elliptic problem, where the sign condition $c \geq 0$ fails and coercivity is lost, is obtained by writing the equation as a compact perturbation of the identity and invoking the Riesz-Schauder theory of compact operators; the existence chapter carries this out. And the spectral theory of the operator L, the existence of a discrete sequence of Dirichlet eigenvalues $\lambda_1 \leq \lambda_2 \leq \cdots \to \infty$ with an orthonormal basis of eigenfunctions, follows from the spectral theorem for compact self-adjoint operators applied to L^{-1}; the spectral chapter carries this out. In both cases the analytic input is precisely the compactness established here, and the abstract functional-analytic machinery gives the rest.

Example 2.2.2: The Resolvent of the Dirichlet Laplacian

Take $A = I$ and $c = 0$, so that $L = -\Delta$ on a bounded open $U \subseteq \mathbb{R}^n$ with zero Dirichlet data. For each $f \in L^2(U)$ the Poisson problem $-\Delta u = f$, $u = 0$ on ∂U, has a unique weak solution $u \in H_0^1(U)$ by theorem 1.3.4, and the resulting operator $(-\Delta)^{-1} \colon L^2(U) \to L^2(U)$ is compact by theorem 2.2.1. It is also self-adjoint, since the Dirichlet form $a(u,v) = \int_U \nabla u \cdot \nabla v \, dx$ is symmetric: for $f, h \in L^2(U)$ with solutions $u = (-\Delta)^{-1} f$ and $w = (-\Delta)^{-1} h$,

$$\big((-\Delta)^{-1} f, h\big) = (u, h) = a(u, w) = a(w, u) = (w, f) = \big(f, (-\Delta)^{-1} h\big)$$

where the middle equality is the symmetry $a(u,w) = a(w,u)$ of the Dirichlet form. Thus $(-\Delta)^{-1}$ is a compact self-adjoint operator on $L^2(U)$, and the spectral theorem produces an orthonormal basis (φ_k) of $L^2(U)$ of eigenfunctions, $(-\Delta)^{-1}\varphi_k = \mu_k \varphi_k$ with $\mu_k > 0$ and $\mu_k \to 0$. Inverting, the φ_k are the Dirichlet eigenfunctions of $-\Delta$ itself, with eigenvalues $\lambda_k = 1/\mu_k$ satisfying $0 < \lambda_1 \leq \lambda_2 \leq \cdots \to \infty$. The compactness proved above is exactly what forces the spectrum to be this discrete, unbounded sequence rather than a continuum.

The remaining loose end of the setup is the boundary condition. The existence theorem of chapter 1 solved $Lu = f$ subject to $u = 0$ on ∂U, encoded by seeking the solution in $H_0^1(U)$, the space of functions with vanishing trace. A genuine boundary-value problem prescribes instead a nonzero datum $u = g$ on ∂U, and the question is what class g may range over and how the inhomogeneous problem connects to the homogeneous one. Both answers come from the trace theorem. The trace operator, imported from the functional-analysis volume, restricts a function on U to its boundary values, and its precise target is the fractional space $H^{1/2}(\partial U)$: the trace maps $H^1(U)$ *onto* $H^{1/2}(\partial U)$, so a datum g is admissible exactly when it is the trace of some H^1 function on U. Surjectivity of the trace is the key: it guarantees a function inside the domain that realizes the prescribed boundary values, and subtracting it turns the inhomogeneous problem into a homogeneous one.

Theorem 2.2.3: Reduction of Inhomogeneous Dirichlet Data

Let $U \subseteq \mathbb{R}^n$ be bounded and open with Lipschitz boundary, let $L = -\operatorname{div}(A\nabla\cdot) + c$ be uniformly elliptic in divergence form as in chapter 1 with A bounded and $c \in L^\infty(U)$, $c \geq 0$, so that the coercive existence theory applies, let $f \in H^{-1}(U)$, and let $g \in H^{1/2}(\partial U)$. Then the inhomogeneous Dirichlet problem

$$Lu = f \text{ in } U, \qquad u = g \text{ on } \partial U$$

admits a unique solution $u \in H^1(U)$ with $\operatorname{trace} u = g$ and $Lu = f$ in $H^{-1}(U)$. It is obtained as follows. By surjectivity of the trace operator ([2]), choose a lift $G \in H^1(U)$ with $\operatorname{trace} G = g$. Then $w = u - G$ solves the homogeneous problem

$$Lw = f - LG \text{ in } U, \qquad w = 0 \text{ on } \partial U$$

in $H_0^1(U)$, which theorem 1.3.4 solves, and $u = w + G$. The reduction is exact: for G fixed with $\operatorname{trace} G = g$, a function $u \in H^1(U)$ has $\operatorname{trace} u = g$ if and only if $w = u - G$ lies in $H_0^1(U)$.

***Proof*:**
The proof has two parts: the lift exists and translates the boundary condition, and the residual equation for w falls to the homogeneous theory.

Step 1: The lift and the exact reduction. Because $g \in H^{1/2}(\partial U)$ lies in the range of the trace operator, which maps $H^1(U)$ onto $H^{1/2}(\partial U)$ by trace surjectivity ([2]), there is a $G \in H^1(U)$ with $\operatorname{trace} G = g$. Set $w = u - G$. The trace operator is linear, so $\operatorname{trace} w = \operatorname{trace} u - \operatorname{trace} G = \operatorname{trace} u - g$. Since $H_0^1(U)$ is the kernel of the trace operator on $H^1(U)$, the membership $w \in H_0^1(U)$ holds if and only if $\operatorname{trace} w = 0$, that is, if and only if $\operatorname{trace} u = g$. This is the exactness of the reduction: prescribing the boundary value g on u is the same as demanding zero boundary value on w, with no loss in either direction. In particular the choice of lift G is immaterial to whether a candidate u satisfies the boundary condition; two lifts differ by an element of $H_0^1(U)$, which is absorbed into w.

Step 2: The residual equation. With G fixed, seek u in the form $u = w + G$ with $w \in H_0^1(U)$; by Step 1 any such u automatically has $\operatorname{trace} u = g$. Because L is linear, the equation $Lu = f$ becomes $Lw + LG = f$, that is,

$$Lw = f - LG$$

The right-hand side lies in $H^{-1}(U)$: the datum f is in $H^{-1}(U)$ by hypothesis, and $LG \in H^{-1}(U)$ because LG is by definition the functional $v \mapsto a(G, v)$ on $H_0^1(U)$, which is bounded since the elliptic form a is bounded on $H^1(U) \times H^1(U)$: for $G \in H^1(U)$ and $v \in H_0^1(U)$,

$$|a(G, v)| \leq M \, \|G\|_{H^1} \|v\|_{H^1}$$

with $M = \Lambda + \|c\|_{L^\infty}$ the bound on a from chapter 1. (Only test functions $v \in H_0^1(U)$ enter, so this is exactly what it means for LG to be an element of $H^{-1}(U) = (H_0^1(U))^*$; the lift G need not itself lie in $H_0^1(U)$.) Thus $f - LG$ is an admissible right-hand side for the homogeneous problem

$$Lw = f - LG \text{ in } U, \qquad w = 0 \text{ on } \partial U$$

and theorem 1.3.4 produces a $w \in H_0^1(U)$ solving it weakly, meaning $a(w,v) = \langle f - LG, v\rangle$ for all $v \in H_0^1(U)$. Setting $u = w + G$ then gives a function in $H^1(U)$ with $\operatorname{trace} u = g$ and $Lu = f$ in $H^{-1}(U)$, establishing existence.

Step 3: Uniqueness. Suppose $u_1, u_2 \in H^1(U)$ both solve the problem, so that $\operatorname{trace} u_1 = \operatorname{trace} u_2 = g$ and $Lu_1 = Lu_2 = f$ in $H^{-1}(U)$. Their difference $d = u_1 - u_2$ has $\operatorname{trace} d = 0$, hence $d \in H_0^1(U)$, and $Ld = 0$ in $H^{-1}(U)$, that is, $a(d,v) = 0$ for all $v \in H_0^1(U)$. By theorem 1.3.4 the homogeneous problem $Ld = 0$, $d = 0$ on ∂U has a unique weak solution in $H_0^1(U)$, namely $d = 0$; equivalently, testing $a(d,v) = 0$ against $v = d$ and using coercivity gives $\theta\|d\|_{H^1}^2 \le a(d,d) = 0$. Thus $u_1 = u_2$, and the solution is unique, completing the proof. The choice of lift G is immaterial: two lifts differ by an element of $H_0^1(U)$, which is absorbed into w so that $u = w + G$ is the same function.

The reduction is the standard device for carrying a boundary condition into the Sobolev framework, and it isolates cleanly what each ingredient gives. Trace surjectivity provides the lift G: without a function inside U realizing the prescribed boundary values, there would be nothing to subtract, and the inhomogeneous problem could not be reached from the homogeneous one. The linearity of L turns the subtraction $u = w + G$ into a genuine simplification, moving the boundary datum onto the right-hand side as the correction $-LG$, and the boundedness of the elliptic form a keeps that correction in $H^{-1}(U)$, the space where the homogeneous theory operates. The solution u is not unique as a lift decomposition, since G may be chosen in many ways, but the solution u itself is unique: different lifts produce different w, yet $u = w + G$ is the same, because the coercive homogeneous problem has a unique solution. The construction shows that the essential difficulty of a linear boundary-value problem is entirely in the interior equation. Once the homogeneous problem is solved, any admissible boundary datum is handled for free.

Example 2.2.4: Prescribed Boundary Values for the Laplacian

Consider the inhomogeneous Dirichlet problem for the Laplacian on a bounded Lipschitz domain $U \subseteq \mathbb{R}^n$,

$$-\Delta u = f \text{ in } U, \qquad u = g \text{ on } \partial U$$

with $f \in L^2(U) \subseteq H^{-1}(U)$ and $g \in H^{1/2}(\partial U)$. By trace surjectivity there is a lift $G \in H^1(U)$ with $\operatorname{trace} G = g$; a concrete choice, when g extends to a function $\tilde g \in H^1(\mathbb{R}^n)$, is the restriction $G = \tilde g|_U$. Set $w = u - G$. Then w solves

$$-\Delta w = f + \Delta G \text{ in } U, \qquad w = 0 \text{ on } \partial U$$

where the right-hand side $f + \Delta G$ is read in $H^{-1}(U)$: even though ΔG need not lie in $L^2(U)$, it is the functional $v \mapsto -\int_U \nabla G \cdot \nabla v\,dx$ on $H_0^1(U)$, which is bounded because $G \in H^1(U)$. Theorem 1.3.4 yields a unique $w \in H_0^1(U)$ with

$$\int_U \nabla w \cdot \nabla v\,dx = \int_U f v\,dx - \int_U \nabla G \cdot \nabla v\,dx \qquad \text{for all } v \in H_0^1(U)$$

and $u = w + G$ is the finite-energy solution taking the boundary value g. The harmonic case $f = 0$ is the classical Dirichlet problem: u is the function of least Dirichlet energy $\int_U |\nabla u|^2$ among all $H^1(U)$ functions with trace g, and the equation $-\Delta u = 0$ is the Euler-Lagrange condition for that minimization.

Taken together, the Lax-Milgram existence theory of chapter 1 and the Sobolev packaging of this chapter have set up the linear boundary-value problem for the elliptic setting, and it is worth recording which imported tool did which job. The Poincaré inequality was the coercivity input: it converted the ellipticity lower bound on the gradient into a bound on the full H^1 norm, and without it the Lax-Milgram theorem of chapter 1 would have had no coercive form to invert. The Rellich-Kondrachov theorem was the compactness input, promoting the bounded solution operator to a compact one and so opening the Fredholm and spectral theories to come. The trace theorem was what carried the boundary condition into the Sobolev setting, defining the Dirichlet space $H_0^1(U)$ as the kernel of the trace and, through its surjectivity onto $H^{1/2}(\partial U)$, reducing inhomogeneous data to the homogeneous case. All three are imported from the analysis volumes and used; the work of this chapter has been to assemble them, with the negative space $H^{-1}(U)$ and the Gelfand triple, into the exact package the existence, regularity, and spectral chapters use.

3

Existence by Variational Methods

Chapter 1 solved the Dirichlet problem for a coercive operator by the Lax-Milgram theorem; this chapter completes the existence theory in two directions. When the operator is symmetric, the weak solution is not only produced by an abstract theorem but characterized as the minimizer of an energy, and the direct method of the calculus of variations produces that minimizer from coercivity and lower semicontinuity alone. When the operator is not coercive, because a lower-order term carries the wrong sign, existence and uniqueness can fail together, and both are then governed by the Fredholm alternative, which the compact solution operator of chapter 2 places at our disposal. The chapter closes with worked boundary-value problems that exercise both routes.

3.1 The Energy Functional and the Direct Method

For a symmetric coercive operator the weak equation $a(u, v) = \langle f, v\rangle$ is not the only way to pin down its solution. The symmetric case of Lax-Milgram (proposition 1.2.4) already recorded that the weak solution is the minimizer of an energy, but it produced that minimizer by representing f in the inner product a, which is Riesz representation in disguise and leans on the abstract theorem of chapter 1. This section gives the minimizer a second, intrinsic construction that never invokes Lax-Milgram: the *direct method* of the calculus of variations, which manufactures a minimizer out of coercivity and lower semicontinuity alone. The payoff is twofold. The direct method is the method that survives into the nonlinear and constrained problems where no bilinear form and no Riesz theorem are available, so it is worth seeing in the linear case where every step can be checked by hand; and it exhibits the three properties of J that any existence-by-minimization argument must verify, isolated one at a time. The section closes by turning the same energy on the Neumann problem, where the test space widens to $H^1(U)$ and the boundary condition emerges from the variational formulation rather than being imposed on it.

Throughout, a is the elliptic form $a(u, v) = \int_U (A\nabla u) \cdot \nabla v + c\, uv\, dx$ of definition 1.1.2 attached to $L = -\operatorname{div}(A\nabla\cdot) + c$, now assumed *symmetric* ($A(x)$ symmetric for a.e. x, so $a(u,v) = a(v,u)$), and $f \in H^{-1}(U)$ is the datum, paired against test functions through the H^{-1}-H_0^1 bracket $\langle\cdot,\cdot\rangle$ of definition 2.1.1.

Definition 3.1.1: The Energy Functional

Let $U \subseteq \mathbb{R}^n$ be bounded and open, let $a\colon H_0^1(U) \times H_0^1(U) \to \mathbb{R}$ be a symmetric bounded coercive bilinear form, and let $f \in H^{-1}(U)$. The *energy functional* associated to a and f is the map $J\colon H_0^1(U) \to \mathbb{R}$ defined by

$$J(v) = \tfrac{1}{2}\, a(v,v) - \langle f, v\rangle$$

Its first term $\frac{1}{2}\, a(v,v)$ is the *internal energy* of v, and its second term $\langle f, v\rangle$ is the *work* done by the datum f against v.

The functional is the sum of a quadratic form and a linear functional, so it is the exact analog of the scalar $\frac{1}{2}At^2 - bt$ whose minimizer is $t = b/A$: minimizing J is the infinite-dimensional version of completing the square, and the coercivity constant α plays the role of the positive leading coefficient A that makes the parabola open upward. The two terms compete. The internal energy $\frac{1}{2}a(v,v)$ grows quadratically as v leaves the origin and penalizes large gradients, while the work $\langle f, v\rangle$ grows only linearly and rewards alignment with the datum. A minimizer balances the two, and the balance condition, that no first-order perturbation lowers J, is precisely the weak equation. The definition asks nothing of f beyond membership in $H^{-1}(U)$, so the same functional covers an L^2 source, where $\langle f, v\rangle = \int_U f v\, dx$, and a rougher datum such as a point source.

Example 3.1.2: The Dirichlet Energy on the Interval

Take $n = 1$, $U = (0,1)$, and $L = -\Delta$, so $a(u,v) = \int_0^1 u'v'\, dx$ and, for $f \equiv 1$,

$$J(v) = \tfrac{1}{2}\int_0^1 (v')^2\, dx - \int_0^1 v\, dx$$

on $H_0^1(0,1)$. This is the classical Dirichlet energy of v minus the work of the unit source. Evaluate it on the classical solution $u(x) = \frac{1}{2}x(1-x)$ of example 1.1.3, for which $u'(x) = \frac{1}{2} - x$: direct calculation gives $\int_0^1 (u')^2\, dx = \int_0^1 (\frac{1}{2} - x)^2\, dx = \frac{1}{12}$ and $\int_0^1 u\, dx = \int_0^1 \frac{1}{2}x(1-x)\, dx = \frac{1}{12}$, so $J(u) = \frac{1}{2}\cdot\frac{1}{12} - \frac{1}{12} = -\frac{1}{24}$. To see that no competitor does better, perturb by $h \in H_0^1(0,1)$ and expand, using $\int_0^1 u'h'\, dx = \int_0^1 h\, dx$ (the weak equation for u, verified in example 1.1.3):

$$J(u+h) = J(u) + \Big(\int_0^1 u'h'\, dx - \int_0^1 h\, dx\Big) + \tfrac{1}{2}\int_0^1 (h')^2\, dx = J(u) + \tfrac{1}{2}\int_0^1 (h')^2\, dx$$

The cross term vanishes by the weak equation, and the remaining term is nonnegative and zero only when $h' \equiv 0$, that is $h \equiv 0$ since $h(0) = 0$. So u is the strict minimizer, with value $-\frac{1}{24}$, and J falls below zero exactly because the work of the source outweighs half the energy it costs to accommodate it.

The example computed the minimizer from a solution already in hand and confirmed after the fact that it minimizes J. The existence theory must run the other way: it must produce a minimizer without knowing a solution in advance, and then read the weak equation off the minimization.

Chapter 1 already gave one half of the bridge between the two problems, the direction that requires the abstract theorem; stating both directions together as a single equivalence is the natural starting point, and the direct method then discharges the existence of the minimizer on its own terms.

The next result is Dirichlet's principle in the concrete elliptic setting, with the existence of the minimizer proved from scratch by the direct method. The equivalence of the two problems, weak solution versus energy minimizer, is inherited from proposition 1.2.4; the new content is the construction of the minimizer through the three properties of J that the direct method isolates, coercivity, reflexivity, and weak lower semicontinuity, each proved in its own step.

Theorem 3.1.3: Dirichlet's Principle and the Direct Method

Let $U \subseteq \mathbb{R}^n$ be bounded and open, let a be a symmetric bounded coercive bilinear form on $H_0^1(U)$ with bound M and coercivity constant α, and let $f \in H^{-1}(U)$. Let J be the energy functional of definition 3.1.1. Then:

1. A function $u \in H_0^1(U)$ minimizes J over $H_0^1(U)$ if and only if u is the weak solution of $Lu = f$, that is $a(u, v) = \langle f, v\rangle$ for all $v \in H_0^1(U)$.
2. J attains its infimum on $H_0^1(U)$: there exists $u \in H_0^1(U)$ with $J(u) = \inf_{v\in H_0^1(U)} J(v)$, and this u is unique.

***Proof*:**
Part (1): minimizer equals weak solution. Fix $u \in H_0^1(U)$ and let $h \in H_0^1(U)$ be arbitrary. Expanding $J(u + h)$ with bilinearity and symmetry of a,

$$J(u+h) = \tfrac{1}{2}\, a(u,u) + a(u,h) + \tfrac{1}{2}\, a(h,h) - \langle f,u\rangle - \langle f,h\rangle = J(u) + \big(a(u,h) - \langle f,h\rangle\big) + \tfrac{1}{2}\, a(h,h)$$

Suppose first u is the weak solution, so $a(u, h) = \langle f, h\rangle$ for every h. The middle term vanishes and $J(u + h) = J(u) + \frac{1}{2}a(h, h) \geq J(u)$ by coercivity, with equality only when $a(h, h) = 0$, that is $h = 0$; so u minimizes J. Conversely, suppose u minimizes J. For fixed h and $t \in \mathbb{R}$ the scalar function $\varphi(t) = J(u + th) = J(u) + t\big(a(u, h) - \langle f, h\rangle\big) + \frac{t^2}{2}\, a(h, h)$ is a parabola in t with a minimum at $t = 0$, so $\varphi'(0) = a(u, h) - \langle f, h\rangle = 0$. As h was arbitrary, $a(u, h) = \langle f, h\rangle$ for all $h \in H_0^1(U)$, so u is the weak solution. This identity $\varphi'(0) = 0$ is the vanishing of the first variation of J, and the weak equation is its Euler-Lagrange equation.

It remains to prove part (2), the existence and uniqueness of the minimizer, by the direct method. The method has a fixed shape: bound a minimizing sequence using coercivity, extract a weakly convergent subsequence using reflexivity, and pass the infimum through the limit using weak lower semicontinuity. Each is a separate step.

Step 1: The infimum is finite and every minimizing sequence is bounded. Write $m = \inf_{v\in H_0^1(U)} J(v)$. Coercivity and the definition of the dual norm give, for any $v \in H_0^1(U)$,

$$J(v) = \tfrac{1}{2}\, a(v,v) - \langle f,v\rangle \geq \tfrac{\alpha}{2}\, \|v\|_{H_0^1}^2 - \|f\|_{H^{-1}}\|v\|_{H_0^1}$$

The right-hand side is a quadratic in $t = \|v\|_{H_0^1} \geq 0$ with positive leading coefficient $\alpha/2$, so it is bounded below over $t \geq 0$; its minimum, attained at $t = \|f\|_{H^{-1}}/\alpha$, equals $-\|f\|_{H^{-1}}^2/(2\alpha)$. Hence

$$J(v) \geq -\frac{\|f\|_{H^{-1}}^2}{2\alpha} \qquad \text{for all } v \in H_0^1(U)$$

so $m \geq -\|f\|_{H^{-1}}^2/(2\alpha) > -\infty$ is finite. Now let (u_k) be a *minimizing sequence*, meaning $u_k \in H_0^1(U)$ and $J(u_k) \to m$. Then the numbers $J(u_k)$ are bounded, say $J(u_k) \leq m+1$ for large k, and the lower bound above rearranges to

$$\tfrac{\alpha}{2}\,\|u_k\|_{H_0^1}^2 \leq J(u_k) + \|f\|_{H^{-1}}\|u_k\|_{H_0^1} \leq (m+1) + \|f\|_{H^{-1}}\|u_k\|_{H_0^1}$$

which is again a quadratic inequality forcing $\|u_k\|_{H_0^1}$ to stay bounded: if $t_k = \|u_k\|_{H_0^1}$ then $\frac{\alpha}{2}t_k^2 - \|f\|_{H^{-1}}t_k - (m+1) \leq 0$, and a quadratic with positive leading coefficient is nonpositive only on a bounded interval of t_k. Thus $\sup_k \|u_k\|_{H_0^1} =: R < \infty$. Coercivity is exactly what makes the minimizing sequence bounded: a functional that could stay small while its argument runs off to infinity would give no such control.

Step 2: A weakly convergent subsequence. The space $H_0^1(U)$ is a Hilbert space, hence reflexive, so every bounded sequence has a weakly convergent subsequence [2, Weak Sequential Compactness Of Bounded Sets]: reflexivity is the property that closed bounded sets are weakly sequentially compact, the substitute for the Bolzano-Weierstrass theorem that norm-boundedness loses in infinite dimensions. Applying this to the bounded sequence (u_k) of Step 1 yields a subsequence, still written (u_k), and a limit $u \in H_0^1(U)$ with

$$u_k \rightharpoonup u \quad \text{in } H_0^1(U)$$

that is, $\langle \Lambda, u_k\rangle \to \langle \Lambda, u\rangle$ for every bounded linear functional Λ on $H_0^1(U)$. Weak convergence is genuinely weaker than norm convergence in infinite dimensions, and the whole difficulty of the method is concentrated in the next step: passing the infimum through a limit that need not be a norm limit.

Step 3: J is weakly lower semicontinuous. The claim is that

$$u_k \rightharpoonup u \quad \Longrightarrow \quad J(u) \leq \liminf_{k\to\infty} J(u_k)$$

The linear part is the easy half. The functional $v \mapsto \langle f, v\rangle$ is bounded and linear, so by the definition of weak convergence $\langle f, u_k\rangle \to \langle f, u\rangle$; a linear functional is weakly continuous, not lower semicontinuous. The quadratic part carries the work. Because a is symmetric bounded coercive it is an inner product on $H_0^1(U)$ whose norm $\|v\|_a = a(v,v)^{1/2}$ is equivalent to $\|\cdot\|_{H_0^1}$ (proposition 1.2.4), so weak convergence in $\|\cdot\|_{H_0^1}$ is weak convergence in $\|\cdot\|_a$, and it suffices to show the a-norm is weakly lower semicontinuous. Expanding the nonnegative quantity $a(u_k - u, u_k - u) \geq 0$ and using bilinearity,

$$a(u_k, u_k) \geq 2\,a(u, u_k) - a(u,u)$$

Now $v \mapsto a(u, v)$ is a bounded linear functional (it is $\langle Au, \cdot\rangle$ in the notation of theorem 1.2.3), so $a(u, u_k) \to a(u, u)$ by weak convergence. Taking lim inf across the inequality,

$$\liminf_{k\to\infty} a(u_k, u_k) \geq 2\, a(u, u) - a(u, u) = a(u, u)$$

which is the weak lower semicontinuity of $v \mapsto a(v, v)$. Combining the two parts,

$$J(u) = \tfrac{1}{2}\, a(u, u) - \langle f, u\rangle \leq \liminf_{k\to\infty} \Big(\tfrac{1}{2}\, a(u_k, u_k) - \langle f, u_k\rangle\Big) = \liminf_{k\to\infty} J(u_k)$$

using that lim inf is superadditive and that the linear part converges. This is the property claimed, and it is where the convexity of J does its work: a continuous convex functional is weakly lower semicontinuous, and J is convex because its quadratic part is a nonnegative quadratic form.

Step 4: The weak limit attains the infimum, and is unique. The subsequence is minimizing, $J(u_k) \to m$, so $\liminf_k J(u_k) = m$; Step 3 then gives $J(u) \leq m$. But $u \in H_0^1(U)$, so $J(u) \geq m$ by the definition of the infimum. Hence $J(u) = m$, and u is a minimizer. Uniqueness is the strict convexity given by coercivity: if u_1, u_2 both minimize J, then by part (1) both are weak solutions, so $a(u_1 - u_2, v) = 0$ for all v; taking $v = u_1 - u_2$ and using coercivity, $\alpha\|u_1 - u_2\|^2_{H_0^1} \leq a(u_1 - u_2, u_1 - u_2) = 0$, forcing $u_1 = u_2$. The minimizer is therefore unique, completing the proof.

The theorem is the variational counterpart of the weak-solvability theorem of chapter 1: where that route inverted the form abstractly through Lax-Milgram, this one exhibits the same solution as the minimizer of an energy functional and reaches it by a compactness argument. The two produce the same u, since both characterize it by the weak equation, but they package different information. The direct method makes visible the mechanism behind the abstract theorem, that coercivity is what keeps a minimizing sequence from escaping to infinity, and it degrades gracefully: drop symmetry and there is no energy to minimize, but the coercivity-plus-lower-semicontinuity template survives essentially unchanged into nonlinear functionals, where it is the standard existence tool of the calculus of variations. The three ingredients are not interchangeable. Coercivity confines the minimizing sequence, reflexivity extracts a candidate limit, and lower semicontinuity certifies that the candidate is optimal; remove any one and the argument fails, which is why each earned its own step. The role of *convexity* deserves a second look. It entered only through the sign in Step 3, the inequality $a(u_k - u, u_k - u) \geq 0$, and this is the general principle: for functionals more complicated than a quadratic, convexity of the integrand in the gradient is the hypothesis that gives weak lower semicontinuity, and its failure is the source of nonexistence in genuinely nonconvex variational problems.

The Dirichlet problem fixed the boundary values in advance by working in $H_0^1(U)$, where every function vanishes on ∂U. The complementary boundary condition prescribes not the values of u but its *flux* across ∂U, and it behaves in the opposite way under the variational formulation: rather than being built into the space, it falls out of the boundary term that integration by parts leaves behind. This is the Neumann problem, and the shift in the boundary condition forces a shift in the test space, from $H_0^1(U)$ to all of $H^1(U)$.

For a classical solution $u \in C^2(\overline{U})$ of $Lu = f$ and any $v \in C^\infty(\overline{U})$, Green's identity for the divergence-form operator reads

$$\int_U \big(-\operatorname{div}(A\nabla u) + cu\big)v\,dx = \int_U (A\nabla u)\cdot\nabla v + cuv\,dx - \int_{\partial U} (A\nabla u\cdot\nu)\,v\,dS$$

where ν is the outward unit normal on ∂U. The bulk integral on the right is exactly the elliptic form $a(u,v)$; the boundary integral involves the *conormal derivative* $\partial u/\partial\nu_A := (A\nabla u)\cdot\nu$, the flux of u through ∂U measured in the metric of A. If one prescribes $\partial u/\partial\nu_A = 0$ on ∂U, the boundary integral vanishes for every test function v, and the classical equation $Lu = f$ becomes the identity $a(u,v) = \int_U fv$, now required for all $v \in H^1(U)$ rather than only for v vanishing on the boundary. The vanishing of the flux is not imposed on the test functions; it is read off from the formulation, which is why it is called a *natural* boundary condition, in contrast to the *essential* Dirichlet condition that must be built into the space.

Proposition 3.1.4: The Neumann Problem and the Compatibility Condition

Let $U \subseteq \mathbb{R}^n$ be bounded, open, and connected with C^1 boundary, and let $f \in L^2(U)$. The *weak Neumann problem* for $L = -\operatorname{div}(A\nabla\cdot) + c$ asks for $u \in H^1(U)$ with

$$a(u,v) = \int_U fv\,dx \qquad \text{for all } v \in H^1(U) \tag{3.1}$$

where $a(u,v) = \int_U (A\nabla u)\cdot\nabla v + cuv\,dx$; a sufficiently smooth solution of (3.1) solves $Lu = f$ in U with the natural boundary condition $\partial u/\partial\nu_A = (A\nabla u)\cdot\nu = 0$ on ∂U. For the pure second-order operator $L = -\operatorname{div}(A\nabla\cdot)$ (the case $c \equiv 0$), the following hold.

1. *Compatibility is necessary.* If (3.1) has a solution, then the datum satisfies the *compatibility condition* $\int_U f\,dx = 0$.
2. *Solvability on the quotient.* With A symmetric and uniformly elliptic, a is symmetric, bounded, and coercive on the quotient space $\dot H^1(U) = H^1(U)/\mathbb{R}$ of H^1 functions modulo additive constants. Consequently, for every $f \in L^2(U)$ with $\int_U f = 0$, the problem (3.1) has a solution $u \in H^1(U)$, unique up to an additive constant.

***Proof*:**

The natural boundary condition. Suppose $u \in H^1(U)$ solves (3.1) and is smooth enough to run Green's identity in reverse, $u \in C^2(\overline{U})$. For $v \in C^\infty(\overline{U})$, subtract the identity $\int_U (A\nabla u)\cdot\nabla v\,dx = \int_U (Lu)v\,dx + \int_{\partial U}(A\nabla u\cdot\nu)v\,dS$ from (3.1) (with $c \equiv 0$) to get

$$\int_U (f - Lu)v\,dx = \int_{\partial U} (A\nabla u\cdot\nu)\,v\,dS \qquad \text{for all } v \in C^\infty(\overline{U})$$

Testing first against $v \in C_c^\infty(U)$, which kill the boundary term, forces $Lu = f$ in U; the bulk integral then vanishes, leaving $\int_{\partial U}(A\nabla u\cdot\nu)v\,dS = 0$ for all $v \in C^\infty(\overline{U})$, and letting v range over a family whose traces are dense on ∂U forces $(A\nabla u)\cdot\nu = 0$ on ∂U. So the natural condition $\partial u/\partial\nu_A = 0$ is recovered from the weak formulation, not assumed in it.

Part (1): necessity of the compatibility condition. Take $c \equiv 0$, so $a(u,v) = \int_U (A\nabla u)\cdot\nabla v\,dx$. The constant function $v \equiv 1$ lies in $H^1(U)$ because U is bounded, and it is an admissible test function since the Neumann test space is all of $H^1(U)$, not $H^1_0(U)$. Its gradient is zero, so $a(u,1) = \int_U (A\nabla u)\cdot\nabla 1\,dx = 0$. Substituting $v \equiv 1$ into (3.1),

$$0 = a(u,1) = \int_U f\cdot 1\,dx = \int_U f\,dx$$

Hence $\int_U f = 0$ is forced by the mere existence of a solution. The condition is the signature of the natural boundary condition: with the flux vanishing on ∂U, the divergence theorem $\int_U \operatorname{div}(A\nabla u)\,dx = \int_{\partial U}(A\nabla u)\cdot\nu\,dS = 0$ says $Lu = f$ can have zero total integral only if f does.

Part (2): coercivity on the quotient and existence. The obstruction to coercivity on all of $H^1(U)$ is exactly the constants: $a(v,v) = \int_U (A\nabla v)\cdot\nabla v\,dx$ vanishes whenever $\nabla v = 0$, which on the connected set U means v is constant, so a cannot control the full H^1 norm of v (a nonzero constant has positive H^1 norm but zero energy). The remedy is to quotient the constants out. Let $\dot H^1(U) = H^1(U)/\mathbb{R}$, with norm the L^2 norm of the gradient, $\|[v]\|_{\dot H^1} = \left(\int_U |\nabla v|^2\,dx\right)^{1/2}$, which is well defined on cosets since ∇v is unchanged by adding a constant, and which is a genuine norm because it vanishes only when $\nabla v = 0$, that is when $[v] = 0$ in the quotient. On this space the Poincaré-Wirtinger inequality of [2, Poincaré-Wirtinger Inequality] gives a constant C_P with

$$\int_U \left|v - \overline{v}\right|^2 dx \le C_P \int_U |\nabla v|^2\,dx \qquad \text{where } \overline{v} = \frac{1}{|U|}\int_U v\,dx$$

so that on the coset representative $v - \overline{v}$ of mean zero the full H^1 norm is comparable to the gradient norm, and $\|[v]\|_{\dot H^1}$ is equivalent to the quotient H^1 norm. Uniform ellipticity then gives coercivity:

$$a(v,v) = \int_U (A\nabla v)\cdot\nabla v\,dx \ge \theta\int_U |\nabla v|^2\,dx = \theta\,\|[v]\|^2_{\dot H^1}$$

while boundedness $|a(v,w)| \le \Lambda\,\|[v]\|_{\dot H^1}\|[w]\|_{\dot H^1}$ is immediate from $|A(x)\xi| \le \Lambda|\xi|$ and Cauchy-Schwarz. The right-hand side $v \mapsto \int_U fv\,dx$ descends to a bounded linear functional on $\dot H^1(U)$ precisely when $\int_U f = 0$: adding a constant t to v changes the functional by $t\int_U f$, which is zero under compatibility, so the functional is well defined on cosets, and it is bounded there by Cauchy-Schwarz and the norm equivalence. The Lax-Milgram theorem (theorem 1.2.3), applied on the Hilbert space $\dot H^1(U)$ with the symmetric coercive form a and this functional, produces a unique $[u] \in \dot H^1(U)$ with $a(u,v) = \int_U fv\,dx$ for all $v \in H^1(U)$. Lifting $[u]$ to a representative $u \in H^1(U)$ gives a solution of (3.1); two solutions differ by an element of the kernel, which is the constants, so u is unique up to an additive constant, as we sought to show.

Three features of the Neumann problem stand out against the Dirichlet case. First, the boundary condition is natural: nothing in the test space $H^1(U)$ enforces a boundary value, and the condition $\partial u/\partial\nu_A = 0$ is a consequence of the weak equation rather than a constraint on it. This is the general pattern for variational boundary conditions, that the ones absorbed into the space are essential and the ones surviving in the boundary term are natural, and it is why the correct test space must be chosen before the boundary condition can even be stated. Second, the pure Neumann Laplacian is not solvable for every datum: the constants sit in the kernel of the operator, uniqueness fails by an additive constant, and existence fails unless $\int_U f = 0$. The two failures are dual faces of the same degeneracy, the one-dimensional kernel spanned by the constants, and the compatibility condition $\int_U f = 0$ is the orthogonality of the datum to that kernel. This is the first appearance of the Fredholm-alternative pattern that the next section makes general: a non-coercive elliptic problem is solvable exactly when the datum is orthogonal to the kernel of the adjoint, and here the operator is its own adjoint and the kernel is the constants. Third, the repair is to quotient out the kernel and recover coercivity on the quotient, where Lax-Milgram again applies verbatim; the lower-order term $c \geq 0$, when present and not identically zero, breaks the degeneracy from the other side by making a coercive on all of $H^1(U)$ with no quotient needed, since then $a(v, v) \geq \theta \int_U |\nabla v|^2 + \int_U cv^2$ controls both the gradient and the function.

3.2 The Fredholm Alternative for Elliptic Operators

The coercive theory of chapter 1 required the sign condition $c \geq 0$, and it is exactly this condition that makes the elliptic form positive enough for Lax-Milgram to invert it outright. Many operators of interest violate it: the zeroth-order coefficient may be negative on part of U, or the operator may carry first-order drift terms, and then coercivity is lost and the equation $Lu = f$ can fail to be solvable for some f while admitting nontrivial solutions of $Lu = 0$. The instrument for this regime is the Fredholm alternative. It does not restore unconditional solvability; instead it pins down precisely how solvability can fail, tying the solvability of $Lu = f$ to a finite list of orthogonality conditions and equating the dimension of the solution space of the homogeneous problem with that of an adjoint problem. The mechanism is to write L as a coercive operator plus a lower-order perturbation, so that inverting the coercive part turns $Lu = f$ into a compact perturbation of the identity, where the Riesz-Schauder theory of compact operators applies.

Fix once and for all the splitting. Write

$$L = L_0 + B$$

where L_0 is a coercive uniformly elliptic operator to which the theory of chapter 1 applies, and B collects the remaining lower-order terms. Concretely, if $Lu = -\operatorname{div}(A\nabla u) + b\cdot\nabla u + cu$ with A bounded and uniformly elliptic, $b \in L^\infty(U; \mathbb{R}^n)$, and $c \in L^\infty(U)$ of arbitrary sign, take $L_0 u = -\operatorname{div}(A\nabla u) + c_0 u$ with c_0 a large enough nonnegative constant that L_0 is coercive, and $Bu = b \cdot \nabla u + (c - c_0)u$. The perturbation B is a first-order operator, one order below the second-order L_0. The choice of L_0 is not canonical, and nothing below depends on it beyond coercivity of L_0 and the requirement that B be a genuine lower-order term: a bounded operator $B : L^2(U) \to H^{-1}(U)$.

The lower-order operator B is bounded from $L^2(U)$ into $H^{-1}(U)$ because it drops one order of differentiation. For the potential this is immediate: multiplication by the bounded function $c-c_0$ sends $u \in L^2(U)$ to $(c-c_0)u \in L^2(U)$ with $\|(c-c_0)u\|_{L^2} \leq \|c-c_0\|_{L^\infty}\|u\|_{L^2}$, and $L^2(U)$ sits inside $H^{-1}(U)$ continuously through the Gelfand triple. For the drift, writing $b \cdot \nabla u = \operatorname{div}(bu) - (\operatorname{div} b)\, u$ shows that when $b \in L^\infty$ has bounded divergence, $u \in L^2(U)$ is carried into $H^{-1}(U)$, since $\operatorname{div}(bu)$ acts on $v \in H^1_0(U)$ by $v \mapsto -\int_U bu\cdot\nabla v$, a functional bounded by $\|b\|_{L^\infty}\|u\|_{L^2}\|v\|_{H^1}$. So $B : L^2(U) \to H^{-1}(U)$ is bounded, with $\|Bu\|_{H^{-1}} \leq C\|u\|_{L^2}$. What matters is that B accepts an L^2 argument and drops it just one space, into $H^{-1}(U)$: it maps $L^2(U)$, not just $H^1_0(U)$, into $H^{-1}(U)$, and it is this one-space drop, fed back up by L_0^{-1}, that the compactness will exploit.

Applying L_0^{-1} to the equation $Lu = f$, written as $L_0 u = f - Bu$ in $H^{-1}(U)$, converts it into a fixed-point equation posed inside $L^2(U)$: L_0^{-1} carries $H^{-1}(U)$ back up to $H^1_0(U)$, which embeds compactly into $L^2(U)$ by Rellich's theorem. This is the step that turns the differential equation into a fixed-point equation for a compact operator, and the compactness is imported wholesale from the solution operator of chapter 2.

Theorem 3.2.1: The Fredholm Alternative for Elliptic Operators

Let $U \subseteq \mathbb{R}^n$ be bounded and open, and let $L = L_0 + B$ where L_0 is a uniformly elliptic operator in divergence form satisfying the coercivity hypotheses of chapter 1 and B is a lower-order perturbation, that is, a bounded operator $B\colon L^2(U) \to H^{-1}(U)$ with $\|Bu\|_{H^{-1}} \leq C\|u\|_{L^2}$. Then exactly one of the following holds.

1. *(Unique solvability.)* The homogeneous problem $Lu = 0$ has only the trivial solution $u = 0$ in $H_0^1(U)$, and then $Lu = f$ has a unique weak solution $u \in H_0^1(U)$ for every $f \in H^{-1}(U)$.

2. *(Fredholm dichotomy.)* The kernel $\ker L = \{u \in H_0^1(U) \mid Lu = 0\}$ is finite-dimensional and nonzero, the adjoint operator L^* has a kernel of the same finite dimension,

$$\dim \ker L = \dim \ker L^* < \infty$$

and $Lu = f$ is weakly solvable if and only if f is orthogonal to $\ker L^*$, that is,

$$\langle f, v \rangle = 0 \qquad \text{for all } v \in \ker L^*$$

Here L^* is the operator of the adjoint bilinear form $a^*(u,v) = a(v,u)$, where a is the form of L.

***Proof*:**
The argument rewrites $Lu = f$ as a compact perturbation of the identity, this time on $L^2(U)$ so that the Hilbert-space adjoint is the plain L^2 adjoint, applies the Fredholm alternative for compact operators, and reads off the solvability and orthogonality statements from it. It is convenient to take the source in $L^2(U)$ throughout; the general case $f \in H^{-1}(U)$ requires only routine changes, noted at the end.

Step 1: A compact operator on L^2. By the coercive existence theory of chapter 1, the operator $L_0\colon H_0^1(U) \to H^{-1}(U)$ is a bounded bijection, so its inverse $L_0^{-1}\colon H^{-1}(U) \to H_0^1(U)$ is bounded. Compose it with the bounded perturbation $B\colon L^2(U) \to H^{-1}(U)$ and define

$$K := L_0^{-1}B\colon L^2(U) \to L^2(U)$$

where the value $L_0^{-1}Bu$, a priori in $H_0^1(U)$, is read as an element of $L^2(U)$ through the inclusion $H_0^1(U) \subseteq L^2(U)$. This K is compact. Indeed it factors as the chain

$$L^2(U) \to H^{-1}(U) \to H_0^1(U) \to L^2(U)$$

with B bounded on the first arrow, L_0^{-1} bounded on the second, and the third the *compact* embedding of Rellich's theorem ([2]), so that K is a bounded operator followed by a compact one. Concretely, if (u_k) is bounded in $L^2(U)$, then (Bu_k) is bounded in $H^{-1}(U)$ and $(L_0^{-1}Bu_k)$ is bounded in $H_0^1(U)$, whence the compact embedding extracts a subsequence along which (Ku_k) converges in $L^2(U)$. This is exactly the compactness of the solution operator recorded in theorem 2.2.1, now carried by the range of L_0^{-1} rather than by any smoothing of B; the argument uses nothing about B beyond its boundedness $L^2(U) \to H^{-1}(U)$, so a first-order drift is admitted on the same footing as a potential. Hence K is a compact operator on $L^2(U)$.

Step 2: Reduction to $u + Ku = g$. Apply L_0^{-1} to $Lu = f$, written as $L_0 u = f - Bu$. For $u \in H_0^1(U)$ this gives $u = L_0^{-1}(f - Bu) = L_0^{-1} f - Ku$, that is,

$$u + Ku = g, \qquad g := L_0^{-1} f$$

Conversely, a solution $u \in L^2(U)$ of $u + Ku = g$ lies in $H_0^1(U)$, since $u = g - Ku = L_0^{-1}(f - Bu)$ is the value of the solution operator L_0^{-1}, whose range is $H_0^1(U)$, and it satisfies $L_0 u = f - Bu$, that is $Lu = f$. So u solves $Lu = f$ if and only if $u + Ku = g$, with identical solution sets; in particular $\ker L = \ker(I + K)$, taking $f = 0$.

Step 3: The Fredholm alternative for $I + K$. $L^2(U)$ is a Hilbert space and K is compact on it, so the Riesz-Schauder theory of [2, Fredholm Alternative, Atkinson's Theorem, Index Invariant Under Compact Operator, Adjoint Properties I] applies to $I + K$ and gives the dichotomy: either $I + K$ is a bijection of $L^2(U)$, in which case $u + Ku = g$ has a unique solution for every g; or $I + K$ is not injective, in which case $\ker(I + K)$ is finite-dimensional, the range of $I + K$ is closed with

$$\operatorname{ran}(I + K) = \ker(I + K^*)^\perp$$

the orthogonal complement in $L^2(U)$, and $\dim\ker(I + K) = \dim\ker(I + K^*) < \infty$, where K^* is the L^2 adjoint of K. In the first case, for every f the datum $g = L_0^{-1} f$ is defined and $u = (I + K)^{-1} g$ is the unique solution of $u + Ku = g$, so by Step 2 the equation $Lu = f$ has a unique solution for every f; this is alternative (1).

Step 4: The adjoint kernel has dimension $\dim\ker L^*$. The operator L^* is the one represented by the adjoint form $a^*(u, v) = a(v, u)$: a function $v \in H_0^1(U)$ lies in $\ker L^*$ when $a(u, v) = a^*(v, u) = 0$ for all $u \in H_0^1(U)$, the weak form of $L^* v = 0$ with zero boundary data. Here $L^* = L_0^* + B^*$, where L_0^* is coercive (its form $a_0^*(u, v) = a_0(v, u)$ satisfies $a_0^*(v, v) = a_0(v, v) \geq \alpha\|v\|_{H^1}^2$) and B^* is the lower-order part of the adjoint. To make the reduction of Steps 1 and 2 applicable to L^*, let $B^* \colon H_0^1(U) \to L^2(U)$ denote the Banach-space adjoint of $B \colon L^2(U) \to H^{-1}(U)$, characterized by $\langle Bu, v\rangle = (u, B^* v)$ for $u \in L^2(U)$ and $v \in H_0^1(U)$; being bounded $H_0^1(U) \to L^2(U)$ it is in particular bounded $L^2(U) \to H^{-1}(U)$ (through $H_0^1(U) \subseteq L^2(U) \subseteq H^{-1}(U)$), so Steps 1 and 2 run verbatim for L^* and give the equivalence

$$\ker L^* = \ker(I + \widetilde{K}), \qquad \widetilde{K} := (L_0^*)^{-1} B^*$$

with $(L_0^*)^{-1} \colon H^{-1}(U) \to H_0^1(U)$ the bounded inverse of the coercive L_0^*. Now compute the L^2 adjoint of $K = L_0^{-1} B$. Fix $u, w \in L^2(U)$ and set $p := L_0^{-1} Bu \in H_0^1(U)$ and $q := (L_0^*)^{-1} w \in H_0^1(U)$. From $L_0^* q = w$ one has $a(z, q) = a^*(q, z) = \langle L_0^* q, z\rangle = (w, z)$ for every $z \in H_0^1(U)$, while from $L_0 p = Bu$ one has $a(p, z') = \langle Bu, z'\rangle$ for every $z' \in H_0^1(U)$. Applying the first with $z = p$ and the second with $z' = q$ identifies the two with $a(p, q)$, so

$$(Ku, w) = (p, w) = a(p, q) = \langle Bu, q\rangle = (u, B^* q) = (u, B^*(L_0^*)^{-1} w)$$

the fourth equality by the defining property of B^*. Hence $K^* = B^*(L_0^*)^{-1}$. This differs from $\widetilde{K} = (L_0^*)^{-1} B^*$ only in the order of the two factors: with $S := (L_0^*)^{-1} \colon L^2(U) \to H_0^1(U)$ and $T := B^* \colon H_0^1(U) \to L^2(U)$ one has $\widetilde{K} = ST$ (on $H_0^1(U)$) and $K^* = TS$ (on $L^2(U)$). For the eigenvalue $-1 \neq 0$ the map S carries $\ker(I + TS)$ into $\ker(I + ST)$ injectively, since $(I + TS)x = 0$ forces $(I + ST)(Sx) = S(x + TSx) = 0$ and $Sx = 0$ would give $x = -TSx = 0$; the map T runs the other way, so the two kernels have equal dimension. Hence $\dim\ker(I + K^*) = \dim\ker(I + \widetilde{K}) = \dim\ker L^*$. Combining with Step 2 and the equality of dimensions from Step 3,

$$\dim\ker L = \dim\ker(I + K) = \dim\ker(I + K^*) = \dim\ker L^*$$

all finite. Finiteness together with $\ker L \neq 0$ is alternative (2).

Step 5: The solvability condition. The condition $\langle f, v\rangle = 0$ for all $v \in \ker L^*$ is read off directly from the form, without unwinding the reduction. For necessity, suppose $u \in H_0^1(U)$ solves $Lu = f$ and let $v \in \ker L^*$. Testing the weak equation against v and using that v annihilates the adjoint form,

$$\langle f, v\rangle = a(u, v) = a^*(v, u) = 0$$

the middle equality being the weak formulation of $Lu = f$ and the last the definition of $v \in \ker L^*$. So every solvable f satisfies the orthogonality conditions $\langle f, v\rangle = 0$ for all $v \in \ker L^*$.

For sufficiency, suppose $\langle f, v\rangle = 0$ for all $v \in \ker L^*$; the goal is to place $g = L_0^{-1} f$ in $\operatorname{ran}(I+K) = \ker(I + K^*)^\perp$, which by Step 3 delivers a solution of $u + Ku = g$ and hence of $Lu = f$. Take any $w \in \ker(I + K^*)$ and set $\varphi := (L_0^*)^{-1} w \in H_0^1(U)$. From $w = -K^* w = -B^*(L_0^*)^{-1} w = -B^*\varphi$ one gets $\widetilde{K}\varphi = (L_0^*)^{-1} B^*\varphi = -(L_0^*)^{-1} w = -\varphi$, so $(I + \widetilde{K})\varphi = 0$ and $\varphi \in \ker(I + \widetilde{K}) = \ker L^*$ by Step 4. Now

$$(g, w) = (L_0^{-1} f, w) = (f, (L_0^{-1})^* w) = (f, (L_0^*)^{-1} w) = \langle f, \varphi\rangle = 0$$

the last equality by hypothesis, since $\varphi \in \ker L^*$. As w was arbitrary, $g \perp \ker(I + K^*)$, so $g \in \operatorname{ran}(I + K)$ and $Lu = f$ is solvable. Together with necessity, $Lu = f$ is solvable if and only if $\langle f, v\rangle = 0$ for all $v \in \ker L^*$, the orthogonality statement of alternative (2).

Step 6: The general source $f \in H^{-1}(U)$. The steps above took $f \in L^2(U)$; the extension to an arbitrary $f \in H^{-1}(U)$ is routine, because the reduction runs through $g = L_0^{-1} f$, and $L_0^{-1}\colon H^{-1}(U) \to H_0^1(U)$ is a bounded bijection by the coercive theory of chapter 1, so $g \in H_0^1(U) \subseteq L^2(U)$ is defined for every such f. For alternative (1), when $I+K$ is a bijection of $L^2(U)$ the equation $u + Ku = g$ has a unique solution $u = (I + K)^{-1} g$ for this g, and Step 2 identifies it with the unique solution of $Lu = f$; the equivalence of Step 2 was stated for $u \in H_0^1(U)$ and $f - Bu \in H^{-1}(U)$ and never used $f \in L^2(U)$, so it applies verbatim. For the sufficiency direction of alternative (2), the only place the source entered was the pairing $(g, w) = \langle f, \varphi\rangle$ of Step 5. Read in the general case, this identity is $(g, w) = (L_0^{-1} f, w) = \langle f, (L_0^*)^{-1} w\rangle = \langle f, \varphi\rangle$, now the H^{-1}-H_0^1 pairing rather than the L^2 inner product, using that $(L_0^*)^{-1}\colon L^2(U) \to H_0^1(U)$ is the adjoint of $L_0^{-1}\colon H^{-1}(U) \to L^2(U)$; the hypothesis $\langle f, \varphi\rangle = 0$ is exactly the orthogonality condition as stated in the theorem for $f \in H^{-1}(U)$, and the necessity computation of Step 5, $\langle f, v\rangle = a(u, v) = 0$, was already the H^{-1}-H_0^1 pairing. So both horns hold for every $f \in H^{-1}(U)$, completing the proof.

The theorem is the exact structural analogue of a fact about square matrices. For a matrix M acting on $\mathbb{R}^N$, the system $Mx = b$ is either uniquely solvable for every b, when M is invertible, or else $\ker M \neq 0$, and then $Mx = b$ is solvable precisely when $b \perp \ker M^\top$, with $\dim\ker M = \dim\ker M^\top$ forced by the rank-nullity theorem. The Fredholm alternative says that an elliptic operator on a bounded domain behaves like such a matrix, even though it acts on an infinite-dimensional space, and the reason is compactness: writing $Lu = f$ as $u + Ku = g$ with K compact places the problem inside the class of operators for which infinite dimension is no obstruction, since a compact operator is a norm-limit of finite-rank ones and $I + K$ inherits the rigid solvability theory of a finite matrix. The one feature with no finite-dimensional counterpart is the appearance of the adjoint *problem* rather than the transpose matrix: the solvability obstruction lives in $\ker L^*$, the null space of the operator built from the transposed form $a^*(u, v) = a(v, u)$, and identifying that kernel with concrete solutions of an adjoint boundary-value problem is where the analytic work of applying the theorem usually lies.

Two points about the hypotheses are worth making explicit, because they mark the boundary of the method. The splitting $L = L_0 + B$ is available for any second-order operator whose top-order part is uniformly elliptic: the leading term $-\operatorname{div}(A\nabla\cdot)$ gives the coercive L_0 once its zeroth-order coefficient is shifted up by a constant, and every lower-order term, whether a sign-changing potential cu or a first-order drift $b \cdot \nabla u$, goes into B. What the argument needs from B is not smallness but that it be genuinely lower-order: a bounded operator $B\colon L^2(U) \to H^{-1}(U)$, dropping one order below the second-order L_0, so that $L_0^{-1}B$ factors through the compact embedding $H_0^1(U) \hookrightarrow L^2(U)$ given by L_0^{-1}'s range. This is the precise sense in which ellipticity plus boundedness of the domain, and nothing about the size of the lower-order terms, is what powers the Fredholm theory; a large negative potential is handled as readily as a small one, and only the dichotomy's two horns record whether the particular operator happens to have a kernel.

Example 3.2.2: A Sign-Changing Potential

Take $Lu = -\Delta u + cu$ on a bounded open U, with $c \in L^\infty(U)$ of arbitrary sign, and Dirichlet boundary data $u = 0$ on ∂U. The form is $a(u, v) = \int_U \nabla u \cdot \nabla v + cuv$, which need not be coercive: where c is negative and large in magnitude, the term $\int_U cu^2$ can overwhelm the Dirichlet energy $\int_U |\nabla u|^2$ and drive $a(u, u)$ below zero.

Split $L = L_0 + B$ by choosing the constant $c_0 = \|c^-\|_{L^\infty}$, the sup norm of the negative part $c^- = \max(-c, 0)$, and setting

$$L_0 u = -\Delta u + c_0 u, \qquad Bu = (c - c_0)u$$

Then $c_0 \geq 0$ is constant, so L_0 meets the coercivity hypotheses of chapter 1 and L_0^{-1} exists and is compact by theorem 2.2.1. The coefficient $c - c_0 = c - \|c^-\|_{L^\infty}$ is bounded, so B is multiplication by a bounded function; it sends $u \in L^2(U)$ to $(c - c_0)u \in L^2(U) \subseteq H^{-1}(U)$ with $\|Bu\|_{H^{-1}} \leq C\|c - c_0\|_{L^\infty}\|u\|_{L^2}$, meeting the perturbation hypothesis. Theorem 3.2.1 therefore applies verbatim: either $-\Delta u + cu = f$ is uniquely solvable in $H_0^1(U)$ for every $f \in L^2(U)$, or the homogeneous problem $-\Delta u + cu = 0$ has a finite-dimensional space of solutions in $H_0^1(U)$ and $-\Delta u + cu = f$ is solvable exactly when $\int_U fv = 0$ for every solution v of the homogeneous problem. Here L is formally self-adjoint, since the form $a(u, v) = \int_U \nabla u \cdot \nabla v + cuv$ is symmetric, so $L^* = L$ and $\ker L^* = \ker L$: the solvability condition is orthogonality to the homogeneous solutions themselves. Which horn of the alternative holds depends on c, and the borderline is a spectral one: a nonzero kernel appears exactly when 0 is a Dirichlet eigenvalue of $-\Delta + c$.

When the operator is self-adjoint, as in the example, the two kernels coincide and the alternative reads most cleanly, but the theorem is stated for the general case because the nonself-adjoint operators are exactly the ones the energy method of chapter 1 could not reach. A first-order drift term $b \cdot \nabla u$ destroys symmetry of the form, so there is no energy to minimize and no way to identify $\ker L^*$ with $\ker L$; the Fredholm alternative still governs solvability, now genuinely referring to a separate adjoint problem, and the orthogonality condition is against the solutions of that adjoint problem rather than against the homogeneous solutions of L itself.

The first horn of the alternative converts a uniqueness statement into an existence statement at no additional cost. For a coercive operator, existence and uniqueness both came from Lax-Milgram at once; for the noncoercive operators now in scope, uniqueness is often the easier of the two to verify directly, by a maximum principle or an energy argument, and the alternative then hands over existence for free.

Corollary 3.2.3: Uniqueness Implies Existence

Let $L = L_0 + B$ be as in theorem 3.2.1. If the homogeneous Dirichlet problem

$$Lu = 0 \text{ in } U, \qquad u = 0 \text{ on } \partial U$$

has only the trivial weak solution $u = 0$ in $H_0^1(U)$, then for every $f \in H^{-1}(U)$ the problem $Lu = f$, $u = 0$ on ∂U, has a unique weak solution $u \in H_0^1(U)$, and this solution obeys an a priori bound

$$\|u\|_{H^1} \leq C\,\|f\|_{H^{-1}}$$

with C independent of f.

***Proof*:**
The hypothesis is that $\ker L = 0$, which is precisely the case that excludes alternative (2) of theorem 3.2.1: a nonzero kernel is required there, so its absence forces alternative (1), under which $Lu = f$ has a unique solution for every $f \in H^{-1}(U)$. It remains to produce the bound. In the notation of the theorem's proof, the reduced operator $I + K$ is a bijection of the Hilbert space $L^2(U)$, and K is bounded (being compact), so by the open mapping theorem of [2, Open Mapping Theorem, Bijective And Continuous] its inverse $(I + K)^{-1}$ is bounded, say with norm M. For $f \in H^{-1}(U)$ the reduced datum $g = L_0^{-1} f$ satisfies $\|g\|_{L^2} \leq \|L_0^{-1}\|_{H^{-1}\to L^2}\|f\|_{H^{-1}}$, and the solution $u = (I+K)^{-1}g$ obeys $\|u\|_{L^2} \leq M\|g\|_{L^2}$. To reach the H^1 norm, use $u = g - Ku = L_0^{-1}(f - Bu)$: since $L_0^{-1}\colon H^{-1}(U) \to H_0^1(U)$ is bounded and $\|Bu\|_{H^{-1}} \leq C_1\|u\|_{L^2}$,

$$\|u\|_{H^1} \leq \|L_0^{-1}\|_{H^{-1}\to H^1}\big(\|f\|_{H^{-1}} + C_1\|u\|_{L^2}\big)$$

Substituting the L^2 bound on $\|u\|_{L^2}$ gives $\|u\|_{H^1} \leq C\|f\|_{H^{-1}}$ with C a constant built from M, C_1, and the norms of L_0^{-1}, independent of f, as we sought to show.

The corollary trades a proof of existence, which for a noncoercive operator has no direct variational route, for a proof of uniqueness, which is frequently within reach: a maximum principle, or a coercivity estimate valid on the kernel alone, can force $\ker L = 0$, and the alternative then gives existence together with the stability bound $\|u\|_{H^1} \leq C\|f\|_{H^{-1}}$ automatically. The bound is not a separate hypothesis but a consequence, delivered by the open mapping theorem the instant $I + K$ is known to be a bijection; the compactness that reduced L to $I + K$ is again the enabling fact, since it is what placed the problem in a setting where injectivity alone yields boundedly invertible.

3.3 Worked Boundary-Value Problems

The two existence routes of this chapter are best understood by running them on concrete problems. This section works two boundary-value problems in full. The first is the Neumann problem for the Laplacian, where coercivity fails on the whole of $H^1(U)$ because the constants are annihilated by the operator, and the variational reduction of proposition 3.1.4 forces a solvability condition on the datum. The second is the shifted operator $-\Delta - \lambda$, an indefinite problem whose existence theory is governed entirely by the Fredholm alternative of theorem 3.2.1, and whose failure of invertibility marks exactly the Dirichlet eigenvalues. Both examples introduce no new theory: they exercise the direct method and the Fredholm dichotomy already established.

The Neumann problem is where the natural boundary condition and the compatibility condition of proposition 3.1.4 become a concrete arithmetic constraint on f. For the pure Laplacian the kernel of the operator is exactly the constants, and the solvability condition inherited from the quotient construction reads as the vanishing of the average of f.

Example 3.3.1: Solvability of the Neumann Problem

Let $U \subseteq \mathbb{R}^n$ be bounded, open, and connected, with Lipschitz boundary, and let $f \in L^2(U)$. Consider the Neumann problem for the Laplacian,

$$-\Delta u = f \text{ in } U, \qquad \partial u/\partial \nu = 0 \text{ on } \partial U$$

whose weak formulation, by proposition 3.1.4, seeks $u \in H^1(U)$ with

$$\int_U \nabla u \cdot \nabla v = \int_U fv \qquad \text{for all } v \in H^1(U)$$

the Neumann condition being natural, that is, encoded by the choice of the full test space $H^1(U)$ rather than imposed on the trial functions. This problem is solvable if and only if

$$\int_U f = 0$$

and when it is solvable the solution is unique up to an additive constant.

Step 1: The compatibility condition is necessary. Suppose $u \in H^1(U)$ solves the weak problem. The constant function $v \equiv 1$ lies in $H^1(U)$ because U is bounded, so it is an admissible test function, and $\nabla v = 0$. Substituting $v \equiv 1$ into the weak formulation gives

$$0 = \int_U \nabla u \cdot \nabla 1 = \int_U f \cdot 1 = \int_U f$$

Thus $\int_U f = 0$ is forced by the existence of a solution. This is the concrete form of the general principle behind proposition 3.1.4: testing the equation against the kernel of the operator reads off a linear constraint on the datum. Here the kernel contains the constants, so the constraint is that f integrate to zero.

Step 2: The quotient space and its coercive form. Let $V = H^1(U)/\mathbb{R}$ be the quotient of $H^1(U)$ by the constants, whose elements are functions determined up to an additive constant. Because U is connected, a function in $H^1(U)$ with $\nabla u = 0$ is constant, so the seminorm $\|u\|_V = \left(\int_U |\nabla u|^2\right)^{1/2}$ is a genuine norm on V: it vanishes only on the zero class. The Poincaré-Wirtinger inequality ([2]) gives the reverse bound,

$$\int_U |u - \bar{u}|^2 \leq C_W \int_U |\nabla u|^2$$

where $\bar{u} = |U|^{-1} \int_U u$ is the mean of u and $|U|$ the volume of U, so that on the quotient V, where a representative may be taken with mean zero, the seminorm $\|\cdot\|_V$ is equivalent to the full H^1 norm. The Dirichlet form $a(u,v) = \int_U \nabla u \cdot \nabla v$ descends to V, since it does not see additive constants, and on V it is coercive:

$$a(u,u) = \int_U |\nabla u|^2 = \|u\|_V^2$$

with coercivity constant 1 in the quotient norm.

Step 3: Existence on the quotient by Lax-Milgram. Assume now $\int_U f = 0$. The functional $v \mapsto \int_U fv$ descends to V: if v_1 and v_2 differ by a constant c, then $\int_U f(v_1 - v_2) = c\int_U f = 0$, so the value $\int_U fv$ depends only on the class of v in V, precisely because f has mean zero. It is a bounded linear functional on V: choosing the mean-zero representative $v - \bar{v}$ and applying Poincaré-Wirtinger,

$$\left|\int_U fv\right| = \left|\int_U f(v - \bar{v})\right| \leq \|f\|_{L^2} \|v - \bar{v}\|_{L^2} \leq \|f\|_{L^2} C_W^{1/2} \|v\|_V$$

where the first equality again uses $\int_U f = 0$ to subtract the mean freely. The bilinear form a is bounded and coercive on the Hilbert space V by Step 2, so the Lax-Milgram theorem produces a unique class $u \in V$ with $a(u, v) = \int_U fv$ for all $v \in V$. Lifting u to any representative in $H^1(U)$ and testing against an arbitrary $v \in H^1(U)$, decomposed as its mean $\bar{v}$ plus its mean-zero part, gives $a(u, v) = a(u, v - \bar{v}) = \int_U f(v - \bar{v}) = \int_U fv$, where the last step is once more the mean-zero property of f; so the representative solves the full weak Neumann problem.

Step 4: Uniqueness up to a constant. If $u_1, u_2 \in H^1(U)$ both solve the weak problem, their difference $d = u_1 - u_2$ satisfies $\int_U \nabla d \cdot \nabla v = 0$ for all $v \in H^1(U)$. Testing against $v = d$ gives $\int_U |\nabla d|^2 = 0$, so $\nabla d = 0$, and since U is connected d is constant. Thus any two solutions differ by an additive constant, and the solution class in V is unique, as we sought to show.

The Neumann example makes visible the mechanism that proposition 3.1.4 states in the abstract. The obstruction to solvability and the non-uniqueness both come from the kernel of the operator: the constants sit in the kernel of $-\Delta$ under Neumann conditions, and this single fact produces both the compatibility condition $\int_U f = 0$, obtained by testing against the kernel, and the ambiguity of the solution by an additive element of the kernel. The problem is not that the theory has broken down but that the operator is not injective, and the Fredholm-type bookkeeping (a solvability condition matched to a non-trivial kernel) is already present in this simplest indefinite-looking case. Connectedness of U is what pins the kernel down to a one-dimensional space of constants; on a domain with several components each component carries its own constant, and the compatibility condition becomes one integral constraint per component.

The second worked problem turns from a solvability condition imposed by the operator's kernel to one imposed by a spectral parameter. Shifting the Laplacian by a constant, $-\Delta - \lambda$, keeps the operator elliptic but destroys coercivity once λ is large enough to overwhelm the Poincaré lower bound, and the Fredholm alternative of theorem 3.2.1 is exactly the tool that determines solvability in the regime where the direct method no longer applies.

Example 3.3.2: The Shifted Operator and the Fredholm Alternative

Let $U \subseteq \mathbb{R}^n$ be bounded and open, fix $\lambda \in \mathbb{R}$, and consider the Dirichlet problem for the shifted operator $-\Delta - \lambda$,

$$-\Delta u - \lambda u = f \text{ in } U, \qquad u = 0 \text{ on } \partial U$$

with $f \in L^2(U)$, whose weak formulation seeks $u \in H_0^1(U)$ with

$$\int_U \nabla u \cdot \nabla v - \lambda \int_U uv = \int_U fv \qquad \text{for all } v \in H_0^1(U)$$

Write $L = -\Delta - \lambda$, splitting it as $L = L_0 + B$ with $L_0 = -\Delta$ the coercive Dirichlet Laplacian of chapter 1 and $B = -\lambda\,\mathrm{Id}$ the zeroth-order perturbation. The point of the splitting is that B is a lower-order term, so theorem 3.2.1 applies verbatim.

Step 1: The Fredholm reduction. By theorem 3.2.1 the equation $Lu = f$ is equivalent to

$$u + Ku = L_0^{-1} f, \qquad K = L_0^{-1} B = -\lambda\, L_0^{-1}$$

where $L_0^{-1} \colon L^2(U) \to L^2(U)$ is the compact solution operator of the Dirichlet Laplacian (theorem 2.2.1). Since K is a scalar multiple of a compact operator it is compact, and the Fredholm alternative for $\mathrm{Id} + K$ governs the problem: either $\mathrm{Id} + K$ is invertible, in which case $Lu = f$ has a unique solution for every f, or $\ker L$ is finite-dimensional and $Lu = f$ is solvable exactly when f is orthogonal to $\ker L^*$.

Step 2: The dichotomy in terms of λ. A number λ is called a Dirichlet eigenvalue of $-\Delta$ on U if the homogeneous problem $-\Delta u = \lambda u$, $u = 0$ on ∂U, has a nonzero weak solution $u \in H_0^1(U)$, that is, if $\ker L \neq \{0\}$. Translating through the reduction of Step 1, λ is a Dirichlet eigenvalue exactly when $u = \lambda L_0^{-1} u$ for some $u \neq 0$, that is, when $\mu = 1/\lambda$ is an eigenvalue of the compact self-adjoint operator L_0^{-1}; the value $\lambda = 0$ is never an eigenvalue, since $-\Delta u = 0$ with $u \in H_0^1(U)$ forces $u = 0$ by coercivity. The two horns of the alternative are therefore:

- If λ is *not* a Dirichlet eigenvalue, then $\ker L = \{0\}$, so $\mathrm{Id} + K$ is injective, hence invertible; the problem $-\Delta u - \lambda u = f$ has a unique solution $u \in H_0^1(U)$ for every $f \in L^2(U)$. For $\lambda \leq 0$ this is the coercive case: the form $a(u, v) = \int_U \nabla u \cdot \nabla v - \lambda \int_U uv$ satisfies $a(u, u) = \int_U |\nabla u|^2 - \lambda \int_U u^2 \geq \int_U |\nabla u|^2$, so coercivity is retained and the direct method or Lax-Milgram already gives the solution without invoking Fredholm.

- If λ *is* a Dirichlet eigenvalue, then $\ker L$ is the associated eigenspace, finite-dimensional and nonzero. The operator $-\Delta - \lambda$ is symmetric, so the adjoint form $a^*(u, v) = a(v, u)$ equals a, and $\ker L^* = \ker L$ is the same eigenspace. The problem $-\Delta u - \lambda u = f$ is then solvable if and only if

$$\int_U f w = 0 \qquad \text{for every eigenfunction } w \text{ with } -\Delta w = \lambda w$$

  and when solvable the solution is unique only up to the addition of an element of the eigenspace.

Step 3: A concrete instance. On the interval $U = (0, \pi) \subseteq \mathbb{R}$ the Dirichlet eigenfunctions of $-\Delta = -d^2/dx^2$ are $w_k(x) = \sin(kx)$ with eigenvalues $\lambda_k = k^2$ for $k = 1, 2, 3, \ldots$. For $\lambda = 1$, an eigenvalue with eigenfunction $w_1(x) = \sin x$, the problem $-u'' - u = f$, $u(0) = u(\pi) = 0$, is solvable precisely when $\int_0^\pi f(x) \sin x \, dx = 0$; a datum such as $f \equiv 1$, for which $\int_0^\pi \sin x \, dx = 2 \neq 0$, admits no solution, while $f(x) = \sin(2x)$, which is orthogonal to $\sin x$ on $(0, \pi)$, does. For $\lambda = 2$, which is not of the form k^2, the operator is invertible and $-u'' - 2u = f$ has a unique solution for every $f \in L^2(0, \pi)$.

The shifted operator exhibits the Fredholm alternative in its purest form: as λ increases past the coercivity threshold, the problem is invertible for all λ except a discrete set, and at each exceptional value the loss of invertibility is exactly matched by a finite-dimensional obstruction. This is the same accounting seen for the Neumann problem, with the kernel now given by a spectral resonance rather than by the constants: solvability requires the datum to be orthogonal to the kernel of the adjoint, and the operator being symmetric, that kernel is the eigenspace itself. The exceptional values are the Dirichlet eigenvalues, and the fact that they form a discrete sequence $0 < \lambda_1 < \lambda_2 \leq \lambda_3 \leq \cdots \to \infty$ rather than a continuum is precisely the compactness of L_0^{-1} at work: a compact self-adjoint operator has a discrete spectrum accumulating only at zero, and inverting the eigenvalues sends that accumulation point to infinity. The systematic study of this eigenvalue sequence, its variational characterization through the Rayleigh quotient, and the orthonormal basis of eigenfunctions it furnishes is the subject of the spectral chapter; the Fredholm alternative used here is the doorway to it, identifying the eigenvalues as the exceptional set where existence and uniqueness pass from the generic horn of the alternative to the resonant one.

4

Schauder Regularity

The existence theory produced weak solutions, functions in H^1 whose derivatives are controlled only in an L^2 sense. Regularity theory asks how much more can be said: when the data and coefficients are smooth, is the weak solution smooth? Schauder theory answers this in the Hölder scale. Its engine is a single a priori estimate, the interior Schauder estimate, bounding the $C^{2,\alpha}$ norm of a solution by the C^α norm of the data, and its method is to compare a variable-coefficient operator with the constant-coefficient one obtained by freezing coefficients at a point, where the explicit Newtonian potential gives the estimate directly. This chapter proves the interior estimate, extends it to the boundary for global $C^{2,\alpha}$ regularity of the Dirichlet problem, and closes the existence theory by upgrading weak solutions to classical ones through the method of continuity.

4.1 Hölder Spaces and the Interior Schauder Estimate

The chapter's engine is an a priori estimate in the Hölder scale, and the scale must be built before the estimate can be stated. The L^2-based Sobolev spaces of the existence theory measure a derivative only in an averaged sense and cannot see pointwise regularity, so they are the wrong home for an estimate that promises a genuine second derivative. The Hölder spaces replace the averaged control by a pointwise continuity modulus on the derivatives, and it is in this scale that the constant-coefficient model problem admits an explicit and sharp bound. This section builds the scale, proves that bound for the Newtonian potential, and then runs the freezing-coefficients argument that carries the model estimate to a variable-coefficient operator on the interior of the domain.

The operator throughout is written in *non-divergence form*

$$Lu = -a^{ij}\partial_{ij}u + b^i\partial_i u + cu$$

with summation over repeated indices, uniformly elliptic in the sense that the coefficient matrix $(a^{ij}(x))$ satisfies $\theta|\xi|^2 \le a^{ij}(x)\xi_i\xi_j \le \Lambda|\xi|^2$ for all $\xi \in \mathbb{R}^n$ and all x, with $0 < \theta \le \Lambda$. Non-divergence form is the natural setting for Schauder theory: the estimate bounds the individual second derivatives $\partial_{ij}u$, which appear undifferentiated in Lu, and asking the coefficients to be Hölder continuous is exactly enough to run the comparison with the frozen operator.

Definition 4.1.1: Hölder Spaces

Let $U \subseteq \mathbb{R}^n$ be open and let $0 < \alpha \leq 1$. The *Hölder seminorm* of exponent α of a function $u\colon U \to \mathbb{R}$ is

$$[u]_{\alpha;U} = \sup_{\substack{x,y\in U\\ x\neq y}} \frac{|u(x)-u(y)|}{|x-y|^\alpha}$$

A function u is *Hölder continuous of exponent* α on $\overline{U}$, written $u \in C^{0,\alpha}(\overline{U})$, if it is bounded and $[u]_{\alpha;U} < \infty$; for $\alpha = 1$ this is Lipschitz continuity. For a nonnegative integer k, the space $C^{k,\alpha}(\overline{U})$ consists of the functions u whose derivatives up to order k are bounded and continuous on $\overline{U}$ and whose derivatives of order exactly k lie in $C^{0,\alpha}(\overline{U})$. It carries the norm

$$\|u\|_{C^{k,\alpha}(\overline{U})} = \sum_{|\beta|\leq k} \sup_U |\partial^\beta u| + \sum_{|\beta|=k} [\partial^\beta u]_{\alpha;U}$$

Where the exponent alone is displayed, C^α abbreviates $C^{0,\alpha}$, and the domain is dropped from the seminorm when it is understood.

The definition splits the norm into a boundedness part and a continuity part. The finitely many supremum terms control the size of u and its derivatives, while the seminorm terms control the oscillation of the top derivatives at the scale of a fractional power of the distance. The exponent α interpolates between the two integer regimes it separates: exponent near 0 imposes almost nothing beyond continuity, while exponent near 1 approaches Lipschitz control and hence, off a null set, a bounded gradient. The reason the scale is indexed by a strict fraction $0 < \alpha < 1$ rather than by integers is that the estimate to come fails at the endpoints. At $\alpha = 0$ the second-derivative operator maps continuous data to functions that need not be continuous, and at $\alpha = 1$ the same operator loses a derivative in a logarithm; only strictly between the endpoints does the estimate close. The space $C^{k,\alpha}(\overline{U})$ is a Banach space under this norm, a routine consequence of the completeness of the uniform norm on the derivatives together with the fact that a uniform limit of functions with a common Hölder bound inherits that bound.

The seminorm sees exactly the regularity that the sup norm misses, and the cleanest illustration of the gap is a function that is Hölder but not Lipschitz.

Example 4.1.2: The Cusp $|x|^\alpha$

Fix $0 < \alpha < 1$ and let $u(x) = |x|^\alpha$ on the unit ball $U = \{|x| < 1\} \subseteq \mathbb{R}^n$. This u lies in $C^{0,\alpha}(\overline{U})$ but not in $C^{0,1}(\overline{U})$: it is Hölder of exponent α yet not Lipschitz.

For the Hölder bound it suffices to work in one variable along the ray through two points, since $\big||x|^\alpha - |y|^\alpha\big| \leq \big||x|-|y|\big|^\alpha \leq |x-y|^\alpha$; the first inequality is the elementary bound $|s^\alpha - t^\alpha| \leq |s-t|^\alpha$ for $s, t \geq 0$ and $0 < \alpha < 1$, which holds because $t \mapsto t^\alpha$ is concave with $0^\alpha = 0$, and the second is the reverse triangle inequality. Hence $[u]_{\alpha;U} \leq 1$, and u is bounded by 1, so $u \in C^{0,\alpha}(\overline{U})$.

That u is not Lipschitz is the failure of the same estimate at exponent 1. Along the segment from

0 to a point x with $|x| = r$,

$$\frac{|u(x) - u(0)|}{|x - 0|} = \frac{r^\alpha}{r} = r^{\alpha - 1}$$

which tends to ∞ as $r \to 0$ because $\alpha - 1 < 0$. The gradient $\nabla u = \alpha|x|^{\alpha-2}x$ blows up at the origin at the rate $|x|^{\alpha-1}$, so no finite Lipschitz constant controls the difference quotient near 0. The cusp is the model of a function that has a definite fractional continuity modulus without having a bounded first derivative, and it shows that the inclusion $C^{0,1} \subseteq C^{0,\alpha}$ is strict for every $\alpha < 1$.

The exponent α records the precise rate at which a difference quotient may grow, and Schauder theory is the statement that an elliptic operator preserves this rate: apply L to a function whose second derivatives grow at rate α, and the output grows at rate α, and conversely. The forward direction is immediate from the definition of $C^{2,\alpha}$; the content is the converse, that control of Lu in C^α forces control of the second derivatives in C^α. That converse is an a priori estimate, and it is proved in two movements: first for the constant-coefficient model $-\Delta$, where the solution is an explicit integral of the data, and then for a general variable-coefficient L by comparison with the frozen model.

The first movement is the model problem $-\Delta w = f$ on all of $\mathbb{R}^n$, whose solution is an integral of f against the fundamental solution of the Laplacian. Recall the *Newtonian potential*: for $n \geq 3$ the fundamental solution of $-\Delta$ is

$$N(x) = \frac{1}{n(n-2)\omega_n}\,|x|^{2-n}$$

where ω_n is the volume of the unit ball, normalized so that $-\Delta N = \delta_0$ in the sense of distributions, and $N(x) = -\frac{1}{2\pi}\log|x|$ in dimension $n = 2$. Convolution with N inverts $-\Delta$ on compactly supported data. The next result is the exact statement that this inverse gains two derivatives on the Hölder scale: the second derivatives of the potential are controlled in C^α by the data in C^α, with a constant that depends only on the dimension and the exponent. Every later estimate in the chapter reduces to it.

Lemma 4.1.3: Interior Hölder Estimate for the Newtonian Potential

Let $0 < \alpha < 1$ and let $f \in C^\alpha(\mathbb{R}^n)$ have compact support. Then the Newtonian potential

$$w(x) = (N * f)(x) = \int_{\mathbb{R}^n} N(x-y)f(y)\,dy$$

is twice continuously differentiable, solves $-\Delta w = f$ in $\mathbb{R}^n$, and its second derivatives are Hölder continuous of exponent α with

$$[\,\partial_{ij}w\,]_\alpha \leq C(n,\alpha)\,[f]_\alpha$$

for a constant $C(n,\alpha)$ depending only on n and α. In particular $w \in C^{2,\alpha}_{\mathrm{loc}}(\mathbb{R}^n)$.

The full argument is a long but elementary estimation of a singular integral, and its length is out of proportion to the ideas it uses; the proof below is condensed to those ideas, naming each step, with the difference-quotient bookkeeping indicated rather than carried out in full. The one point that must not be lost is the mechanism by which the estimate closes, the cancellation in the second-derivative kernel, and that point is made explicit.

Proof Key ideas:
Step 1: The second derivatives form a singular integral. Differentiating under the integral once is harmless, since $\partial_i N$ is locally integrable, and gives $\partial_i w = (\partial_i N) * f$. Differentiating a second time cannot be done under the integral, because $\partial_{ij} N$ has a nonintegrable singularity of order $|x|^{-n}$ at the origin. The correct formula, obtained by integrating by parts against the difference $f(y) - f(x)$ to exploit the Hölder modulus, is

$$\partial_{ij} w(x) = \int_{\mathbb{R}^n} \partial_{ij} N(x-y)\big(f(y) - f(x)\big)\, dy + f(x) \int_{|z|=1} \partial_i N(z)\, \nu_j \, dS(z)$$

where ν is the outward normal, the boundary term being written here over the unit sphere as the ball-domain special case; for a general domain $\Omega \supseteq \operatorname{supp} f$ it runs over $\partial\Omega$, and its value is independent of that choice because $\partial_i N$ is homogeneous of degree $1-n$, which makes the surface integral scale-invariant across spheres centered at x. The kernel $K_{ij}(z) = \partial_{ij} N(z)$ is homogeneous of degree $-n$ and has mean zero over spheres, the two defining properties of a Calderón-Zygmund singular integral. The subtraction of $f(x)$ tames the singularity: near the diagonal the factor $f(y) - f(x)$ is $O(|x-y|^\alpha)$, which upgrades the nonintegrable $|x-y|^{-n}$ to the integrable $|x-y|^{\alpha-n}$. This is already enough to see that $\partial_{ij} w$ is continuous and that $\Delta w = \sum_i \partial_{ii} w = -f$, i.e. $-\Delta w = f$, since $\sum_i K_{ii} = \Delta N = 0$ away from the origin and the trace of the boundary term collapses to $-f(x)$: on the unit sphere $\nabla N \cdot \nu = -1/(n\omega_n)$, so $\int_{|z|=1} \nabla N \cdot \nu \, dS = -1$ and $\sum_i f(x) \int_{|z|=1} \partial_i N(z)\, \nu_i \, dS = -f(x)$.

Step 2: The annular decomposition of a difference. To bound $[\partial_{ij} w]_\alpha$, fix two points x_1, x_2, set $h = |x_1 - x_2|$ and let $\bar{x}$ be their midpoint. Split the integration domain by a sphere of radius $R = \frac{3}{2} h$ about $\bar{x}$ into a *near* region $B = \{|y - \bar{x}| < R\}$ and a *far* region $\mathbb{R}^n \setminus B$, and estimate $\partial_{ij} w(x_1) - \partial_{ij} w(x_2)$ separately on the two. The scale R is chosen comparable to the separation h so that on the near region the singularity is the dominant effect and on the far region the two evaluation points are close relative to their distance to the source.

Step 3: The near region uses the Hölder modulus. On B each of the two second-derivative integrals is estimated by itself, without differencing, using the representation of Step 1 centered at the respective point. The integrand is bounded by $[f]_\alpha \, |y - x_k|^\alpha \, |K_{ij}(x_k - y)| \le [f]_\alpha \, C \, |y - x_k|^{\alpha - n}$, and integrating $|y - x_k|^{\alpha-n}$ over a ball of radius comparable to h produces a factor h^α. Thus the near contribution to the difference is at most $C(n,\alpha)\, [f]_\alpha \, h^\alpha$.

Step 4: The far region uses kernel cancellation and the mean value theorem. On the far region the two singularities are absent, so the difference is estimated by differencing the kernel:

$$\big|K_{ij}(x_1 - y) - K_{ij}(x_2 - y)\big| \le |x_1 - x_2| \sup_z |\nabla K_{ij}(z - y)| \le C \, \frac{h}{|\bar{x} - y|^{\,n+1}}$$

by the mean value theorem and the homogeneity $|\nabla K_{ij}(z)| \le C|z|^{-n-1}$, valid because $|x_k - y|$ is comparable to $|\bar{x} - y|$ throughout the far region. What is being estimated is the integral against the *difference* of kernels $K_{ij}(x_1 - y) - K_{ij}(x_2 - y)$; subtracting the constant $f(\bar{x})$ contributes only $f(\bar{x}) \int_{\mathbb{R}^n \setminus B} \big(K_{ij}(x_1 - y) - K_{ij}(x_2 - y)\big)\, dy$, which the same $h/|\bar{x} - y|^{n+1}$ bound controls, so the datum may be replaced by $f(y) - f(\bar{x})$ and again enters only through its oscillation:

$$\int_{\mathbb{R}^n \setminus B} \big|K_{ij}(x_1 - y) - K_{ij}(x_2 - y)\big| \, |f(y) - f(\bar{x})| \, dy \le C \, h \, [f]_\alpha \int_{|\bar{x} - y| \ge R} \frac{|\bar{x} - y|^\alpha}{|\bar{x} - y|^{\,n+1}} \, dy$$

The remaining integral is $\int_R^\infty \rho^{\alpha-2} \, d\rho$ in polar coordinates: writing $\rho = |\bar{x} - y|$ and $dy = \rho^{n-1} \, d\rho \, dS$, the integrand $\rho^{\alpha-n-1}$ times the volume factor ρ^{n-1} leaves $\rho^{\alpha-2}$, which converges precisely because $\alpha < 1$ and evaluates to a multiple of $R^{\alpha-1} \sim h^{\alpha-1}$. Multiplying by the prefactor h leaves $C(n,\alpha)\,[f]_\alpha\, h^\alpha$.

Adding the near and far contributions gives $|\partial_{ij}w(x_1) - \partial_{ij}w(x_2)| \le C(n,\alpha)\,[f]_\alpha\,|x_1 - x_2|^\alpha$, which is exactly $[\partial_{ij}w]_\alpha \le C(n,\alpha)[f]_\alpha$, completing the proof. The convergence of the far integral at the upper endpoint is where the restriction $\alpha < 1$ is spent, and the convergence of the near integral at the singularity is where $\alpha > 0$ is spent; the estimate lives strictly between the endpoints, as promised.

The lemma is the entire Schauder theory in the flat, constant-coefficient case: the operator $-\Delta$ gains two derivatives on the Hölder scale, with a scale-invariant constant. Two features generalize. The estimate is on the *seminorm* of the second derivatives alone, not on the full norm, and this is what makes it a statement about oscillation that is untouched by the passage to variable coefficients. And the constant depends on nothing but n and α, in particular not on the support or size of f, a homogeneity that reflects the invariance of $-\Delta$ under translation and scaling. Everything that follows extracts the variable-coefficient estimate from this one by making a general operator look, on a small enough ball, like $-\Delta$ with constant coefficients.

The second movement carries the model estimate to a variable-coefficient operator. A variable-coefficient operator is not solved by an explicit potential, so the lemma does not apply to it directly. The device that repairs this is to *freeze* the leading coefficients at a single point. Near a fixed x_0 the matrix $(a^{ij}(x))$ differs from its value $(a^{ij}(x_0))$ by an amount that the Hölder continuity of the coefficients makes as small as one likes, provided the neighborhood is small enough. On that neighborhood L is a small perturbation of the constant-coefficient operator $L_0 = -a^{ij}(x_0)\partial_{ij}$, and L_0, being a constant-coefficient elliptic operator, is reducible to $-\Delta$ by a linear change of variables and so inherits the estimate of lemma 4.1.3. The variable part of L contributes an error term, and the Hölder modulus of the coefficients makes this error absorbable: it carries a factor that shrinks with the radius, and choosing the radius small forces the error below half the quantity being estimated, whereupon it moves to the left-hand side. The smallness of the ball is where the C^α hypothesis on the coefficients is used.

Theorem 4.1.4: Interior Schauder Estimate

Let $U \subseteq \mathbb{R}^n$ be bounded and open, and let

$$Lu = -a^{ij}\partial_{ij}u + b^i\partial_i u + cu$$

be uniformly elliptic with ellipticity constants $0 < \theta \le \Lambda$ and coefficients $a^{ij}, b^i, c \in C^\alpha(\overline{U})$ with $0 < \alpha < 1$. Suppose $u \in C^{2,\alpha}(U)$ satisfies $Lu = f$ in U with $f \in C^\alpha(\overline{U})$. Then $u \in C^{2,\alpha}_{\mathrm{loc}}(U)$, and for every $V \Subset U$ there is a constant $C = C\big(n, \alpha, \theta, \Lambda, \|a^{ij}\|_{C^\alpha}, \|b^i\|_{C^\alpha}, \|c\|_{C^\alpha}, \mathrm{dist}(V, \partial U)\big)$ with

$$\|u\|_{C^{2,\alpha}(V)} \le C\big(\|u\|_{C^0(U)} + \|f\|_{C^\alpha(U)}\big)$$

***Proof*:**
The argument bounds the top-order seminorm $[\partial^2 u]_\alpha$ locally, then trades the localization for the sup norm of u through a cutoff, and finally recovers the lower-order pieces of the $C^{2,\alpha}$ norm by interpolation. Write $L_0u = -a^{ij}\partial_{ij}u$ for the leading part, so that $Lu = L_0u + b^i\partial_i u + cu$.

Step 1: The frozen model on a small ball. Fix $x_0 \in U$ and, for a radius r to be chosen, let $B = B_r(x_0) \Subset U$. Freeze the leading coefficients at x_0, setting $A_0 = (a^{ij}(x_0))$ and $L_0^{\mathrm{fr}} = -a^{ij}(x_0)\partial_{ij}$.

On B the equation $Lu = f$ rearranges to

$$L_0^{\mathrm{fr}} u = f + \big(a^{ij}(x) - a^{ij}(x_0)\big)\partial_{ij} u - b^i \partial_i u - cu =: F$$

so u solves a constant-coefficient elliptic equation on B with right-hand side F. The matrix A_0 is symmetric positive definite with eigenvalues in $[\theta, \Lambda]$, so a linear change of variables $y = A_0^{-1/2} x$ turns L_0^{fr} into $-\Delta$ and distorts Hölder seminorms by factors controlled by θ and Λ alone. Lemma 4.1.3, applied to a suitable cutoff of u so that the datum has compact support and then read back through the linear change of variables, yields the frozen estimate

$$[\partial^2 u]_{\alpha;B} \le C_0\big([F]_{\alpha;B} + \|u\|_{C^0(B)}\big) \tag{4.1}$$

with $C_0 = C_0(n, \alpha, \theta, \Lambda)$; the $\|u\|_{C^0}$ term is the price of the cutoff, which introduces lower-order commutator terms supported where the cutoff varies.

Step 2: The oscillation error and its absorption. It remains to bound $[F]_{\alpha;B}$. The lower-order terms $b^i\partial_i u$ and cu have C^α seminorms controlled by the product rule in terms of $\|b^i\|_{C^\alpha}, \|c\|_{C^\alpha}$ and the $C^{1,\alpha}$, respectively C^α, norm of u, which are lower order than $[\partial^2 u]_\alpha$ and will be handled by interpolation. The dangerous term is the oscillation term $\big(a^{ij}(x) - a^{ij}(x_0)\big)\partial_{ij} u$, because it contains the top derivative $\partial_{ij} u$. Estimating a product by the Hölder Leibniz rule,

$$\big[(a^{ij} - a^{ij}(x_0))\, \partial_{ij} u\big]_{\alpha;B} \le \Big(\sup_B |a^{ij} - a^{ij}(x_0)|\Big)[\partial_{ij} u]_{\alpha;B} + [a^{ij}]_{\alpha;B} \sup_B |\partial_{ij} u|$$

On the ball $B = B_r(x_0)$ the first supremum is bounded by $\sup_B |a^{ij}(x) - a^{ij}(x_0)| \le [a^{ij}]_\alpha\, r^\alpha$, which is the whole point: the Hölder continuity of the coefficients makes the multiplier in front of the top-order seminorm as small as $[a^{ij}]_\alpha\, r^\alpha$, and this tends to 0 with r. Choose r small enough that $C_0\,[a^{ij}]_\alpha\, r^\alpha \le \frac{1}{2}$. Then the term $C_0 \sup_B |a^{ij} - a^{ij}(x_0)|\,[\partial^2 u]_{\alpha;B}$ appearing on the right of (4.1) is at most $\frac{1}{2}[\partial^2 u]_{\alpha;B}$ and can be subtracted from the left, leaving

$$[\partial^2 u]_{\alpha;B} \le C_1\Big(\|f\|_{C^\alpha(B)} + \|\partial^2 u\|_{C^0(B)} + \|u\|_{C^{1,\alpha}(B)} + \|u\|_{C^0(B)}\Big) \tag{4.2}$$

where the lower-order coefficient norms have been folded into C_1. Two families of terms survive on the right besides f: the second piece $[a^{ij}]_\alpha \sup_B |\partial_{ij} u|$ of the oscillation term, which carries two derivatives but only in the sup norm and so contributes $\|\partial^2 u\|_{C^0}$, and the lower-order terms $b^i\partial_i u$ and cu, which carry at most one derivative and are collected in $\|u\|_{C^{1,\alpha}}$. The radius r now depends only on α, θ, Λ and $[a^{ij}]_\alpha$, not on the solution, so the same r works at every center x_0.

Step 3: Interpolation of the lower-order norms. The bound (4.2) still contains the two intermediate quantities $\|\partial^2 u\|_{C^0}$ and $\|u\|_{C^{1,\alpha}}$, each of which sits between $\|u\|_{C^0}$ and the top-order seminorm being estimated. The interpolation inequality for Hölder norms states that for every $\varepsilon > 0$ there is a constant C_ε with

$$\|\partial^2 u\|_{C^0(B)} + \|u\|_{C^{1,\alpha}(B)} \le \varepsilon\,[\partial^2 u]_{\alpha;B} + C_\varepsilon \|u\|_{C^0(B)}$$

the intermediate derivatives being controlled by a small multiple of the top seminorm plus a large multiple of the function itself. For a single term this is the elementary convexity estimate $\|\partial u\|_{C^0} \le \varepsilon[\partial u]_\alpha + C_\varepsilon\|u\|_{C^0}$, obtained by writing a first difference as an average of second differences over a segment of length tuned to ε and iterating up the derivative ladder; the sup of $\partial^2 u$ enters through the same estimate applied to ∂u in place of u. Inserting this into (4.2) with

ε chosen so that $C_1\varepsilon \leq \frac{1}{2}$ absorbs both intermediate quantities into the left side a second time, and yields the local estimate on the ball,

$$\|u\|_{C^{2,\alpha}(B_{r/2}(x_0))} \leq C_2\big(\|f\|_{C^\alpha(U)} + \|u\|_{C^0(U)}\big) \tag{4.3}$$

where passing to the half ball $B_{r/2}(x_0)$ absorbs the cutoff commutators of Step 1 into the right side, and C_2 depends only on the stated quantities. In particular $u \in C^{2,\alpha}$ near x_0, and since $x_0 \in U$ was arbitrary, $u \in C^{2,\alpha}_{\mathrm{loc}}(U)$.

Step 4: Patching to a compact subdomain. Fix $V \Subset U$ and set $d = \operatorname{dist}(V, \partial U) > 0$. The balls $\{B_{r/2}(x_0)\}_{x_0\in V}$, with r the radius fixed in Step 2 taken no larger than $d/2$ so that every $B_r(x_0) \Subset U$, cover the compact set $\overline{V}$, so finitely many of them do. On each the local estimate (4.3) holds with the same constant C_2, and the $C^{2,\alpha}(V)$ norm is bounded by the sum over the finite subcover, the number of whose members depends only on n, $\operatorname{diam}(V)$, and r. Hence

$$\|u\|_{C^{2,\alpha}(V)} \leq C\big(\|u\|_{C^0(U)} + \|f\|_{C^\alpha(U)}\big)$$

with C depending on $n, \alpha, \theta, \Lambda$, the C^α norms of the coefficients, and $\operatorname{dist}(V, \partial U)$ through r and the number of balls, as we sought to show.

The estimate says that the two derivatives lost in passing from a classical solution to a weak one are recoverable in the Hölder scale, at the price only of the sup norm of the solution and the C^α norm of the data. Three of its features are worth isolating. It is an *a priori* estimate: it bounds a solution assumed to exist, and does not by itself produce one. The existence of a classical solution to fill it in comes later, from the method of continuity, and the estimate is exactly the uniform bound that argument needs. It is *interior*: the constant degenerates as V approaches ∂U, since a solution can be irregular at the boundary no matter how smooth the data, and recovering regularity up to the boundary requires a separate argument that flattens ∂U, taken up in section 4.2. And the sup norm $\|u\|_{C^0(U)}$ on the right is not decoration: without a bound on u itself the estimate is empty, since adding a harmonic function changes neither Lu nor the left side by a controlled amount; it is the maximum principle that will later convert $\|u\|_{C^0}$ into a bound by the data alone.

The mechanism deserves a second look, because it recurs in every regularity theory. The variable operator was never estimated directly; it was compared, on a scale small enough that its leading coefficients could not vary much, with a constant-coefficient operator that could be estimated explicitly. The error of the comparison carried a factor r^α from the Hölder modulus of the coefficients, and that factor, made small by shrinking the ball, was what let the error be absorbed into the quantity it was perturbing. The exponent α enters twice, once in the potential estimate of lemma 4.1.3 and once in the coefficient oscillation, and both times it is the strict inequality $0 < \alpha < 1$ that keeps the argument alive: the same fraction that indexed the Hölder scale is the fraction that measures how fast the frozen model degrades as the ball grows.

4.2 Boundary Schauder Estimates and Global Regularity

The interior estimate of theorem 4.1.4 controls $\|u\|_{C^{2,\alpha}}$ on any set $V \Subset U$ compactly contained in the domain, and its constant blows up as V approaches ∂U. That degeneration is not an artifact of the proof: a solution can genuinely lose regularity at the boundary if the boundary itself is rough or if the prescribed data are rough, so no estimate up to ∂U can hold without hypotheses on both. This section states those hypotheses and the estimate they give. With a $C^{2,\alpha}$ boundary and Dirichlet data $g \in C^{2,\alpha}(\partial U)$, the same $C^{2,\alpha}$ control persists across ∂U, and combining the interior and boundary estimates over a finite cover yields the global bound on $\overline{U}$ that the existence theory of the next section will use as its a priori estimate.

The mechanism is a change of coordinates. Near a boundary point the graph of ∂U is straightened to a piece of the hyperplane $\{x_n = 0\}$ by a $C^{2,\alpha}$ diffeomorphism, which carries L to another uniformly elliptic operator with C^α coefficients on a half-ball and carries the Dirichlet condition to a condition on the flat face. On the half-space the constant coefficient model can be solved explicitly, exactly as the Newtonian potential solved it in the interior, and the flat estimate transfers back through the inverse map. Two reductions make the argument clean: first flatten the boundary, then subtract off the data to reduce to homogeneous boundary values. Both are developed in full below; only the half-space potential computation itself, the boundary analog of lemma 4.1.3, is condensed, since it repeats the interior estimate with the reflected kernel.

Throughout, L is taken in non-divergence form $Lu = -a^{ij}\partial_{ij}u + b^i\partial_i u + cu$ with $C^\alpha(\overline{U})$ coefficients, uniformly elliptic in the sense of definition 1.3.1 with the same ellipticity constants used in the interior estimate, and $\alpha \in (0,1)$ is fixed. The half-space is $\mathbb{R}^n_+ = \{x = (x', x_n) \in \mathbb{R}^n \mid x_n > 0\}$ with $x' = (x_1, \ldots, x_{n-1})$, and for $r > 0$ the half-ball is $B_r^+ = B_r(0) \cap \mathbb{R}^n_+$ with flat face $\Gamma_r = B_r(0) \cap \{x \mid x_n = 0\}$.

Definition 4.2.1: $C^{2,\alpha}$ Boundary

A bounded open set $U \subseteq \mathbb{R}^n$ has $C^{2,\alpha}$ *boundary* if for every point $x_0 \in \partial U$ there is a radius $r > 0$, an orthonormal coordinate system centred at x_0, and a function $\gamma\colon \mathbb{R}^{n-1} \to \mathbb{R}$ of class $C^{2,\alpha}$ with $\gamma(0) = 0$ and $\nabla\gamma(0) = 0$ such that, within the ball $B_r(x_0)$,

$$U \cap B_r(x_0) = \{x \in B_r(x_0) \mid x_n > \gamma(x')\}$$

so that ∂U is locally the graph $x_n = \gamma(x')$ and U lies locally above it.

The definition asks that ∂U look, near each of its points and after a rotation, like the graph of a function whose second derivatives are Hölder continuous of exponent α. The normalization $\gamma(0) = 0$, $\nabla\gamma(0) = 0$ places the point at the origin with the tangent hyperplane horizontal, so the boundary is already flat to first order and the graph map γ records the second-order and finer bending that the flattening must undo. The class $C^{2,\alpha}$ of the boundary is exactly the class of regularity being proved for u: to control two Hölder derivatives of a solution up to ∂U the boundary must itself carry two Hölder derivatives, since the flattening map inherits its smoothness from γ and any deficit in γ would appear as a deficit in the transformed coefficients.

Example 4.2.2: A Ball Has $C^{2,\alpha}$ Boundary

Take $U = B_1(0) \subseteq \mathbb{R}^n$, the open unit ball, and the boundary point $x_0 = -e_n$ (the south pole), with coordinates chosen so that the inward normal points along $+e_n$. Near x_0 the sphere $\{x \mid |x| = 1\}$ is the graph of the lower hemisphere; writing y for coordinates centred at x_0, the boundary is $y_n = \gamma(y')$ with

$$\gamma(y') = 1 - \sqrt{1 - |y'|^2} = \tfrac{1}{2}|y'|^2 + \tfrac{1}{8}|y'|^4 + \cdots$$

for $|y'| < 1$. This γ is real-analytic on $|y'| < 1$, so in particular C^∞ and a fortiori $C^{2,\alpha}$ for every $\alpha \in (0,1)$, and it satisfies $\gamma(0) = 0$ and $\nabla\gamma(0) = 0$ since the expansion starts at second order. The Hessian at the origin is $\partial_{ij}\gamma(0) = \delta_{ij}$ for $i, j < n$, the second fundamental form of the sphere, which records that the ball curves away from its tangent plane at unit rate. Every point of ∂B_1 is equivalent to the south pole under a rotation, so the whole sphere has $C^{2,\alpha}$ boundary, and the same computation with radius R scales the Hessian to $R^{-1}\delta_{ij}$.

The reduction to homogeneous boundary data is the first of the two simplifications and requires no regularity theory at all, only the extendability of Hölder data off the boundary. Given $g \in C^{2,\alpha}(\partial U)$ there is a function $G \in C^{2,\alpha}(\overline{U})$ with $G = g$ on ∂U and $\|G\|_{C^{2,\alpha}(\overline{U})} \le C\|g\|_{C^{2,\alpha}(\partial U)}$, obtained by extending g off the graph in the flattening coordinates and patching with a partition of unity; this is the standard Whitney-type extension for Hölder functions on a $C^{2,\alpha}$ surface. Replacing u by $u - G$ turns the problem $Lu = f$, $u = g$ on ∂U into $L(u - G) = f - LG$, $u - G = 0$ on ∂U, where the new right-hand side $f - LG$ still lies in $C^{\alpha}(\overline{U})$ with $\|f - LG\|_{C^{\alpha}} \le \|f\|_{C^{\alpha}} + C\|G\|_{C^{2,\alpha}} \le \|f\|_{C^{\alpha}} + C\|g\|_{C^{2,\alpha}}$. So it suffices to prove the boundary estimate for solutions vanishing on ∂U, with the extra $\|g\|_{C^{2,\alpha}}$ term that then appears in the final bound. This step is recorded once here and used silently in the proof.

The half-space model is the boundary analog of the Newtonian potential. On $\mathbb{R}^n_+$ the constant coefficient operator $-\Delta$ with the zero Dirichlet condition on $\{x_n = 0\}$ is inverted by the Green's function built from the fundamental solution and its reflection across the boundary hyperplane, and this reflected potential enjoys the same Hölder estimate on second derivatives as the free Newtonian potential, uniformly up to the flat face.

Lemma 4.2.3: The Half-Space Potential Estimate

Let $f \in C^{\alpha}(\overline{B_1^+})$ and let w solve $-\Delta w = f$ in B_1^+ with $w = 0$ on the flat face Γ_1. Then $w \in C^{2,\alpha}(\overline{B_{1/2}^+})$ and

$$\|w\|_{C^{2,\alpha}(\overline{B_{1/2}^+})} \le C(n,\alpha)\big(\|w\|_{C^0(B_1^+)} + [f]_{\alpha;B_1^+} + \|f\|_{C^0(B_1^+)}\big)$$

In particular the second-derivative Hölder seminorm satisfies $[D^2 w]_{\alpha;\overline{B_{1/2}^+}} \le C(n,\alpha)\big([f]_{\alpha;B_1^+} + \|w\|_{C^0(B_1^+)}\big)$, the estimate holding uniformly as $x_n \to 0$, including for the tangential and normal second derivatives at the flat face.

***Proof Key ideas*:**
The estimate is the interior estimate of lemma 4.1.3 carried to the flat boundary by odd reflection.

The reflected representation. Write $\widetilde{x} = (x', -x_n)$ for the reflection of $x = (x', x_n)$ across $\{x_n = 0\}$, and let N be the fundamental solution of $-\Delta$ on $\mathbb{R}^n$. The Green's function of the half-space with Dirichlet data on Γ is $G_+(x,y) = N(x-y) - N(x-\widetilde{y})$, the free kernel minus its mirror image, which vanishes when $x_n = 0$ because then $|x-y| = |x-\widetilde{y}|$. The potential $w(x) = \int_{B_1^+} G_+(x,y) f(y)\,dy$ vanishes on Γ by construction, and it is w, not the right-hand side, that reflects cleanly: its *odd* extension $\widetilde{w}(x', x_n) = -w(x', -x_n)$ for $x_n < 0$ is continuous across $\{x_n = 0\}$ precisely because w vanishes there, and $\widetilde{w}$ solves $-\Delta\widetilde{w} = \widetilde{f}$ on the full ball B_1, where $\widetilde{f}(x', x_n) = -f(x', -x_n)$ for $x_n < 0$ is the odd extension of the source. The substitution $y \mapsto \widetilde{y}$ on the mirror term converts $-N(x-\widetilde{y})$ into the free kernel $N(x-y)$ integrated against $\widetilde{f}$ over $\widetilde{B_1^+}$, so $\widetilde{w}$ equals *exactly* the free Newtonian potential of $\widetilde{f}$ on B_1. The odd extension $\widetilde{f}$ need not be Hölder across the hyperplane since f need not vanish on Γ; but this is harmless, because $D^2\widetilde{w}$ is recovered below without ever bounding $[\widetilde{f}]_{\alpha}$ across the face.

The estimate by annular decomposition. The tangential second derivatives are estimated first, because they are the ones that reflect. Since $\widetilde{w}$ is odd in x_n, each purely tangential second derivative $\partial_{x_i x_j}\widetilde{w}$ with $i,j<n$ is again odd, hence continuous across $\{x_n=0\}$, and on either half-ball it equals the corresponding singular integral of the source against the reflected kernel $D^2_{x_i x_j}G_+$. That kernel's second singularity, at $y=\widetilde{x}$, never meets the closed half-space for $x\in\overline{\mathbb{R}^n_+}$, so it contributes only a smooth term; the difference-quotient estimate on annuli $\{y \mid 2^{-k-1}\le |x-y|\le 2^{-k}\}$ that produced $[D^2 w]_\alpha\le C[f]_\alpha$ in the interior lemma applies to the remaining $N(x-y)$ term with f restricted to B_1^+, giving $[\partial_{x'x'}w]_{\alpha;B^+_{1/2}}\le C[f]_{\alpha;B_1^+}$ uniformly up to Γ. The mixed derivatives $\partial_{x'x_n}\widetilde{w}$ are handled the same way; being one tangential and one normal derivative of an odd function they are even in x_n, hence again continuous across $\{x_n=0\}$, and the same annular estimate on the $N(x-y)$ term applies. The normal second derivative $\partial_{x_n x_n}w$ is then recovered not from a singular integral but from the equation itself, $\partial_{x_n x_n}w=-f-\sum_{i<n}\partial_{x_i x_i}w$, so its Hölder seminorm is controlled by $[f]_\alpha$ and the tangential seminorms already estimated. This is where the boundary condition is used: it is precisely the vanishing of w on Γ that lets w be reflected oddly with $\widetilde{w}$ still solving $-\Delta\widetilde{w}=\widetilde{f}$, so that the tangential derivatives are governed by a genuine whole-space singular integral even though the source $\widetilde{f}$ itself has a jump across the face. Interpolating the lower-order norms of w against $\|w\|_{C^0}$ and $[f]_\alpha$ as in the interior case gives the stated full $C^{2,\alpha}$ bound on the smaller half-ball, completing the argument.

With the flat model and the two reductions in hand, the boundary estimate follows the interior template exactly: freeze the coefficients at the boundary point, invoke the flat model for the frozen operator, and absorb the coefficient oscillation on a small half-ball where the C^α modulus of continuity of the coefficients makes the error term small. The one new ingredient is the flattening map, which must be shown to preserve uniform ellipticity and the C^α class of the coefficients.

Theorem 4.2.4: The Boundary Schauder Estimate

Let $U\subseteq\mathbb{R}^n$ be bounded with $C^{2,\alpha}$ boundary, let $L=-a^{ij}\partial_{ij}+b^i\partial_i+c$ be uniformly elliptic with coefficients in $C^\alpha(\overline{U})$, and let $x_0\in\partial U$. Suppose $u\in C^{2,\alpha}(\overline{U}\cap B_r(x_0))$ solves

$$Lu=f \quad \text{in } U\cap B_r(x_0), \qquad u=g \quad \text{on } \partial U\cap B_r(x_0)$$

with $f\in C^\alpha(\overline{U})$ and $g\in C^{2,\alpha}(\partial U)$. Then there are a smaller radius $\rho\in(0,r)$ and a constant C, depending on n,α, the ellipticity constants, the C^α norms of the coefficients, and the $C^{2,\alpha}$ norm of the boundary chart, such that

$$\|u\|_{C^{2,\alpha}(\overline{U}\cap B_\rho(x_0))}\le C\big(\|f\|_{C^\alpha(U)}+\|g\|_{C^{2,\alpha}(\partial U)}+\|u\|_{C^0(U)}\big)$$

***Proof*:**
Fix $x_0\in\partial U$. By the reduction to homogeneous data recorded above, subtracting an extension $G\in C^{2,\alpha}(\overline{U})$ of g reduces to the case $g=0$ and leaves the term $C\|g\|_{C^{2,\alpha}}$ in the bound, so assume $u=0$ on $\partial U\cap B_r(x_0)$; the new right-hand side is still called f and still satisfies $\|f\|_{C^\alpha}\le\|f\|_{C^\alpha}+C\|g\|_{C^{2,\alpha}}$.

Step 1: Flatten the boundary. By definition 4.2.1 there is, after a rotation centering x_0 at the origin with the inward normal along $+e_n$, a $C^{2,\alpha}$ function γ with $\gamma(0) = 0$, $\nabla\gamma(0) = 0$, and $U \cap B_r(x_0) = \{x \mid x_n > \gamma(x')\}$. Define the map $\Phi\colon x \mapsto y$ by

$$y' = x', \qquad y_n = x_n - \gamma(x')$$

which sends the boundary graph $x_n = \gamma(x')$ to the flat face $y_n = 0$ and the domain to the region $y_n > 0$. Because $\gamma \in C^{2,\alpha}$, the map Φ is a $C^{2,\alpha}$ diffeomorphism onto its image with $C^{1,\alpha}$ Jacobian, and $\nabla\gamma(0) = 0$ makes $D\Phi(0)$ the identity, so on a small enough half-ball B_R^+ the Jacobian stays close to the identity and Φ is invertible with Φ^{-1} of the same class. Set $v = u \circ \Phi^{-1}$, so v is defined on B_R^+ and vanishes on the flat face Γ_R. By the chain rule v solves a transformed equation $\widetilde{L}v = \widetilde{f}$ in B_R^+, where

$$\widetilde{L} = -\widetilde{a}^{ij}\partial_{y_i y_j} + \widetilde{b}^i\partial_{y_i} + \widetilde{c}, \qquad \widetilde{a}^{ij}(y) = a^{kl}(\Phi^{-1}y)\,\partial_{x_k}y_i\,\partial_{x_l}y_j$$

and $\widetilde{f} = f \circ \Phi^{-1}$. The transformed leading coefficient $\widetilde{a}^{ij}$ is a product of the original $a^{kl} \in C^\alpha$ with entries of $D\Phi$, which are $C^{1,\alpha}$, hence itself $C^\alpha(\overline{B_R^+})$; and $\widetilde{a}$ is uniformly elliptic because $D\Phi$ is invertible with bounded inverse, so the conjugated matrix has eigenvalues bounded away from 0 and ∞ by the ellipticity constants of a times the condition number of $D\Phi$. The first-order coefficients $\widetilde{b}^i$ pick up the second derivatives of Φ, which are C^α since $\Phi \in C^{2,\alpha}$, so $\widetilde{b}^i \in C^\alpha$; and $\widetilde{c} = c \circ \Phi^{-1} \in C^\alpha$. The flattening therefore lands in exactly the class of the interior proof: a uniformly elliptic non-divergence operator with C^α coefficients on a half-ball, and homogeneous Dirichlet data on the flat face. This is where the $C^{2,\alpha}$ regularity of ∂U is spent: any less smoothness of γ would drop $\widetilde{a}$ below C^α and break the frozen-coefficient argument to come.

Step 2: Freeze the coefficients at the boundary point. Work now with v and $\widetilde{L}$ on B_R^+, and freeze the leading coefficients at the origin, writing $\widetilde{L}_0 = -\widetilde{a}^{ij}(0)\partial_{y_i y_j}$ for the constant-coefficient elliptic operator with the boundary-point values. A linear change of coordinates $y \mapsto z = Ty$ turns $\widetilde{L}_0$ into $-\Delta$: taking $T = Q\,\widetilde{a}(0)^{-1/2}$ with Q orthogonal gives $T\widetilde{a}(0)T^\top = QQ^\top = I$, so the frozen operator becomes the Laplacian in the z-variables. This T does not fix $\{y_n = 0\}$ (the factor $\widetilde{a}(0)^{-1/2}$ mixes the normal direction with the tangential ones) but it need not: being linear and invertible, T carries the half-space $\{y_n > 0\}$ to another half-space, whose boundary is again a flat hyperplane through the origin. Since $-\Delta$ is invariant under the orthogonal factor Q, that Q is free to be chosen so as to rotate this image hyperplane back onto $\{z_n = 0\}$, and then the half-space model lemma 4.2.3 applies to $\widetilde{L}_0$ after this normalization. It is the flatness of the face, preserved by any linear map, and not its coordinate description that the model requires. Rewrite the equation as

$$\widetilde{L}_0 v = \widetilde{f} + \big(\widetilde{L}_0 - \widetilde{L}\big)v = \widetilde{f} + \big(\widetilde{a}^{ij}(y) - \widetilde{a}^{ij}(0)\big)\partial_{y_i y_j}v - \widetilde{b}^i\partial_{y_i}v - \widetilde{c}v$$

treating everything but the frozen principal part as a right-hand side. Applying the half-space estimate of lemma 4.2.3 on B_s^+ (for $s \le R$) to v against this right-hand side, and using that the tangential and normal second derivatives are controlled uniformly up to Γ_s,

$$[D^2v]_{\alpha;B^+_{s/2}} \le C\Big([\widetilde{f}]_\alpha + \big[(\widetilde{a}^{ij}(\cdot) - \widetilde{a}^{ij}(0))\partial_{y_i y_j}v\big]_\alpha + \text{lower-order}\Big)$$

Step 3: Absorb the coefficient oscillation on a small half-ball. The crux is the term $\big[(\widetilde{a}^{ij}(\cdot) - \widetilde{a}^{ij}(0))\partial_{y_i y_j} v\big]_\alpha$. The Hölder seminorm of a product obeys $[gh]_\alpha \leq \|g\|_{C^0}[h]_\alpha + [g]_\alpha \|h\|_{C^0}$, and here $g = \widetilde{a}^{ij} - \widetilde{a}^{ij}(0)$ is the oscillation of the leading coefficient from its boundary value. On the half-ball B_s^+,

$$\|\widetilde{a}^{ij} - \widetilde{a}^{ij}(0)\|_{C^0(B_s^+)} \leq [\widetilde{a}^{ij}]_\alpha \, s^\alpha =: \omega(s)$$

which is the C^α-continuity of the coefficients made quantitative: the modulus $\omega(s)$ tends to 0 as $s \to 0$ precisely because $\widetilde{a}^{ij} \in C^\alpha$ vanishes at the origin after subtracting its value there. Therefore

$$\big[(\widetilde{a}^{ij} - \widetilde{a}^{ij}(0))\partial_{ij} v\big]_\alpha \leq \omega(s)\, [D^2 v]_{\alpha; B_s^+} + [\widetilde{a}^{ij}]_\alpha \, \|D^2 v\|_{C^0(B_s^+)}$$

The first term carries a factor $\omega(s)$ that can be made as small as desired by shrinking s; choosing s small enough that $C\,\omega(s) \leq \frac{1}{2}$ lets this term be absorbed into the left-hand side. The absorption is legitimate despite the mismatch of half-balls (the left-hand seminorm lives on $B_{s/2}^+$ while the crux term carries $[D^2 v]_{\alpha; B_s^+}$ on the full half-ball) because it is run, exactly as in the interior case, on the weighted interior Hölder seminorms whose scaling absorbs the ball-shrinkage through the same covering-and-iteration step used there; passing back to unweighted seminorms on $B_{s/2}^+$ afterward is what yields the displayed bound. The smallness of s is the whole point: the boundary point gives a value $\widetilde{a}^{ij}(0)$ from which the coefficient cannot stray far on a small enough neighborhood, and it is only on that small neighborhood that the constant-coefficient model is a good enough approximation for the error to be reabsorbed. After absorption,

$$[D^2 v]_{\alpha; B_{s/2}^+} \leq C\big([\widetilde{f}]_\alpha + \|D^2 v\|_{C^0(B_s^+)} + \text{lower-order}\big)$$

and the remaining lower-order terms $\|D^2 v\|_{C^0}$, $\|\nabla v\|_{C^0}$, $\|v\|_{C^0}$ are handled by interpolation of Hölder norms exactly as in the interior estimate: for every $\varepsilon > 0$ there is C_ε with $\|D^2 v\|_{C^0} \leq \varepsilon [D^2 v]_\alpha + C_\varepsilon \|v\|_{C^0}$, so a second absorption of the $\varepsilon[D^2 v]_\alpha$ piece leaves

$$\|v\|_{C^{2,\alpha}(\overline{B_{s/2}^+})} \leq C\big([\widetilde{f}]_\alpha + \|v\|_{C^0(B_s^+)}\big)$$

Step 4: Transfer back. Undo the coordinate changes. The diagonalizing map T and the flattening Φ are $C^{2,\alpha}$ with $C^{2,\alpha}$ inverses on the relevant half-balls, and composition with such a map changes a $C^{2,\alpha}$ norm only by a constant factor depending on the $C^{2,\alpha}$ norm of the map, so $\|u\|_{C^{2,\alpha}}$ near x_0 is comparable to $\|v\|_{C^{2,\alpha}}$ on the half-ball. Likewise $\|\widetilde{f}\|_{C^\alpha} \leq C\|f\|_{C^\alpha}$ and $\|v\|_{C^0} = \|u\|_{C^0}$. Setting ρ so that $B_\rho(x_0)$ maps into $B_{s/2}^+$ under Φ, and restoring the $C\|g\|_{C^{2,\alpha}}$ term deferred from the homogeneous reduction, the transferred estimate reads

$$\|u\|_{C^{2,\alpha}(\overline{U} \cap B_\rho(x_0))} \leq C\big(\|f\|_{C^\alpha(U)} + \|g\|_{C^{2,\alpha}(\partial U)} + \|u\|_{C^0(U)}\big)$$

which is the claimed boundary estimate, completing the proof.

The boundary estimate is the same result as the interior one seen through a $C^{2,\alpha}$ change of coordinates: every step, freezing, the potential model, absorbing the oscillation, and interpolating, recurs, and the only genuinely new work is the flattening map and the odd-reflection model that keeps the flat Dirichlet condition compatible with the elliptic equation. Three points deserve emphasis. First, the boundary must carry two Hölder derivatives because the flattening spends exactly that smoothness:

a C^2 boundary without the Hölder modulus would leave $\widetilde{a}$ continuous and the frozen-coefficient absorption would have no quantitative modulus $\omega(s)$ to exploit. Second, the Dirichlet datum enters through the homogeneous reduction, which is why g is asked to lie in $C^{2,\alpha}(\partial U)$, the same class as the solution: the estimate can be no better than the data it is fed, and $g \in C^{2,\alpha}$ is the sharp hypothesis for a $C^{2,\alpha}$ solution up to the boundary. Third, the estimate is local near a single boundary point, with a constant depending on the chart there; globalizing requires patching finitely many such neighborhoods together with the interior estimate, which is the content of the next result.

Assembling the interior and boundary estimates over a finite cover of $\overline{U}$ produces the global bound. Since $\overline{U}$ is compact and ∂U has $C^{2,\alpha}$ charts at every point, finitely many boundary balls cover a neighborhood of ∂U, and the interior estimate covers the compact remainder; adding the finitely many local estimates gives control of $\|u\|_{C^{2,\alpha}}$ on all of $\overline{U}$ by the same three quantities.

Theorem 4.2.5: The Global Schauder Estimate

Let $U \subseteq \mathbb{R}^n$ be bounded with $C^{2,\alpha}$ boundary, let $L = -a^{ij}\partial_{ij} + b^i\partial_i + c$ be uniformly elliptic with coefficients in $C^\alpha(\overline{U})$, and let $u \in C^{2,\alpha}(\overline{U})$ solve

$$Lu = f \quad \text{in } U, \qquad u = g \quad \text{on } \partial U$$

with $f \in C^\alpha(\overline{U})$ and $g \in C^{2,\alpha}(\partial U)$. Then there is a constant C, depending on n, α, the ellipticity constants, the C^α norms of the coefficients, and the $C^{2,\alpha}$ regularity of ∂U, such that

$$\|u\|_{C^{2,\alpha}(\overline{U})} \le C\big(\|f\|_{C^\alpha(\overline{U})} + \|g\|_{C^{2,\alpha}(\partial U)} + \|u\|_{C^0(\overline{U})}\big)$$

***Proof*:**
The estimate is assembled from the local estimates by a compactness and patching argument.

Step 1: A finite cover. Each boundary point $x_0 \in \partial U$ has, by theorem 4.2.4, a ball $B_{\rho(x_0)}(x_0)$ on which the boundary estimate holds. The collection $\big\{B_{\rho(x_0)/2}(x_0) \bigm| x_0 \in \partial U\big\}$ is an open cover of the compact set ∂U, so a finite subfamily $B_1, \dots, B_N$ of these half-radius balls, centred at boundary points, already covers ∂U. Write B_j^* for the double of B_j, the full-radius ball $B_{\rho(x_0)}(x_0)$ on which the boundary estimate is actually available; so $B_j \subseteq B_j^*$ and the estimate on B_j^* in particular controls u on $\overline{U} \cap B_j$. Let $K = \overline{U} \setminus \bigcup_j B_j$ be the part of $\overline{U}$ left uncovered; K is compact and disjoint from ∂U, hence $K \Subset U$, and the interior estimate theorem 4.1.4 applies on a slightly larger V with $K \Subset V \Subset U$.

Step 2: Add the local estimates. On each boundary ball the boundary estimate gives

$$\|u\|_{C^{2,\alpha}(\overline{U}\cap B_j)} \le C_j\big(\|f\|_{C^\alpha(U)} + \|g\|_{C^{2,\alpha}(\partial U)} + \|u\|_{C^0(U)}\big)$$

and on the interior piece the interior estimate gives $\|u\|_{C^{2,\alpha}(K)} \le C_0(\|u\|_{C^0(U)} + \|f\|_{C^\alpha(U)})$. Because $\overline{U}$ is covered by K together with the finitely many $\overline{U} \cap B_j$, and a $C^{2,\alpha}$ norm on a finite union is bounded by the sum of the norms on the pieces (the pieces overlap, and Hölder seminorms across two overlapping sets are controlled by the seminorms on each together with the C^0 norm on the overlap), taking $C = \sum_{j=0}^N C_j$ gives

$$\|u\|_{C^{2,\alpha}(\overline{U})} \le C\big(\|f\|_{C^\alpha(\overline{U})} + \|g\|_{C^{2,\alpha}(\partial U)} + \|u\|_{C^0(\overline{U})}\big)$$

which is the global estimate, as we sought to show.

The global estimate is the a priori bound the existence theory needs: it says that a $C^{2,\alpha}$ solution of the Dirichlet problem is controlled in $C^{2,\alpha}(\overline{U})$ by the three data of the problem, its source f, its boundary values g, and its own supremum $\|u\|_{C^0}$. The dependence on $\|u\|_{C^0}$ is the one blemish on the estimate, since it bounds the solution partly in terms of itself, but it is removable. A separate principle, the maximum principle, bounds the supremum of a solution by its data alone: for an operator with $c \geq 0$ the estimate $\|u\|_{C^0(\overline{U})} \leq C\|f\|_{C^0(U)} + \|g\|_{C^0(\partial U)}$ holds, so the last term on the right can be absorbed into the first two and dropped from the estimate entirely. That principle is developed in a later chapter of this book, from the comparison of a solution with suitable barriers, and is logically independent of the Schauder theory; the Schauder estimate is stated here with the $\|u\|_{C^0}$ term present so that it stands on its own, and the maximum principle is what later removes it to leave the clean bound

$$\|u\|_{C^{2,\alpha}(\overline{U})} \leq C\big(\|f\|_{C^\alpha(\overline{U})} + \|g\|_{C^{2,\alpha}(\partial U)}\big)$$

in terms of the data only. In this form the estimate does two jobs at once. Read as a regularity statement it says a solution is exactly two Hölder derivatives smoother than its right-hand side, the gain of two orders that is the signature of a second-order elliptic operator. Read as a stability statement it says the solution map $(f, g) \mapsto u$ is bounded from $C^\alpha \times C^{2,\alpha}$ into $C^{2,\alpha}$, and this boundedness, applied along the family of operators interpolating between L and the Laplacian, is precisely the uniform a priori bound that drives the method of continuity in the next section.

4.3 Classical Solutions by the Method of Continuity

The global estimate of theorem 4.2.5 bounds the $C^{2,\alpha}$ norm of a solution by the data, but presupposes that a solution exists. This section closes that gap. An a priori bound can be converted into an existence proof, and the device that does the conversion is the *method of continuity*: deform the operator L along a straight path to the Laplacian, for which the Dirichlet problem is already solved, and use the estimate, held uniformly along the path, to carry solvability across. The section first isolates the deformation argument as a statement about bounded operators between Banach spaces, where nothing analytic intrudes and the whole mechanism is a connectedness argument on $[0, 1]$, and then feeds the Schauder estimate into it to produce a classical solution of the variable-coefficient Dirichlet problem.

Throughout, L is the non-divergence operator $Lu = -a^{ij}\partial_{ij}u + b^i\partial_i u + cu$ with $a^{ij} = a^{ji}$ uniformly elliptic and coefficients $a^{ij}, b^i, c \in C^\alpha(\overline{U})$, the form in which the Schauder estimates were proved.

The deformation lives entirely at the level of linear operators. Suppose L_0 and L_1 are bounded linear maps between the same pair of Banach spaces, and interpolate them linearly by $L_t = (1-t)L_0 + tL_1$. If every L_t obeys one and the same a priori bound $\|x\| \leq C\|L_t x\|$, then solvability of $L_0 x = y$ propagates to solvability of $L_1 x = y$. The bound does two things at once: it forces each L_t to be injective with closed range, so "surjective" and "bijective" coincide, and it makes the inverse operators uniformly bounded, so the solution of $L_t x = y$ cannot escape to infinity as t moves. These are exactly the ingredients that make the solvable set of parameters both open and closed.

Theorem 4.3.1: Method of Continuity

Let X be a Banach space, Y a normed vector space, and let $L_0, L_1 \colon X \to Y$ be bounded linear operators. For $t \in [0,1]$ set

$$L_t = (1-t)L_0 + tL_1$$

Suppose there is a constant $C > 0$, independent of t, such that

$$\|x\|_X \leq C\,\|L_t x\|_Y \qquad \text{for all } x \in X \text{ and all } t \in [0,1] \tag{4.4}$$

Then L_1 is surjective if and only if L_0 is surjective. More precisely, the set

$$S = \{t \in [0,1] \mid L_t \colon X \to Y \text{ is a bijection}\}$$

is either empty or all of $[0,1]$.

***Proof*:**
The bound (4.4) first gives injectivity: if $L_t x = 0$ then $\|x\|_X \leq C \cdot 0 = 0$, so $x = 0$, and every L_t is one-to-one. Thus for each t, "L_t is a bijection" is the same as "L_t is onto", and a bijective L_t has a bounded inverse of norm $\|L_t^{-1}\| \leq C$ directly from (4.4) applied to $x = L_t^{-1}y$. The claim is that S is open and closed in $[0,1]$; a connected interval has no proper nonempty subset that is both, so S is $\emptyset$ or $[0,1]$, and since it contains 0 exactly when L_0 is onto, the theorem follows.

Step 1: S is open. Fix $s \in S$, so L_s is a bijection with $\|L_s^{-1}\| \leq C$. For any $t \in [0,1]$ write L_t as a perturbation of L_s: since $L_t - L_s = (t-s)(L_1 - L_0)$,

$$L_t = L_s + (t-s)(L_1 - L_0) = L_s\bigl(I + (t-s)L_s^{-1}(L_1 - L_0)\bigr)$$

Set $M = \|L_1 - L_0\|$. When $|t-s| < 1/(2CM)$ the operator $E = (t-s)L_s^{-1}(L_1 - L_0)$ has norm $\|E\| \leq |t-s|\,C\,M < \frac{1}{2} < 1$, so $I + E$ is invertible by the Neumann series $\sum_{k \geq 0}(-E)^k$, which converges in the operator norm because X is complete. Hence L_t, the composite of the bijection L_s with the bijection $I + E$, is itself a bijection, and the whole interval $(s - 1/(2CM), s + 1/(2CM)) \cap [0,1]$ lies in S. The radius $1/(2CM)$ does not depend on s, so S is open.

Step 2: S is closed. Let $t_k \in S$ with $t_k \to t_* \in [0,1]$; the goal is $t_* \in S$, that is, L_{t_*} is onto. Fix $y \in Y$. For each k there is a unique $x_k \in X$ with $L_{t_k} x_k = y$, and the sequence (x_k) is the candidate solution to chase. First it is bounded: (4.4) gives $\|x_k\|_X \leq C\|L_{t_k} x_k\|_Y = C\|y\|_Y$. Next it is Cauchy. Subtracting the defining equations and using $L_{t_k} - L_{t_j} = (t_k - t_j)(L_1 - L_0)$,

$$L_{t_k}(x_k - x_j) = L_{t_k}x_k - L_{t_k}x_j = y - \bigl(L_{t_j}x_j + (t_k - t_j)(L_1 - L_0)x_j\bigr) = -(t_k - t_j)(L_1 - L_0)x_j$$

Applying (4.4) to $x_k - x_j$ and bounding the right side by $M\|x_j\|_X \leq CM\|y\|_Y$,

$$\|x_k - x_j\|_X \leq C\,\|L_{t_k}(x_k - x_j)\|_Y \leq C\,|t_k - t_j|\,M\,\|x_j\|_X \leq C^2 M \|y\|_Y\,|t_k - t_j|$$

Since (t_k) converges it is Cauchy, so (x_k) is Cauchy, and completeness of X gives $x_k \to x_*$ for some $x_* \in X$. The operator L_{t_*} is bounded, and $L_{t_k}x_k \to L_{t_*}x_*$ because $\|L_{t_k}x_k - L_{t_*}x_*\|_Y \leq \|L_{t_k}(x_k - x_*)\|_Y + |t_k - t_*|\,M\,\|x_*\|_X$ and both terms vanish in the limit. Since $L_{t_k}x_k = y$ for every k, the limit gives $L_{t_*}x_* = y$. As y was arbitrary, L_{t_*} is onto and $t_* \in S$, so S is closed, completing the proof.

The two steps use the a priori bound in two different registers, and it is worth separating them. Openness is a perturbation fact: an invertible operator stays invertible under a small enough perturbation, and the bound (4.4) makes the permissible perturbation size $1/(2CM)$ *uniform* in the base point, which is what lets a single Neumann-series radius cover the whole interval rather than shrinking to nothing as t varies. Closedness is a compactness surrogate: the bound pins the solutions x_k inside a ball and controls how far apart the solutions of nearby equations can be, so the solutions of $L_{t_k}x = y$ converge to a solution of the limit equation. Neither step needs Y complete, since the solutions are built in X; it is the completeness of the domain X that both the Neumann series and the Cauchy argument use. The theorem reduces solving a hard equation to solving an easy one along a path, provided the path never loses the estimate, and this is precisely how a Schauder estimate, which is an a priori bound and nothing more, is converted into an existence theorem.

The mechanism is visible already in finite dimensions, where it recovers a familiar fact about matrices and shows what the bound (4.4) is doing geometrically.

Example 4.3.2: Continuity for a Path of Matrices

Take $X = Y = \mathbb{R}^n$ with the Euclidean norm, $L_0 = I$ the identity, and $L_1 = A$ an invertible $n \times n$ matrix, and interpolate by $A_t = (1-t)I + tA$. Suppose every A_t satisfies $|x| \le C|A_t x|$ for a single C; equivalently A_t is invertible with $\|A_t^{-1}\| \le C$ uniformly, so the smallest singular value of A_t stays bounded below by $1/C$ along the whole segment. The theorem then concludes that $A = A_1$ is invertible, deforming the trivially solvable system $x = y$ into $Ax = y$. Concretely, for

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}, \qquad A_t = \begin{pmatrix} 1+t & t \\ 0 & 1+t \end{pmatrix}$$

the eigenvalues of A_t are both $1 + t \ge 1$, and $\det A_t = (1+t)^2 \ge 1$ never vanishes, so no A_t is singular and the uniform bound holds. The content of the finite-dimensional statement is exactly that the determinant, a continuous function of t, cannot pass through 0 once the bound keeps it away from 0: the path of invertible matrices stays inside the invertible group. In infinite dimensions there is no determinant to watch, and the a priori bound replaces it, forbidding the "eigenvalue at the origin" that would break invertibility.

The finite-dimensional picture also flags what can go wrong without the *uniform* bound: a path of invertible matrices whose smallest singular value drifts to 0 has each A_t invertible yet the limit singular, and the solutions x_k run off to infinity precisely where the bound fails. The strength of (4.4) is that one constant governs the entire family, and for the Dirichlet problem that constant is given, once and for all, by the global Schauder estimate.

Everything is now in place to solve the variable-coefficient Dirichlet problem in the classical sense. The plan is to read the boundary-value problem as a single bounded operator, deform its top-order part to the Laplacian, verify the endpoint is solvable, and check that the global Schauder estimate holds uniformly along the deformation so that theorem 4.3.1 applies. The one input the Schauder estimate does not itself provide is the C^0 bound: the global estimate of theorem 4.2.5 controls $\|u\|_{C^{2,\alpha}}$ by the data together with $\|u\|_{C^0}$, and to close the argument that last term must be removed. The maximum principle does this, bounding $\|u\|_{C^0(\overline{U})}$ by $\|f\|_{C^0(\overline{U})}$ and $\|g\|_{C^0(\partial U)}$ with a constant depending only on the ellipticity, the coefficient bounds, and $\operatorname{diam} U$; its proof is the business of a later chapter, and it is used here as a cited input. With the C^0 term absorbed, the Schauder estimate becomes a bound by the data alone, uniform across the deformation, which is the hypothesis (4.4) requires.

Theorem 4.3.3: Classical Solvability of the Dirichlet Problem

Let $U \subseteq \mathbb{R}^n$ be bounded with $C^{2,\alpha}$ boundary, $0 < \alpha < 1$, and let $Lu = -a^{ij}\partial_{ij}u + b^i\partial_i u + cu$ be uniformly elliptic with coefficients $a^{ij}, b^i, c \in C^\alpha(\overline{U})$ and $c \geq 0$. For every $f \in C^\alpha(\overline{U})$ and every $g \in C^{2,\alpha}(\partial U)$ the Dirichlet problem

$$Lu = f \ \text{ in } U, \qquad u = g \ \text{ on } \partial U$$

has a unique solution $u \in C^{2,\alpha}(\overline{U})$.

***Proof*:**
Uniqueness comes first and is short. If u_1, u_2 both solve the problem, their difference $w = u_1 - u_2$ satisfies $Lw = 0$ in U and $w = 0$ on ∂U, and the maximum principle for L (with $c \geq 0$) forces $w \equiv 0$. So there is at most one solution, and the work is existence.

Step 1: Reduction to homogeneous boundary data. Because ∂U is $C^{2,\alpha}$ and $g \in C^{2,\alpha}(\partial U)$, there is a function $G \in C^{2,\alpha}(\overline{U})$ with $G = g$ on ∂U (extend g off the boundary using the regularity of ∂U). Writing $u = v + G$, the problem for u becomes $Lv = f - LG$ in U with $v = 0$ on ∂U. Here $LG \in C^\alpha(\overline{U})$ since $G \in C^{2,\alpha}$ and the coefficients are C^α, so the new datum $\tilde{f} = f - LG$ lies in $C^\alpha(\overline{U})$. It therefore suffices to solve

$$Lv = \tilde{f} \ \text{ in } U, \qquad v = 0 \ \text{ on } \partial U \tag{4.5}$$

for every $\tilde{f} \in C^\alpha(\overline{U})$, which is what the operator formulation below encodes.

Step 2: The boundary-value problem as a bounded operator. Let

$$X = \left\{ v \in C^{2,\alpha}(\overline{U}) \;\middle|\; v = 0 \text{ on } \partial U \right\}, \qquad Y = C^\alpha(\overline{U})$$

both Banach spaces in their Hölder norms, and define $\mathcal{L}\colon X \to Y$ by $\mathcal{L}v = Lv$. The map is bounded: for $v \in X$,

$$\|Lv\|_{C^\alpha} \leq \|a^{ij}\|_{C^\alpha}\|\partial_{ij}v\|_{C^\alpha} + \|b^i\|_{C^\alpha}\|\partial_i v\|_{C^\alpha} + \|c\|_{C^\alpha}\|v\|_{C^\alpha} \leq C\|v\|_{C^{2,\alpha}}$$

since the Hölder space $C^\alpha(\overline{U})$ is a Banach algebra under pointwise product. Solving (4.5) is exactly solving $\mathcal{L}v = \tilde{f}$, so $\mathcal{L}$ onto is what must be shown.

Step 3: The deformation and its endpoint. Let $L_1 = L$ and let $L_0 = -\Delta$, and interpolate the operators by

$$L_t = (1-t)(-\Delta) + tL = -\big((1-t)\delta^{ij} + ta^{ij}\big)\partial_{ij} + t\big(b^i\partial_i + c\big)$$

where δ^{ij} is the identity coefficient matrix of $-\Delta$. Each L_t is a non-divergence operator of the same type: its leading coefficients $(1-t)\delta^{ij} + ta^{ij}$ are a convex combination of two symmetric uniformly elliptic matrices, hence symmetric and uniformly elliptic with the *same* ellipticity constants for every t (the ellipticity constant of a convex combination is at least the smaller of the two), its lower-order coefficients tb^i, tc inherit the C^α bounds of L with $tc \geq 0$, and all coefficient C^α norms are bounded by those of $-\Delta$ and L together, uniformly in t. Each L_t therefore defines a bounded operator $X \to Y$ as in Step 2, and $L_t = (1-t)L_0 + tL_1$ is the linear interpolation of theorem 4.3.1. The endpoint $L_0 = -\Delta$ is solvable: for $\tilde{f} \in C^\alpha(\overline{U})$ the Dirichlet problem $-\Delta v = \tilde{f}$ in U, $v = 0$ on ∂U has a weak solution $v \in H^1_0(U)$ by theorem 1.3.4, and the interior and boundary regularity of this chapter (theorems 4.1.4 and 4.2.4) upgrade it to $v \in C^{2,\alpha}(\overline{U})$; thus $L_0 \colon X \to Y$ is onto, and $0 \in S$.

Step 4: The uniform a priori bound. It remains to verify the hypothesis (4.4) of theorem 4.3.1 with a single constant. Fix t and $v \in X$, and set $\tilde{f} = L_t v$. The global Schauder estimate of theorem 4.2.5, applied to the operator L_t, gives

$$\|v\|_{C^{2,\alpha}(\overline{U})} \le C_1\big(\|L_t v\|_{C^{\alpha}(\overline{U})} + \|v\|_{C^0(\overline{U})}\big)$$

and, because $v = 0$ on ∂U, the maximum principle for L_t bounds the last term by the datum, $\|v\|_{C^0(\overline{U})} \le C_2\|L_t v\|_{C^0(\overline{U})} \le C_2\|L_t v\|_{C^{\alpha}(\overline{U})}$. The point is that both constants C_1, C_2 can be taken independent of t: they depend on the ellipticity constants, the coefficient C^{α} norms, the boundary regularity, and $\operatorname{diam} U$, and Step 3 exhibited these as uniform across the family. Combining the two displays,

$$\|v\|_{C^{2,\alpha}(\overline{U})} \le C_1(1 + C_2)\,\|L_t v\|_{C^{\alpha}(\overline{U})} = C\,\|L_t v\|_{C^{\alpha}(\overline{U})}$$

with $C = C_1(1 + C_2)$ independent of t, which is exactly (4.4) for the norms of X and Y.

Step 5: Conclusion. The family $L_t \colon X \to Y$ satisfies the uniform bound of Step 4 and the endpoint L_0 is onto by Step 3. By theorem 4.3.1 the solvable set S equals $[0, 1]$, so $L_1 = \mathcal{L}$ is onto: equation (4.5) has a solution $v \in C^{2,\alpha}(\overline{U})$ for every $\tilde{f}$, hence in particular for $\tilde{f} = f - LG$. Then $u = v + G \in C^{2,\alpha}(\overline{U})$ solves $Lu = f$ in U with $u = g$ on ∂U, and uniqueness was established at the outset, completing the proof.

The theorem is the payoff of the whole chapter, and its structure is worth reading in reverse. The existence of a classical solution rests on the a priori estimate, the a priori estimate rests on the interior and boundary Schauder estimates, and those rest on the single explicit computation for the Newtonian potential in lemma 4.1.3. No fixed-point theorem and no compactness in an infinite-dimensional space is needed; the method of continuity converts the estimate into existence by a connectedness argument on an interval, borrowing solvability from the constant-coefficient model at the far end. The hypothesis $c \ge 0$ enters only through the maximum principle, both for uniqueness and for the C^0 bound that removes the lower-order term from the estimate. When c is allowed to change sign the deformation and the interior estimate survive unchanged, but the C^0 bound can fail exactly when 0 is an eigenvalue of L, and unique solvability then obeys the Fredholm alternative of theorem 3.2.1 rather than holding outright, the Schauder analog of what the weak theory recorded in the L^2 setting.

The gain over the weak theory is a gain in regularity that matches the smoothness of the data. The variational method of chapter 3 produced a solution in $H_0^1(U)$ from a datum in $H^{-1}(U)$, an L^2-scale statement that says nothing pointwise. The Schauder route, run on Hölder data, produces a solution with two classical derivatives that are themselves Hölder continuous, so $Lu = f$ holds at every point of U in the ordinary sense rather than only against test functions. The two theories are the endpoints of a single hierarchy: the weak solution is the object one can always construct, and the Schauder estimate is the machine that promotes it to a classical solution whenever the data are Hölder. The L^p theory of the next chapter interpolates between them, trading the Hölder scale for the $W^{2,p}$ scale and the explicit potential estimate for the Calderón-Zygmund bound on the same second-derivative singular integral.

5

L^p Regularity and the Calderón-Zygmund Theory

Schauder theory upgraded a weak solution to a classical one in the Hölder scale; this chapter runs the parallel theory in the Lebesgue scale, where the estimate is an L^p bound on the second derivatives and the engine is the Calderón-Zygmund theory of singular integrals. The two theories share a method, comparison with the constant-coefficient operator by freezing coefficients, but differ in their analytic core: where Schauder used the explicit Hölder estimate for the Newtonian potential, the L^p theory imports the boundedness of the second-derivative singular integral on L^p from the harmonic-analysis volume. This chapter derives the interior and global $W^{2,p}$ estimates from that imported bound and bootstraps them through the Sobolev embedding to recover the full regularity of solutions with smooth data.

5.1 The Second-Derivative Singular Integral

The L^p theory begins where the Schauder theory did, with the constant-coefficient model $-\Delta w = f$ and its solution by the Newtonian potential. What changes is the scale in which the second derivatives are measured. The Schauder estimate bounded the Hölder seminorm of $\partial_{ij} w$ by the Hölder seminorm of f (lemma 4.1.3); the L^p estimate bounds the L^p norm of $\partial_{ij} w$ by the L^p norm of f. The obstruction is the same in both scales. Writing the second derivative of the potential as an integral against $\partial_{ij} N$ produces a kernel with a nonintegrable singularity at the origin, and the integral makes sense only as a principal value, its convergence gave by a cancellation in the kernel. This section identifies that kernel, records the two structural properties that make it a Calderón-Zygmund kernel, and reads off the L^p bound from the singular-integral theorem imported from harmonic analysis. The analytic work of proving that theorem is not repeated here; it is cited in the form used.

The fundamental solution is the one used throughout chapter 4: for $n \geq 3$,

$$N(x) = \frac{1}{n(n-2)\omega_n}\,|x|^{2-n}$$

where ω_n is the volume of the unit ball in $\mathbb{R}^n$, normalized so that $-\Delta N = \delta_0$ in the sense of distributions, with $N(x) = -\frac{1}{2\pi}\log|x|$ in dimension $n = 2$. Convolution with N inverts $-\Delta$ on compactly supported data. The first derivatives $\partial_i N$ are homogeneous of degree $1-n$ and locally integrable, so the first derivative of the potential is an ordinary convolution; the difficulty is entirely in the second derivative, and it is worth naming the kernel that carries it before the potential is differentiated.

Definition 5.1.1: Calderón-Zygmund Kernel

A *Calderón-Zygmund kernel* on $\mathbb{R}^n$ is a function $K\colon \mathbb{R}^n \setminus \{0\} \to \mathbb{R}$ with the three properties

1. K is smooth on $\mathbb{R}^n \setminus \{0\}$;
2. K is homogeneous of degree $-n$, that is $K(\lambda z) = \lambda^{-n}K(z)$ for every $\lambda > 0$ and every $z \neq 0$;
3. K has mean zero on spheres, $\int_{|z|=1} K(z)\,dS(z) = 0$.

The associated *singular integral operator* acts on $f \in C_c^\infty(\mathbb{R}^n)$ by the principal value

$$Tf(x) = \mathrm{p.v.}\int_{\mathbb{R}^n} K(x-y)f(y)\,dy = \lim_{\varepsilon\to 0}\int_{|x-y|\geq\varepsilon} K(x-y)f(y)\,dy$$

The three conditions are exactly what makes the principal value converge. Homogeneity of degree $-n$ fixes the size of the singularity: $|K(z)| \leq C|z|^{-n}$, so the kernel is on the boundary of local integrability, one power short in every dimension. The excision of a ball of radius ε is forced by that borderline size, since $\int_{\varepsilon\leq|z|\leq 1}|z|^{-n}\,dz$ diverges logarithmically as $\varepsilon \to 0$. The mean-zero condition rescues the limit: on a spherical shell $\varepsilon \leq |z| \leq 1$ the divergent part of $\int K$ is $\left(\int_{|z|=1} K\,dS\right)\log(1/\varepsilon)$, which vanishes identically when the sphere average is zero, so the excised integrals converge as $\varepsilon \to 0$ rather than diverging with the logarithm. Homogeneity and cancellation are therefore not decorative hypotheses but the precise conditions under which a kernel of this borderline size defines an operator at all.

Example 5.1.2: The Hilbert-Transform Kernel and the Riesz Kernels

In dimension $n = 1$ the kernel $K(z) = 1/(\pi z)$ is homogeneous of degree -1 and odd, so its integral over the two-point sphere $\{-1, 1\}$ is $K(1) + K(-1) = 0$; the associated operator is the Hilbert transform. Its higher-dimensional analogues are the Riesz kernels

$$R_j(z) = c_n\,\frac{z_j}{|z|^{n+1}}$$

with c_n a dimensional constant. Each R_j is smooth away from the origin, and under $z \mapsto \lambda z$ the numerator scales by λ and the denominator by λ^{n+1}, so R_j is homogeneous of degree $-n$. It has mean zero on the unit sphere because z_j is an odd function of z and the sphere is symmetric under $z \mapsto -z$: the contributions from antipodal points cancel in pairs. The Riesz kernels are the model Calderón-Zygmund kernels, and the second-derivative kernels of the Newtonian potential belong to the same homogeneity class, arising as a further derivative of a homogeneous fundamental solution.

The kernel that carries the second derivative of the potential is $\partial_{ij}N$, and the content of the next result is twofold: it is the correct kernel, once a local term from differentiating the singularity is separated off, and it is a Calderón-Zygmund kernel in the sense just defined. The local term is not an artifact to be discarded; it is the trace of the delta function that N inverts, and it is what makes $\sum_i \partial_{ii}w = -f$ come out right.

Proposition 5.1.3: Second Derivatives of the Newtonian Potential

Let $f \in C_c^\infty(\mathbb{R}^n)$ and let $w = N * f$ be the Newtonian potential. Then $w \in C^\infty(\mathbb{R}^n)$, and its second derivatives are given by

$$\partial_{ij}w(x) = c_{ij}f(x) + \mathrm{p.v.}\int_{\mathbb{R}^n} K_{ij}(x-y)f(y)\,dy \tag{5.1}$$

where the kernel and the local constant are

$$K_{ij}(z) = \partial_{ij}N(z), \qquad c_{ij} = \int_{|z|=1} \partial_i N(z)\,\nu_j(z)\,dS(z)$$

with $\nu(z) = z/|z|$ the unit vector in the $z = x-y$ variable pointing radially outward from the origin, which in y-space is the inward normal to the excised ball, pointing toward its center. Each K_{ij} is a Calderón-Zygmund kernel: it is smooth on $\mathbb{R}^n \setminus \{0\}$, homogeneous of degree $-n$, and has mean zero on spheres. Moreover $\sum_i c_{ii} = -1$ and $\sum_i K_{ii} = 0$ away from the origin, so that $-\Delta w = f$.

***Proof*:**
Step 1: The kernel and its homogeneity. That $w \in C^\infty(\mathbb{R}^n)$ is routine: $N \in L^1_{\mathrm{loc}}$ and $f \in C_c^\infty$, so writing $w(x) = \int N(z)f(x-z)\,dz$ lets every derivative fall on the smooth compactly supported factor f and pass under the integral. Away from the origin N is smooth, and for $n \geq 3$ a direct differentiation of $N(z) = \gamma_n|z|^{2-n}$ with $\gamma_n = [n(n-2)\omega_n]^{-1}$ gives

$$\partial_i N(z) = \gamma_n(2-n)\,|z|^{-n}z_i, \qquad \partial_{ij}N(z) = \gamma_n(2-n)\Big(|z|^{-n}\delta_{ij} - n\,|z|^{-n-2}z_iz_j\Big)$$

(in dimension $n = 2$ the same computation with $N = -\frac{1}{2\pi}\log|z|$ produces $\partial_{ij}N(z) = \frac{1}{2\pi}\big(2|z|^{-4}z_iz_j - |z|^{-2}\delta_{ij}\big)$, of the same homogeneity). Set $K_{ij} = \partial_{ij}N$. It is smooth on $\mathbb{R}^n \setminus \{0\}$ because N is. Under $z \mapsto \lambda z$ both terms scale by λ^{-n}: the factors $|z|^{-n}$ and $|z|^{-n-2}z_iz_j$ are each homogeneous of degree $-n$. Hence K_{ij} is homogeneous of degree $-n$, on the boundary of local integrability, which is why the integral in (5.1) cannot be taken in the ordinary sense.

Step 2: Mean zero on spheres. Write $K_{ij}(z) = |z|^{-n}\Omega_{ij}(z/|z|)$, where the angular profile is $\Omega_{ij}(\theta) = \gamma_n(2-n)(\delta_{ij} - n\theta_i\theta_j)$ for a unit vector θ. Integrating over the unit sphere,

$$\int_{|z|=1} K_{ij}(z)\,dS(z) = \gamma_n(2-n)\int_{|\theta|=1} \big(\delta_{ij} - n\,\theta_i\theta_j\big)\,dS(\theta)$$

By symmetry of the sphere the off-diagonal moments vanish, $\int_{|\theta|=1}\theta_i\theta_j\,dS = 0$ for $i \neq j$, while the diagonal moments are all equal, $\int_{|\theta|=1}\theta_i^2\,dS = \frac{1}{n}\int_{|\theta|=1}|\theta|^2\,dS = \frac{1}{n}|S^{n-1}|$, since $\sum_i \theta_i^2 = 1$ on the sphere. For $i \neq j$ the integrand's average is $\delta_{ij} - n \cdot 0 = 0$, and for $i = j$ it is $1 - n \cdot \frac{1}{n}|S^{n-1}|/|S^{n-1}| = 1 - 1 = 0$. In both cases the sphere average is zero, so K_{ij} has mean zero on spheres and is a Calderón-Zygmund kernel.

Step 3: The local constant from differentiating the singularity. The formula (5.1) is obtained by differentiating the potential twice while respecting the singularity of the kernel. The first derivative passes under the integral, since $\partial_i N$ is locally integrable:

$$\partial_i w(x) = \int_{\mathbb{R}^n} \partial_i N(x-y) f(y)\, dy$$

The second derivative cannot, because $\partial_{ij} N$ fails to be integrable near the diagonal. Excise a ball $B_\varepsilon(x)$ and differentiate the two remaining regions. On the exterior $\{|x-y| \geq \varepsilon\}$ differentiation passes under the integral and produces $\int_{|x-y|\geq\varepsilon} K_{ij}(x-y) f(y)\, dy$. The boundary of the excised ball contributes a surface term from differentiating the domain of integration: applying ∂_{x_j} to $\int_{|x-y|\geq\varepsilon} \partial_i N(x-y) f(y)\, dy$ moves the ball with x, and the divergence theorem converts the resulting boundary variation into

$$\int_{|x-y|=\varepsilon} \partial_i N(x-y)\, \nu_j\, f(y)\, dS(y)$$

where $\nu = (x-y)/|x-y|$ is the normal pointing toward the moving centre x, that is the inward normal to the excised ball (equivalently the outward normal to the exterior region of integration); the sign is fixed by the boundary moving outward in the direction ν_j as x advances in x_j. As $\varepsilon \to 0$ the factor $f(y)$ may be replaced by $f(x)$ (the error is $O(\varepsilon)$ times a bounded integral), and the surface integral of $\partial_i N\, \nu_j$ over the sphere of radius ε is independent of ε by the homogeneity of $\partial_i N$: rescaling $y = x + \varepsilon\theta$ turns $\partial_i N$ of degree $1-n$ against the surface measure of degree $n-1$ into a scale-invariant integral over the unit sphere. The limit is therefore $c_{ij} f(x)$ with $c_{ij} = \int_{|z|=1} \partial_i N(z)\, \nu_j\, dS(z)$, and the exterior integral converges to the principal value, giving (5.1). The constant c_{ij} is the residue of the singularity, produced precisely because K_{ij} is too singular to differentiate naively.

Step 4: The trace recovers $-\Delta w = f$. Summing (5.1) over $i = j$,

$$\Delta w(x) = \Big(\sum_i c_{ii}\Big) f(x) + \text{p. v.} \int_{\mathbb{R}^n} \Big(\sum_i K_{ii}(x-y)\Big) f(y)\, dy$$

The trace of the kernel is $\sum_i K_{ii} = \Delta N = 0$ on $\mathbb{R}^n \setminus \{0\}$, since N is harmonic away from the origin, so the principal-value term vanishes. For the local constant, on the unit sphere $\partial_i N(z)\, \nu_i = \nabla N \cdot \nu$ is the radial derivative $N'(1)$; from $\partial_i N(z) = \gamma_n (2-n)|z|^{-n} z_i$ and $\nu_i = z_i$ on $|z| = 1$ one gets $\nabla N \cdot \nu = \gamma_n(2-n) = -1/(n\omega_n)$, a constant, so

$$\sum_i c_{ii} = \int_{|z|=1} \nabla N(z) \cdot \nu\, dS(z) = \Big(-\tfrac{1}{n\omega_n}\Big)|S^{n-1}| = -1$$

using $|S^{n-1}| = n\omega_n$. Hence $\Delta w = -f$, that is $-\Delta w = f$, as we sought to show.

The formula (5.1) splits the second derivative into two mechanisms of different character. The principal-value integral is a genuinely nonlocal operator: the value of $\partial_{ij} w$ at x depends on f across all of space, weighted by a kernel that decays only like $|x-y|^{-n}$. The local term $c_{ij} f(x)$ is pointwise, the trace of the delta function that N inverts, and it is what distinguishes $\partial_{ij} N$ as a distribution from its pointwise values away from the origin. Any L^p bound on $\partial_{ij} w$ must control both pieces. The local piece is trivial, a multiplication by the constant c_{ij}, and so bounded on every L^p with norm $|c_{ij}|$; the nonlocal piece is the singular integral, and its L^p boundedness is the substance of the estimate. The cancellation verified in Step 2 is exactly the mechanism that makes that piece an operator: without the mean-zero condition the principal value would not converge, and there would be nothing to bound.

The L^p bound on the singular integral is the theorem the split with the analysis volumes was arranged to import. It is the central fact of the subject, proved in [4] by the Calderón-Zygmund decomposition (a weak-type $(1,1)$ bound from decomposing f at each height into a good part and a bad part supported on a controlled family of cubes, followed by Marcinkiewicz interpolation against the elementary L^2 bound from the Fourier transform). None of that machinery is reproduced here; the result is stated in the form used and cited.

Theorem 5.1.4: Calderón-Zygmund L^p Boundedness

Let K be a Calderón-Zygmund kernel on $\mathbb{R}^n$ and let T be the associated singular integral operator. For every p with $1 < p < \infty$ there is a constant $C_p = C_p(n, K)$ such that

$$\|Tf\|_{L^p(\mathbb{R}^n)} \leq C_p \|f\|_{L^p(\mathbb{R}^n)}$$

for all $f \in C_c^\infty(\mathbb{R}^n)$, and T extends to a bounded operator on $L^p(\mathbb{R}^n)$. The bound fails at the endpoints $p = 1$ and $p = \infty$.

***Proof*:**
This is the Calderón-Zygmund theorem, proved in full in [4]. Under the re-derivation test it is cited: the theorem is the reason the harmonic-analysis foundation sits upstream of this book, and repeating its proof would duplicate that book's central chapter. The interested reader will find there the weak-type $(1,1)$ estimate via the Calderón-Zygmund decomposition and its interpolation with the L^2 bound.

The restriction to the open range $1 < p < \infty$ is not a defect of the proof but a feature of the operators. At $p = 1$ the singular integral does not map L^1 to L^1: the Hilbert transform of an approximate identity has a tail decaying like $1/x$, which is not integrable, so $\|Tf\|_{L^1}$ cannot be controlled by $\|f\|_{L^1}$. What survives is the weak-type bound $|\{|Tf| > \lambda\}| \leq C\lambda^{-1}\|f\|_{L^1}$, the endpoint from which the strong bounds are interpolated, and it is genuinely weaker. At $p = \infty$ the failure is dual: T does not map L^∞ to L^∞, since the logarithmic divergence that the cancellation only just tames leaves Tf unbounded for bounded f, and the correct target is the space BMO of functions of bounded mean oscillation. Both endpoints are developed in [4]; for the elliptic estimates of this chapter the open range is all that is needed, since the data will always be taken in an L^p with p strictly between 1 and ∞.

Applying the theorem to the kernels K_{ij} turns the identification of proposition 5.1.3 into the second-derivative bound that drives the whole L^p theory. The motivating point is that the potential is the exact inverse of $-\Delta$ on compactly supported data, so a bound on $D^2(N * f)$ by f is precisely the statement that inverting the Laplacian gains two derivatives in the L^p scale, the flat model of every interior estimate to come.

Theorem 5.1.5: L^p Bound on the Second Derivatives of the Potential

Let $1 < p < \infty$. There is a constant $C_p = C_p(n,p)$ such that for every $f \in C_c^\infty(\mathbb{R}^n)$ the Newtonian potential $w = N * f$ satisfies

$$\|D^2 w\|_{L^p(\mathbb{R}^n)} \le C_p \, \|f\|_{L^p(\mathbb{R}^n)}$$

where $\|D^2 w\|_{L^p}^p = \sum_{i,j} \|\partial_{ij} w\|_{L^p}^p$. Consequently the map $f \mapsto D^2(N * f)$ extends to a bounded operator from $L^p(\mathbb{R}^n)$ into $L^p(\mathbb{R}^n; \mathbb{R}^{n\times n})$. The bound fails at $p = 1$ and $p = \infty$, where it is replaced by a weak-L^1 estimate and a BMO estimate respectively.

***Proof*:**
By proposition 5.1.3 each second derivative decomposes as

$$\partial_{ij} w = c_{ij} f + T_{ij} f, \qquad T_{ij} f = \mathrm{p.v.} \int_{\mathbb{R}^n} K_{ij}(x-y) f(y)\, dy$$

where K_{ij} is a Calderón-Zygmund kernel and c_{ij} is a constant. The two pieces are bounded separately. The local piece obeys $\|c_{ij} f\|_{L^p} = |c_{ij}| \, \|f\|_{L^p}$, a bounded multiplication on every L^p. The nonlocal piece is the singular integral operator T_{ij} attached to the kernel K_{ij}, and theorem 5.1.4, applied to this kernel, gives $\|T_{ij} f\|_{L^p} \le C_{p,ij} \|f\|_{L^p}$ for $1 < p < \infty$. Adding,

$$\|\partial_{ij} w\|_{L^p} \le \big(|c_{ij}| + C_{p,ij}\big) \|f\|_{L^p}$$

Summing the p-th powers over the finitely many pairs (i,j) and taking the p-th root bounds $\|D^2 w\|_{L^p}$ by $C_p \|f\|_{L^p}$ with C_p the resulting finite constant depending only on n and p. Since $C_c^\infty(\mathbb{R}^n)$ is dense in $L^p(\mathbb{R}^n)$ for $1 < p < \infty$, the bounded linear map $f \mapsto D^2(N * f)$ extends to all of L^p, completing the proof. The failure at the endpoints is inherited from the endpoint behaviour of the singular integrals recorded after theorem 5.1.4.

The estimate is the L^p counterpart of the Hölder potential estimate of lemma 4.1.3, and the parallel is exact. In the Hölder scale the inverse of $-\Delta$ gained two derivatives measured by a fractional continuity modulus; in the L^p scale it gains two derivatives measured by an integral norm, and in both scales the mechanism is a cancellation in the second-derivative kernel. The two scales diverge only at the price paid for the gain: the Schauder estimate holds for every exponent $0 < \alpha < 1$ in the open Hölder range, and this estimate holds for every exponent $1 < p < \infty$ in the open Lebesgue range, each closing strictly between its endpoints and failing at both. The two boundedness results are the same theorem about the Newtonian potential read in two function-space scales, and each is the seed from which its regularity theory grows.

The bound is a statement about the flat Laplacian alone. It is the exact inverse of $-\Delta$ that gains two derivatives, and a general uniformly elliptic L with continuous leading coefficients is not inverted by a convolution, so the estimate does not apply to it directly. The device that repairs this in the L^p scale is the same freezing of coefficients that carried the Schauder model estimate of lemma 4.1.3 to the variable-coefficient estimate theorem 4.1.4: freeze the leading coefficients at a point, apply the constant-coefficient bound, and absorb the coefficient-oscillation error on a small ball. The one difference is what makes the error small. In the Hölder scale the oscillation carried a factor r^α from the Hölder modulus of the coefficients; in the L^p scale it is the uniform continuity of the coefficients that shrinks the perturbation's $L^p \to L^p$ operator norm below one on a small enough ball, whereupon the same absorption applies. The constant-coefficient bound proved here is the model that argument freezes down to.

5.2 Interior $W^{2,p}$ Estimates

The singular-integral bound of the previous section controls the second derivatives of the Newtonian potential, hence of any solution of the constant-coefficient equation $-\Delta w = f$, in L^p. The operator that governs a genuine elliptic problem is not $-\Delta$ but the variable-coefficient

$$Lu = -a^{ij}\partial_{ij}u + b^i\partial_i u + cu$$

in non-divergence form, uniformly elliptic in the sense of definition 1.3.1 with leading matrix $A = (a^{ij})$ and ellipticity constants $0 < \theta \le \Lambda$. This section proves that when A is *continuous*, and the equation $Lu = f$ holds with $f \in L^p(U)$, the solution u inherits two derivatives in L^p on the interior, with the estimate $\|u\|_{W^{2,p}(V)} \le C(\|f\|_{L^p(U)} + \|u\|_{L^p(U)})$ on every $V \Subset U$. The argument runs parallel to theorem 4.1.4: freeze the leading coefficients at a point to reach a constant-coefficient operator estimated by the imported bound, and absorb the coefficient-oscillation error using the continuity of A.

The engine is the second-derivative bound $\|D^2(N * f)\|_{L^p} \le C_p\|f\|_{L^p}$ of theorem 5.1.5, which is stated for $-\Delta$. The frozen operator is a constant-coefficient elliptic operator $-a^{ij}(x_0)\partial_{ij}$ rather than $-\Delta$, so the first task is to transfer the bound across a linear change of variables. The following records that transfer as the estimate the freezing argument will invoke at each center.

Proposition 5.2.1: The Constant-Coefficient Estimate

Let $A_0 = (a_0^{ij})$ be a constant symmetric matrix with eigenvalues in $[\theta, \Lambda]$, and let $L_0 = -a_0^{ij}\partial_{ij}$ be the associated constant-coefficient operator. There is a constant $C = C(n, p, \theta, \Lambda)$ such that every $w \in W^{2,p}(\mathbb{R}^n)$ with compact support satisfies

$$\|D^2 w\|_{L^p(\mathbb{R}^n)} \le C\,\|L_0 w\|_{L^p(\mathbb{R}^n)}$$

for $1 < p < \infty$.

***Proof*:**
Since A_0 is symmetric positive definite with eigenvalues in $[\theta, \Lambda]$, it has a symmetric positive definite square root $A_0^{1/2}$ with $\|A_0^{1/2}\|, \|A_0^{-1/2}\|$ bounded in terms of θ and Λ. The linear change of variables $y = A_0^{-1/2}x$ carries L_0 to $-\Delta$: writing $\tilde{w}(y) = w(A_0^{1/2}y)$, the chain rule gives $a_0^{ij}\partial_{ij}w = \Delta\tilde{w}$ after the substitution, so $L_0 w = g$ becomes $-\Delta\tilde{w} = \tilde{g}$ with $\tilde{g}(y) = g(A_0^{1/2}y)$. Set $g = L_0 w$. Since w has compact support, so does $\tilde{w}$, and $\tilde{g} \in L^p(\mathbb{R}^n)$ has compact support as well. The Newtonian potential $N * \tilde{g}$ also solves $-\Delta(N * \tilde{g}) = \tilde{g}$, so the difference $h = \tilde{w} - N * \tilde{g}$ is harmonic on all of $\mathbb{R}^n$. It is not compactly supported (the potential $N * \tilde{g}$ carries tails decaying like $|y|^{2-n}$), so one cannot conclude $h \equiv 0$; but $\tilde{w} \in W^{2,p}(\mathbb{R}^n)$ and, by theorem 5.1.5, $D^2(N * \tilde{g}) \in L^p(\mathbb{R}^n)$, so $D^2 h \in L^p(\mathbb{R}^n)$. A harmonic function whose second derivatives lie in $L^p(\mathbb{R}^n)$ with $p < \infty$ is affine, by the Liouville-type argument that each entry $\partial_{ij}h$ is itself harmonic and in L^p, hence identically zero (a harmonic L^p function on $\mathbb{R}^n$ vanishes for $p < \infty$). Therefore $D^2\tilde{w} = D^2(N * \tilde{g})$, and theorem 5.1.5 gives $\|D^2\tilde{w}\|_{L^p} = \|D^2(N * \tilde{g})\|_{L^p} \le C_p\|\tilde{g}\|_{L^p}$.

Reading the derivatives and the L^p norm back through the linear map $y = A_0^{-1/2}x$ multiplies each side by Jacobian factors and factors $\|A_0^{\pm 1/2}\|$ bounded by θ and Λ, so

$$\|D^2 w\|_{L^p(\mathbb{R}^n)} \leq C(n, p, \theta, \Lambda)\, \|L_0 w\|_{L^p(\mathbb{R}^n)}$$

which is the claim, completing the proof.

The proposition is the constant-coefficient regularity statement, and it is all the analysis this section imports: a solution of a constant-coefficient elliptic equation has its full second-derivative L^p norm controlled by the L^p norm of the right-hand side, with a constant that sees the operator only through its ellipticity constants. The change of variables is where the two constants θ and Λ enter, and they enter only through the ratio Λ/θ that measures how far A_0 is from a multiple of the identity; the bound degrades as the operator becomes more anisotropic but never depends on the solution. This is the exact Lebesgue-scale counterpart of the frozen estimate that theorem 4.1.4 extracts from the Newtonian potential in the Hölder scale.

With the frozen model in hand, the variable operator is reached by comparison. On a ball small enough that A cannot vary much, L differs from the frozen operator $-a^{ij}(x_0)\partial_{ij}$ by a term carrying the top derivatives with a small coefficient, and that smallness is what lets the error be absorbed. The continuity of A is the whole of the hypothesis: it is what makes the oscillation $\sup_B |A(x) - A(x_0)|$ tend to zero as the ball shrinks, so that the perturbation has $L^p \to L^p$ operator norm below 1.

The subtlety in that absorption is that the frozen estimate controls the second derivatives on an inner ball in terms of a small multiple of the second derivatives on a slightly larger one: the two norms live over different balls, so the dangerous term cannot simply be subtracted. The device that turns such a self-improving inequality into a bound is an iteration over a one-parameter family of concentric balls, recorded once here for use in the proof.

Lemma 5.2.2: Absorption on Concentric Balls

Let $\Phi\colon [\tau_0, \tau_1] \to [0, \infty)$ be bounded, and suppose there are constants $0 \leq \vartheta < 1$, $A \geq 0$, and $\beta > 0$ with

$$\Phi(\rho) \leq \vartheta\, \Phi(\rho') + \frac{A}{(\rho' - \rho)^\beta} \qquad \text{for all} \quad \tau_0 \leq \rho < \rho' \leq \tau_1$$

Then there is a constant $C = C(\vartheta, \beta)$ with $\Phi(\tau_0) \leq C\, A\, (\tau_1 - \tau_0)^{-\beta}$.

***Proof*:**

Fix $0 < \tau < 1$ with $\tau^{-\beta}\vartheta < 1$, and define radii $\rho_0 = \tau_0$ and $\rho_{k+1} = \rho_k + (1-\tau)\tau^k(\tau_1 - \tau_0)$, an increasing sequence in $[\tau_0, \tau_1]$ with $\rho_{k+1} - \rho_k = (1-\tau)\tau^k(\tau_1 - \tau_0)$. Applying the hypothesis on each pair $\rho_k < \rho_{k+1}$ and iterating gives

$$\Phi(\rho_0) \leq \vartheta^m \Phi(\rho_m) + \frac{A}{(1-\tau)^\beta (\tau_1 - \tau_0)^\beta} \sum_{k=0}^{m-1} \vartheta^k \tau^{-\beta k}$$

Since $\vartheta\tau^{-\beta} < 1$ the geometric sum converges as $m \to \infty$, and $\vartheta^m \Phi(\rho_m) \to 0$ because Φ is bounded and $\vartheta < 1$; the limit is the claimed bound with $C = (1-\tau)^{-\beta}(1 - \vartheta\tau^{-\beta})^{-1}$, completing the proof.

The lemma is what makes the absorption work. The coefficient $\vartheta < 1$ in front of the outer-ball quantity is what the smallness of the coefficient oscillation gives, and the factor $(\rho' - \rho)^{-\beta}$ is what the cutoff derivatives cost; the conclusion trades the whole self-improving inequality for a clean bound on the inner ball, at the price of the fixed geometric constant $C(\vartheta, \beta)$.

Theorem 5.2.3: Interior $W^{2,p}$ Estimate

Let $U \subseteq \mathbb{R}^n$ be bounded and open, let $1 < p < \infty$, and let

$$Lu = -a^{ij}\partial_{ij}u + b^i\partial_i u + cu$$

be uniformly elliptic in the sense of definition 1.3.1 with ellipticity constants $0 < \theta \leq \Lambda$, symmetric leading coefficients $a^{ij} = a^{ji} \in C(\overline{U})$, and lower-order coefficients $b^i, c \in L^\infty(U)$. Suppose $u \in W^{2,p}_{\mathrm{loc}}(U)$ satisfies $Lu = f$ almost everywhere in U with $f \in L^p(U)$. Then for every $V \Subset U$ there is a constant

$$C = C\big(n, p, \theta, \Lambda, \omega, \|b^i\|_{L^\infty}, \|c\|_{L^\infty}, \operatorname{dist}(V, \partial U)\big)$$

depending on the coefficients only through the ellipticity constants and the modulus of continuity ω of A, with

$$\|u\|_{W^{2,p}(V)} \leq C\big(\|f\|_{L^p(U)} + \|u\|_{L^p(U)}\big)$$

***Proof*:**
The argument freezes the leading coefficients on a small ball, estimates the constant model by proposition 5.2.1, absorbs the oscillation and lower-order errors, and patches the ball estimates over a finite cover of V. Write $L_0^{\mathrm{fr}}u = -a^{ij}(x_0)\partial_{ij}u$ for the operator frozen at a center x_0, so that $Lu = L_0^{\mathrm{fr}}u + (a^{ij}(x_0) - a^{ij})\partial_{ij}u + b^i\partial_i u + cu$. Let $\omega(r) = \sup\big\{|a^{ij}(x) - a^{ij}(y)| \;\big|\; |x - y| \leq r,\ i, j\big\}$ be the modulus of continuity of A on $\overline{U}$; since A is continuous on the compact set $\overline{U}$ it is uniformly continuous, so $\omega(r) \to 0$ as $r \to 0$.

Step 1: The frozen equation on concentric balls. Fix $x_0 \in U$ and a radius $r > 0$ with $B = B_r(x_0) \Subset U$, to be chosen. On B the equation $Lu = f$ rearranges to

$$L_0^{\mathrm{fr}}u = f + \big(a^{ij}(x) - a^{ij}(x_0)\big)\partial_{ij}u - b^i\partial_i u - cu \tag{5.2}$$

which exhibits u as a solution of the constant-coefficient operator L_0^{fr} with a right-hand side that contains the top derivatives of u through its first term. The absorption of that top-order term needs room to work, so run the estimate not on a single pair of balls but over a one-parameter family of concentric ones. For radii $\frac{r}{2} \leq \rho < \rho' \leq r$ write $B_\rho = B_\rho(x_0)$ and choose a cutoff $\eta \in C_c^\infty(B_{\rho'})$ with $\eta \equiv 1$ on B_ρ and $|\partial^k\eta| \leq C(\rho' - \rho)^{-k}$ for $k \leq 2$; the width $\rho' - \rho$, not r, sets the scale on which η varies. Then $\eta u \in W^{2,p}(\mathbb{R}^n)$ has compact support, and the product rule gives

$$L_0^{\mathrm{fr}}(\eta u) = \eta\, L_0^{\mathrm{fr}}u - a^{ij}(x_0)\big(2\,\partial_i\eta\,\partial_j u + u\,\partial_{ij}\eta\big)$$

The frozen operator sees the constant matrix $A(x_0)$, symmetric with eigenvalues in $[\theta, \Lambda]$; symmetry is no restriction, since $\partial_{ij}u = \partial_{ji}u$ for $u \in W^{2,p}$ means only the symmetric part $\frac{1}{2}(a^{ij} + a^{ji})$ of the leading coefficients acts, so proposition 5.2.1 applies to $w = \eta u$ and yields

$$\|D^2(\eta u)\|_{L^p(\mathbb{R}^n)} \leq C_0\,\|L_0^{\mathrm{fr}}(\eta u)\|_{L^p(\mathbb{R}^n)} \tag{5.3}$$

with $C_0 = C_0(n, p, \theta, \Lambda)$. Because $\eta \equiv 1$ on B_ρ, the left side dominates $\|D^2u\|_{L^p(B_\rho)}$, while η is supported in $B_{\rho'}$, so every term on the right lives on $B_{\rho'}$.

Step 2: The oscillation error and its absorption. Substituting (5.2) into the first term of $L_0^{\mathrm{fr}}(\eta u)$ produces, on the right of (5.3), the dangerous term

$$\left\|\eta\left(a^{ij}(x) - a^{ij}(x_0)\right)\partial_{ij}u\right\|_{L^p(B_{\rho'})} \leq \Big(\sup_{B_{\rho'}}|a^{ij}(x) - a^{ij}(x_0)|\Big)\|D^2u\|_{L^p(B_{\rho'})}$$

which is the only term carrying the full second derivative D^2u in the L^p norm on the right, and it lives on the outer ball $B_{\rho'}$. On $B_{\rho'} \subseteq B_r(x_0)$ the multiplier is bounded by the modulus of continuity, $\sup_{B_{\rho'}}|a^{ij}(x) - a^{ij}(x_0)| \leq \omega(r)$: the continuity of A makes the coefficient in front of the top-order norm as small as $\omega(r)$, which tends to 0 with r. Summing over the n^2 index pairs (i, j), the perturbation $v \mapsto (a^{ij}(x) - a^{ij}(x_0))\partial_{ij}v$ is an operator from $W^{2,p}$ to L^p whose norm on $B_{\rho'}$ is at most $n^2\omega(r)$, and its product with the constant C_0 of (5.3) is the $L^p \to L^p$ factor that must be brought below 1. Choose r small enough that

$$C_0\, n^2\, \omega(r) \leq \tfrac{1}{2} \tag{5.4}$$

Such an r exists because $\omega(r) \to 0$, and it depends only on n, p, θ, Λ and the modulus ω, not on u or on the center x_0; the same r therefore works at every center. With this choice the oscillation term on the right of (5.3) is at most $\frac{1}{2}\|D^2u\|_{L^p(B_{\rho'})}$. The remaining terms of $L_0^{\mathrm{fr}}(\eta u)$ are lower order: the cutoff commutators $2a^{ij}(x_0)\partial_i\eta\,\partial_j u + a^{ij}(x_0)u\,\partial_{ij}\eta$ carry at most one derivative of u and are supported in the annulus $B_{\rho'} \setminus B_\rho$ where $\partial\eta \neq 0$, contributing $C(\rho' - \rho)^{-1}\|\nabla u\|_{L^p(B_{\rho'})} + C(\rho' - \rho)^{-2}\|u\|_{L^p(B_{\rho'})}$, and the lower-order terms $\eta(f - b^i\partial_i u - cu)$ contribute $\|f\|_{L^p(B_r)} + \|b\|_\infty\|\nabla u\|_{L^p(B_{\rho'})} + \|c\|_\infty\|u\|_{L^p(B_r)}$. Collecting these, bounding the f and u norms by their norms over the full ball B_r (using $B_{\rho'} \subseteq B_r$), and keeping the gradient over the outer ball $B_{\rho'}$ where it lives, gives, for every pair $\frac{r}{2} \leq \rho < \rho' \leq r$,

$$\|D^2u\|_{L^p(B_\rho)} \leq \tfrac{1}{2}\,\|D^2u\|_{L^p(B_{\rho'})} + \frac{C_1}{\rho' - \rho}\,\|\nabla u\|_{L^p(B_{\rho'})} + \frac{C_1}{(\rho' - \rho)^2}\left(\|f\|_{L^p(B_r)} + \|u\|_{L^p(B_r)}\right) \tag{5.5}$$

where C_1 absorbs C_0 and the lower-order coefficient bounds. The gradient carries only the weight $(\rho' - \rho)^{-1}$, one power less than the u-term, because the cutoff commutator that produces it differentiates η once; that single power is exactly what lets the scaled interpolation of Step 3 fold the gradient into the top-order term with a constant coefficient. The point of the concentric family is now visible. The dangerous top-order term on the right sits on the *outer* ball $B_{\rho'}$, the controlled quantity on the *inner* ball B_ρ; the two norms are over different balls, so the term cannot simply be subtracted, but the coefficient $\frac{1}{2} < 1$ in front of it is exactly the hypothesis of the following iteration lemma, which converts the self-improving inequality (5.5) into a genuine bound.

Step 3: Folding in the gradient and applying the iteration lemma. Before iterating, remove the gradient term, which still sits on the outer ball $B_{\rho'}$ and would otherwise reintroduce a top-order quantity there. The scaled interpolation inequality for Sobolev norms ([2], the standard Gagliardo-Nirenberg interpolation on a ball, scaled by dilation) states that on a ball of radius σ and for every $\mu \in (0, 1]$,

$$\|\nabla u\|_{L^p(B_\sigma)} \leq C\Big(\mu\sigma\,\|D^2u\|_{L^p(B_\sigma)} + \frac{1}{\mu\sigma}\,\|u\|_{L^p(B_\sigma)}\Big)$$

with $C = C(n,p)$; the small parameter μ gives a small multiple of the second derivatives at the price of a large multiple of the function. Apply it on $B_{\rho'}$ with $\mu = \lambda(\rho' - \rho)/\rho'$ for a constant λ to be fixed, so that $\mu\sigma = \lambda(\rho' - \rho)$ and the gradient term of (5.5) becomes

$$\frac{C_1}{\rho' - \rho} \|\nabla u\|_{L^p(B_{\rho'})} \le CC_1\lambda \|D^2 u\|_{L^p(B_{\rho'})} + \frac{CC_1}{\lambda(\rho' - \rho)^2} \|u\|_{L^p(B_{\rho'})}$$

The single power $(\rho' - \rho)^{-1}$ carried by the gradient is what makes the $\|D^2 u\|$ coefficient $CC_1\lambda$ a *constant*, tunable below any threshold by shrinking λ, rather than a blowing-up weight; the price is only a heavier $(\rho' - \rho)^{-2}$ weight on the harmless u-term. Choose λ with $CC_1\lambda \le \frac{1}{4}$. Substituting into (5.5) and merging the two top-order terms gives, for every pair $\frac{r}{2} \le \rho < \rho' \le r$,

$$\|D^2 u\|_{L^p(B_\rho)} \le \tfrac{3}{4} \|D^2 u\|_{L^p(B_{\rho'})} + \frac{C_1'}{(\rho' - \rho)^2} \big(\|f\|_{L^p(B_r)} + \|u\|_{L^p(B_r)}\big) \tag{5.6}$$

with $C_1' = C_1'(n,p,\theta,\Lambda,\omega,r)$, the combined top-order coefficient $\frac{1}{2} + \frac{1}{4} = \frac{3}{4} < 1$. Now apply lemma 5.2.2 with $\Phi(\rho) = \|D^2 u\|_{L^p(B_\rho)}$ on $[\frac{r}{2}, r]$, bounded since $u \in W^{2,p}_{\text{loc}}$, and with $\vartheta = \frac{3}{4}$, $\beta = 2$, and $A = C_1'(\|f\|_{L^p(B_r)} + \|u\|_{L^p(B_r)})$ held fixed as ρ, ρ' vary. The hypothesis is exactly (5.6), and the conclusion, with inner radius $\tau_0 = r/2$ and $\tau_1 - \tau_0 = r/2$, is the local estimate

$$\|u\|_{W^{2,p}(B_{r/2})} \le C_2\big(\|f\|_{L^p(B_r)} + \|u\|_{L^p(B_r)}\big) \tag{5.7}$$

on the half-ball $B' = B_{r/2}(x_0)$, once $\|\nabla u\|_{L^p(B_{r/2})}$ and $\|u\|_{L^p(B_{r/2})}$ are added back to the left by the scaled interpolation and the trivial bound, with C_2 depending only on the stated quantities. The top-order oscillation and gradient terms are now both genuinely absorbed, on matched balls: the iteration replaced the invalid subtraction of an outer-ball norm from an inner-ball one with a convergent geometric sum over concentric shells, and the coefficient $\frac{3}{4} < 1$ that the smallness (5.4) and the interpolation together give is what makes the sum converge. The right side is finite for every $x_0 \in U$, consistent with the standing assumption $u \in W^{2,p}_{\text{loc}}(U)$: the estimate is a priori, quantifying the local $\|D^2 u\|_{L^p}$ already known to be finite rather than establishing that finiteness.

Step 4: Patching to a compact subdomain. Fix $V \Subset U$ and set $d = \operatorname{dist}(V, \partial U) > 0$. Shrink the radius r fixed in Step 2 if necessary so that $r \le d$, so that $B_r(x_0) \Subset U$ for every $x_0 \in V$. The half-balls $\{B_{r/2}(x_0)\}_{x_0 \in V}$ cover the compact set $\overline{V}$, so finitely many of them do, say N of them, with N depending only on n, $\operatorname{diam}(V)$, and r. On each the local estimate (5.7) holds with the same constant C_2, and summing the p-th powers over the finite subcover, using that each $B_r(x_0) \subseteq U$ so that every right-hand side is bounded by the norms over U, gives

$$\|u\|_{W^{2,p}(V)} \le C\big(\|f\|_{L^p(U)} + \|u\|_{L^p(U)}\big)$$

with $C = C(n,p,\theta,\Lambda,\omega,\|b\|_\infty,\|c\|_\infty,d)$ through r and the number of balls, as we sought to show.

The estimate recovers, in the Lebesgue scale, the two derivatives lost in passing from a strong solution to the equation to the raw datum f, at the price only of the L^p norm of u itself. Its architecture is identical to the Schauder estimate of theorem 4.1.4, and the parallel is worth reading off term by term. Both freeze the leading coefficients and compare with an explicitly solvable constant-coefficient model, the Newtonian potential in each case; both isolate the oscillation term $(a^{ij}(x) - a^{ij}(x_0))\partial_{ij} u$ as the one dangerous term carrying the top derivative; both absorb it by shrinking the ball until the coefficient in front is small; and both then clean up the lower-order remainder by interpolation. The single point of divergence is the analytic engine and the measure of the oscillation. Schauder bounded the frozen model by the Hölder estimate for the potential and paid for the oscillation with the Hölder modulus $[a^{ij}]_\alpha r^\alpha$, which required the coefficients to be Hölder continuous. The L^p theory bounds the frozen model by the singular-integral estimate theorem 5.1.5 and pays for the oscillation with the plain modulus of continuity $\omega(r)$, which requires only that A be continuous. Continuity of the leading coefficients, no more, is exactly the hypothesis that the operator-norm smallness (5.4) requires.

Two features of the estimate deserve isolation, and both mirror the Schauder discussion. It is an *a priori* estimate on a solution assumed to exist in $W^{2,p}_{\text{loc}}$; the production of such a solution, and the removal of the $\|u\|_{L^p}$ term on the right in favor of the data alone, come later from the global theory and the solvability of theorem 1.3.4. And it is *interior*: the constant degenerates as V approaches ∂U, because a solution can lose regularity at the boundary no matter how regular the data, and recovering $W^{2,p}$ control up to ∂U requires the boundary-flattening argument of the next section. The term $\|u\|_{L^p(U)}$ on the right is not decoration either: without it the estimate would be false. The homogeneous equation $Lu = 0$ has nonzero solutions (harmonic functions when $L = -\Delta$), whose L^p norm can be arbitrarily large while $f = Lu = 0$; no bound of $\|u\|_{W^{2,p}(V)}$ by $\|f\|_{L^p(U)}$ alone could hold on such a solution, and the $\|u\|_{L^p(U)}$ term is what carries their size.

The restriction $1 < p < \infty$ is inherited from the singular-integral bound and is genuine. The Calderón-Zygmund estimate theorem 5.1.5 fails at the endpoints $p = 1$ and $p = \infty$: at $p = 1$ the second-derivative singular integral maps L^1 only into weak-L^1, and at $p = \infty$ it maps L^∞ into BMO rather than back into L^∞, so the $W^{2,1}$ and $W^{2,\infty}$ estimates are both false. This is the same endpoint failure that forces the Schauder theory to live in the Hölder scale $C^{0,\alpha}$ with $0 < \alpha < 1$ rather than at the Lipschitz or continuous endpoints: neither regularity theory closes at the ends of its scale, and the open range $1 < p < \infty$ is the price the L^p theory pays for its engine.

The estimate is easiest to see for the model constant-coefficient operator, where the freezing step is vacuous and only the potential bound survives.

Example 5.2.4: The Poisson Equation on a Ball

Take $L = -\Delta$ on $U = B_2(0) \subseteq \mathbb{R}^n$, so $A = I$ is constant with $\theta = \Lambda = 1$ and there are no lower-order terms. Let $f \in L^p(B_2)$ with $1 < p < \infty$, and let $u \in W^{2,p}_{\text{loc}}(B_2)$ solve $-\Delta u = f$. Take $V = B_1(0) \Subset B_2$, so $\operatorname{dist}(V, \partial U) = 1$.

Because A is already constant, the modulus of continuity is $\omega \equiv 0$, and the smallness condition (5.4) holds for every radius r; no shrinking is needed to absorb an oscillation term, because there is none. The proof collapses to Steps 1, 3, and 4: cut off u by $\eta \in C_c^\infty(B_2)$ with $\eta \equiv 1$ on B_1, apply theorem 5.1.5 directly to ηu, which solves $-\Delta(\eta u) = \eta f - 2\nabla\eta \cdot \nabla u - u\Delta\eta$, and read off

$$\|D^2 u\|_{L^p(B_1)} \leq C_p\big(\|f\|_{L^p(B_2)} + \|\nabla u\|_{L^p(B_2)} + \|u\|_{L^p(B_2)}\big)$$

Interpolating the gradient term as in Step 3 and folding the constants gives

$$\|u\|_{W^{2,p}(B_1)} \leq C_p\big(\|f\|_{L^p(B_2)} + \|u\|_{L^p(B_2)}\big)$$

with C_p depending only on n and p. Concretely, for $f \equiv 1$ the radial solution $u(x) = -|x|^2/(2n)$ has $\partial_{ij} u = -\delta_{ij}/n$ constant, so $\|D^2 u\|_{L^p(B_1)}$ is a fixed finite number and the estimate holds with room to spare; the content of the theorem is that the bound persists for every $f \in L^p$, including data too rough for u to be twice differentiable in the classical sense.

5.3 Global $W^{2,p}$ Regularity and the Sobolev Bootstrap

The interior estimate of theorem 5.2.3 controls u on any $V \Subset U$ but says nothing near ∂U, where the second derivatives could in principle blow up. This section carries the second-derivative bound all the way to the boundary, given a Dirichlet condition there: this is the *global* $W^{2,p}$ estimate, and with it comes the existence and uniqueness of a $W^{2,p}$ strong solution of the Dirichlet problem. It then feeds the estimate back into itself. A weak solution produced by the variational theory of chapter 3 lives a priori only in H_0^1; the estimate promotes it to $W^{2,p}$, and if the data are smoother the promotion iterates up the Sobolev scale until, for C^∞ data, the Sobolev embedding forces u to be C^∞.

Throughout, $L = -a^{ij}\partial_{ij} + b^i\partial_i + c$ is uniformly elliptic in the sense of definition 1.3.1, with leading coefficients a^{ij} continuous on $\overline{U}$ and the lower-order coefficients b^i, c bounded, and $1 < p < \infty$.

The boundary estimate rests on the same device that carried the interior estimate: freeze the leading coefficients at a point, reduce to a constant-coefficient operator, and absorb the oscillation. What changes near ∂U is that the model domain is no longer all of $\mathbb{R}^n$ but the half-space, and the Dirichlet condition must be respected by the model. The passage from a curved boundary to a flat one is achieved by a change of variables that straightens ∂U locally; requiring the straightening map and its inverse to be $C^{1,1}$ is exactly what keeps the transformed leading coefficients continuous, so that the frozen-coefficient argument still applies. This is why the natural regularity hypothesis on the domain is $\partial U \in C^{1,1}$: one derivative less than the C^2 boundary of the Schauder theory, matching the fact that the L^p theory asks only for continuous, not Hölder-continuous, leading coefficients.

The half-space model itself is handled by the second-derivative singular-integral bound of theorem 5.1.5, applied not to the Newtonian potential on $\mathbb{R}^n$ but to its *odd reflection* across the boundary hyperplane. If u vanishes on $\{x_n = 0\}$, extending it oddly in x_n produces a function on all of $\mathbb{R}^n$ to which the whole-space estimate applies, and the reflection is arranged so that the tangential and mixed second derivatives, together with $\partial_{nn}u$ recovered from the equation, are all controlled. The following theorem records the outcome and the solvability it yields.

Theorem 5.3.1: Global $W^{2,p}$ Estimate and Solvability

Let $U \subseteq \mathbb{R}^n$ be bounded with boundary ∂U of class $C^{1,1}$, let $1 < p < \infty$, and let L be uniformly elliptic with continuous leading coefficients $a^{ij} \in C(\overline{U})$ and bounded b^i, c. Then:

1. *(A priori estimate.)* There is a constant $C = C(n, p, U, L)$ such that every $u \in W^{2,p}(U) \cap W_0^{1,p}(U)$ satisfies
$$\|u\|_{W^{2,p}(U)} \le C\left(\|Lu\|_{L^p(U)} + \|u\|_{L^p(U)}\right)$$
2. *(Solvability.)* If in addition $c \ge 0$, then for every $f \in L^p(U)$ the Dirichlet problem $Lu = f$ in U, $u = 0$ on ∂U, has a unique solution $u \in W^{2,p}(U) \cap W_0^{1,p}(U)$, and it satisfies $\|u\|_{W^{2,p}(U)} \le C\|f\|_{L^p(U)}$.

***Proof*:**
Step 1: Boundary flattening. Fix $x_0 \in \partial U$. Because $\partial U \in C^{1,1}$, there is a neighbourhood $\mathcal{O}$ of x_0 and a bijection $\Phi \colon \mathcal{O} \to B$, onto a ball $B \subseteq \mathbb{R}^n$, with Φ and Φ^{-1} both of class $C^{1,1}$, carrying $\mathcal{O} \cap U$ to $B^+ = B \cap \{y_n > 0\}$ and $\mathcal{O} \cap \partial U$ to $B \cap \{y_n = 0\}$. Write $y = \Phi(x)$ and $\tilde{u}(y) = u(\Phi^{-1}(y))$.

The chain rule transforms L into an operator $\tilde{L}$ on B^+ of the same second-order form,

$$\tilde{L}\tilde{u} = -\tilde{a}^{k\ell}\partial_{k\ell}\tilde{u} + \tilde{b}^k\partial_k\tilde{u} + c\,\tilde{u}, \qquad \tilde{a}^{k\ell}(y) = a^{ij}(\Phi^{-1}(y))\,\partial_i\Phi^k\,\partial_j\Phi^\ell$$

Since $\Phi \in C^{1,1}$ its first derivatives $\partial_i\Phi^k$ are Lipschitz, hence continuous, so the new leading coefficients $\tilde{a}^{k\ell}$ are continuous on $\overline{B^+}$; uniform ellipticity is preserved because $D\Phi$ is invertible with bounded inverse. The second derivatives of Φ, which are only bounded (again by $\Phi \in C^{1,1}$), enter $\tilde{b}^k$, which is therefore bounded but not necessarily continuous. This is harmless: $\tilde{b}^k$ is a lower-order coefficient, and lower-order terms are absorbed as error, exactly as in theorem 5.2.3. The boundary condition $u = 0$ on $\mathcal{O} \cap \partial U$ becomes $\tilde{u} = 0$ on $B \cap \{y_n = 0\}$.

Step 2: The half-space model by odd reflection. Consider first the constant-coefficient model operator $L_0 = -\tilde{a}^{k\ell}(y_0)\partial_{k\ell}$, the leading part frozen at a boundary point $y_0 \in \{y_n = 0\}$. A linear change of coordinates diagonalizing the constant positive-definite matrix $(\tilde{a}^{k\ell}(y_0))$ turns L_0 into the Laplacian on a half-space, so it suffices to treat $-\Delta$ on $\mathbb{R}^n_+ = \{y_n > 0\}$ with zero data on $\{y_n = 0\}$. Let $v \in W^{2,p}(\mathbb{R}^n_+) \cap W^{1,p}_0(\mathbb{R}^n_+)$ have compact support and set $g = -\Delta v$. Extend g to $\mathbb{R}^n$ by the *odd* reflection $\bar{g}(y', y_n) = -g(y', -y_n)$ for $y_n < 0$, and let $\bar{v} = N * \bar{g}$ be the Newtonian potential of $\bar{g}$ on all of $\mathbb{R}^n$. Because $\bar{g}$ is odd in y_n and the Newtonian kernel is even in y_n, the convolution $\bar{v}$ is odd in y_n, hence vanishes on $\{y_n = 0\}$; and $-\Delta\bar{v} = \bar{g} = g$ on $\mathbb{R}^n_+$. Thus $\bar{v}$ and v solve the same half-space Dirichlet problem, and by uniqueness (which the maximum principle gives, or the energy method of chapter 3 on the half-ball) they agree: $v = \bar{v}|_{\mathbb{R}^n_+}$. The second-derivative bound of theorem 5.1.5 applied to $\bar{v} = N * \bar{g}$ on $\mathbb{R}^n$ gives

$$\|D^2 v\|_{L^p(\mathbb{R}^n_+)} \le \|D^2\bar{v}\|_{L^p(\mathbb{R}^n)} \le C_p\|\bar{g}\|_{L^p(\mathbb{R}^n)} = 2^{1/p}C_p\|g\|_{L^p(\mathbb{R}^n_+)}$$

the last equality because odd reflection doubles the L^p norm of a p-th power. This is the flat-boundary analogue of the whole-space estimate: for the frozen Laplacian on a half-space with zero boundary data, $\|D^2 v\|_{L^p} \le C\|L_0 v\|_{L^p}$.

Step 3: Freezing and absorption at the boundary. Return to the variable-coefficient $\tilde{L}$ on B^+. Fix a boundary point y_0 and write $\tilde{L} = L_0 + (L_0 - \tilde{L}_{\text{lead}}) + (\text{lower order})$, where $L_0 = -\tilde{a}^{k\ell}(y_0)\partial_{k\ell}$ is the frozen leading operator and $\tilde{L}_{\text{lead}} = -\tilde{a}^{k\ell}(y)\partial_{k\ell}$. For $\tilde{u}$ supported in a small half-ball $B^+_r = B^+ \cap \{|y - y_0| < r\}$ and vanishing on the flat part of ∂B^+_r, Step 2 gives

$$\|D^2\tilde{u}\|_{L^p(B^+_r)} \le C\|L_0\tilde{u}\|_{L^p(B^+_r)} \le C\|\tilde{L}\tilde{u}\|_{L^p(B^+_r)} + C\big\|(\tilde{a}^{k\ell}(y_0) - \tilde{a}^{k\ell}(y))\partial_{k\ell}\tilde{u}\big\|_{L^p(B^+_r)} + C\|\text{l.o.t.}\|_{L^p(B^+_r)}$$

The oscillation term is estimated exactly as in the interior argument. By the continuity of $\tilde{a}^{k\ell}$ on the compact $\overline{B^+}$, given any $\varepsilon > 0$ there is $r > 0$ with $\sup_{y \in B^+_r}|\tilde{a}^{k\ell}(y_0) - \tilde{a}^{k\ell}(y)| < \varepsilon$ for all k, ℓ, so the middle term is bounded by $C\varepsilon\|D^2\tilde{u}\|_{L^p(B^+_r)}$. Choosing ε with $C\varepsilon < \frac{1}{2}$ absorbs it into the left-hand side. The lower-order term is controlled by $\|\tilde{u}\|_{W^{1,p}(B^+_r)}$, which the Gagliardo-Nirenberg interpolation inequality of [2] dominates by $\|D^2\tilde{u}\|_{L^p} + \|\tilde{u}\|_{L^p}$ with a small coefficient on the second-derivative part, again absorbed. What survives is a boundary half-ball estimate

$$\|\tilde{u}\|_{W^{2,p}(B^+_r)} \le C\left(\|\tilde{L}\tilde{u}\|_{L^p(B^+_r)} + \|\tilde{u}\|_{L^p(B^+_r)}\right)$$

Undoing the flattening Φ (whose $C^{1,1}$ regularity keeps all $W^{2,p}$ norms comparable) transfers this to a curved half-ball estimate near $x_0 \in \partial U$.

Step 4: Patching. The boundary of U is compact, so finitely many such neighbourhoods $\mathcal{O}_1, \ldots, \mathcal{O}_M$ cover ∂U; together with an interior region $V \Subset U$ they cover $\overline{U}$. Take a smooth partition of unity $\{\zeta_m\}$ subordinate to this cover. For each m, $\zeta_m u$ vanishes on the flat boundary and has small support, so either the interior estimate theorem 5.2.3 (on V) or the boundary half-ball estimate of Step 3 applies to it, with $\tilde{L}(\zeta_m u) = \zeta_m \tilde{L}u + [\tilde{L}, \zeta_m]u$ and the commutator $[\tilde{L}, \zeta_m]u$ a first-order expression in u bounded by $C\|u\|_{W^{1,p}(U)}$. Summing over m and absorbing the resulting $\|u\|_{W^{1,p}(U)}$ by interpolation gives

$$\|u\|_{W^{2,p}(U)} \leq C\left(\|Lu\|_{L^p(U)} + \|u\|_{L^p(U)}\right)$$

which is part (1).

Step 5: Solvability. For part (2) assume $c \geq 0$. Existence at $p = 2$ is given by the weak theory: theorem 1.3.4 produces a unique $u \in H_0^1(U)$ with $Lu = f$ weakly for $f \in L^2(U)$. This weak solution has $Lu = f \in L^p$ for the given f (first taking $f \in L^p \cap L^2$, dense in L^p), so the bootstrap of theorem 5.3.3 places $u \in W^{2,p}(U)$; the a priori bound of part (1), together with the uniqueness from $c \geq 0$ (a solution of $Lu = 0$, $u|_{\partial U} = 0$ vanishes, by the weak maximum principle or by testing the weak form against u and using coercivity), gives $\|u\|_{W^{2,p}} \leq C\|f\|_{L^p}$ with the $\|u\|_{L^p}$ term absorbed. For general $f \in L^p$, approximate by $f_k \to f$ in L^p with $f_k \in L^p \cap L^2$; the solutions u_k form a Cauchy sequence in $W^{2,p}$ by the estimate applied to differences, and the limit $u \in W^{2,p}(U) \cap W_0^{1,p}(U)$ solves $Lu = f$. Uniqueness follows from part (1) applied to the difference of two solutions, completing the proof.

The estimate is the boundary-inclusive twin of the interior estimate, and the two combine into a clean statement: for L elliptic with continuous leading coefficients on a $C^{1,1}$ domain, L is a bounded isomorphism from $W^{2,p} \cap W_0^{1,p}$ onto L^p once the sign condition $c \geq 0$ rules out a kernel. The right-hand side of the a priori bound carries two terms, $\|Lu\|_{L^p}$ and $\|u\|_{L^p}$; the second is a compactness correction and can be dropped precisely when uniqueness holds, as the solvability statement records. This is the L^p counterpart of the global $C^{2,\alpha}$ Schauder estimate: same freezing method, same boundary-flattening apparatus, with the Hölder norm replaced by the L^p norm and the Hölder hypothesis on the coefficients relaxed to continuity. The comparison also marks the two theories' complementary strengths. The Schauder estimate gives classical (twice-differentiable) solutions but requires Hölder data; the L^p estimate accepts integrable data and returns a solution whose second derivatives exist only in the L^p sense, a weaker conclusion from a weaker hypothesis, but one that meshes with the Hilbert-space existence theory, whose native output is a Sobolev function.

The half-space reflection in Step 2 used the zero Dirichlet condition in an essential way: odd reflection is what a homogeneous Dirichlet datum permits. A Neumann condition would call for even reflection, and inhomogeneous boundary data are handled by subtracting a fixed extension of the boundary values, reducing to the homogeneous case treated here.

Example 5.3.2: The Poisson Problem on a Ball
Let $U = B_1 \subseteq \mathbb{R}^n$ be the unit ball and $L = -\Delta$, so $c = 0$ and the leading coefficients are the constants δ^{ij}, trivially continuous. The boundary ∂B_1 is smooth, in particular $C^{1,1}$. Theorem 5.3.1 then asserts that for every $f \in L^p(B_1)$ with $1 < p < \infty$, the Dirichlet problem

$$-\Delta u = f \quad \text{in } B_1, \qquad u = 0 \quad \text{on } \partial B_1$$

has a unique solution $u \in W^{2,p}(B_1) \cap W_0^{1,p}(B_1)$, with $\|u\|_{W^{2,p}(B_1)} \leq C_{n,p}\|f\|_{L^p(B_1)}$. Concretely, take $n = 2$ and $f \equiv 1$. Radial symmetry gives $u(x) = \frac{1}{4}(1 - |x|^2)$, for which $\partial_{ij}u = -\frac{1}{2}\delta_{ij}$ is constant, so $u \in W^{2,p}(B_1)$ for every $p < \infty$ with $\|D^2u\|_{L^p} = \frac{1}{\sqrt{2}}|B_1|^{1/p}$, in accordance with the estimate. The example also shows the estimate is genuinely an L^p statement for finite p:

for a general $f \in L^p$ the solution need not lie in C^2, only in $W^{2,p}$. The Sobolev embedding $W^{2,p}(B_1) \hookrightarrow C^{j,\gamma}(\overline{B_1})$ requires $2 - n/p > j$, so $W^{2,p}$ gives continuity of u itself at $p > n/2$ and of ∇u at $p > n$, but never continuity of $D^2 u$ at any finite p, since $2 - n/p > 2$ would force $n/p < 0$. Thus even the finite-p endpoint $p > n$ delivers only a $C^{1,\alpha}$ solution, which is why an L^p datum cannot be pushed to C^2 by integrability alone.

With the estimate available on the whole domain, the equation feeds regularity back into its own solution. The variational theory delivers a weak solution in H^1_0, which is $W^{1,2}$; the estimate, once its hypotheses are met, upgrades this to $W^{2,p}$; if the data are smoother still, differentiating the equation shows the upgraded solution has an equation of its own with data one derivative better, and the estimate applies again. Each pass gains two derivatives of u for one derivative of f (and one more of the coefficients), climbing the Sobolev scale. The starting rung, that an H^1_0 weak solution with L^p data already lies in $W^{2,p}$, is the substantive step; the rest is induction.

Theorem 5.3.3: The Regularity Bootstrap

Let $U \subseteq \mathbb{R}^n$ be bounded with $C^{1,1}$ boundary, $1 < p < \infty$, and L uniformly elliptic with $c \geq 0$.

1. *(Base step.)* Suppose $u \in H^1_0(U)$ is a weak solution of $Lu = f$ with $f \in L^p(U)$, the leading coefficients $a^{ij} \in C(\overline{U})$, and b^i, c bounded. Then $u \in W^{2,p}(U)$, and $\|u\|_{W^{2,p}(U)} \leq C(\|f\|_{L^p(U)} + \|u\|_{L^p(U)})$.
2. *(Higher regularity.)* If in addition $f \in W^{k,p}(U)$ and the coefficients satisfy $a^{ij} \in C^{k-1,1}(\overline{U})$, $b^i, c \in C^{k-1,1}(\overline{U})$ for some integer $k \geq 1$, then $u \in W^{k+2,p}(U)$, with $\|u\|_{W^{k+2,p}(U)} \leq C(\|f\|_{W^{k,p}(U)} + \|u\|_{L^p(U)})$.
3. *(Smooth data.)* If ∂U, the coefficients, and f are all C^∞, then $u \in C^\infty(\overline{U})$.

***Proof*:**
Step 1: The base step by frozen coefficients. A weak solution lies a priori only in $H^1_0 = W^{1,2}$, which need not be $W^{2,p}$, and the global estimate of theorem 5.3.1 presumes $u \in W^{2,p}$ at the outset. The gap is closed not by that a priori bound but by the regularity theorem behind it: for continuous leading coefficients the second derivatives are produced by the frozen-coefficient argument, not by coefficient difference quotients, which are unavailable because a continuous a^{ij} need not be Lipschitz. Concretely, the interior regularity of theorem 5.2.3 already asserts that a weak solution of $Lu = f$ with $f \in L^p(U)$ lies in $W^{2,p}_{\text{loc}}(U)$: freeze the leading coefficients at a point, treat the frozen operator by the second-derivative bound of theorem 5.1.5, and absorb the oscillation error on a small ball using the continuity (not the differentiability) of a^{ij}. The same frozen-coefficient argument in the flattened boundary coordinates of theorem 5.3.1, applied to the odd reflection of the solution across $\{y_n = 0\}$, carries the $W^{2,p}$ conclusion up to ∂U; patching the interior and boundary pieces as in Steps 3 and 4 of that proof places

$$u \in W^{2,p}(U), \qquad \|u\|_{W^{2,p}(U)} \leq C \left(\|f\|_{L^p(U)} + \|u\|_{L^p(U)}\right)$$

directly, with the a priori bound of theorem 5.3.1(1) then available because u is now known to lie in $W^{2,p}$. This is legitimate for the given p in one pass: the frozen-coefficient regularity is an L^p statement at any $1 < p < \infty$, so no bootstrap in the exponent is needed. What it does require is that the lower-order terms $b^i \partial_i u + cu$ already sit in L^p; since $u \in H^1_0 = W^{1,2}$ gives $\nabla u \in L^2$ and, by the Sobolev embedding $W^{1,2} \hookrightarrow L^{2n/(n-2)}$ of [2, Gagliardo-Nirenberg-Sobolev

Inequality, General Sobolev Embedding] (any L^q, $q < \infty$, if $n \le 2$), $u \in L^{2n/(n-2)}$, one first obtains $u \in W^{2,\min(p,2)}$ from $f - b^i\partial_i u - cu \in L^{\min(p,2)}$, and if $p > 2$ a finite sequence of such passes, each raising the integrability of the lower-order terms by the Sobolev gain, reaches $W^{2,p}$. This establishes (1).

Step 2: Higher regularity by differentiating the equation. Argue by induction on k; the case $k = 0$ is part (1). Suppose the conclusion holds through $k-1$, and let $f \in W^{k,p}$ with coefficients in $C^{k-1,1}$. By the inductive hypothesis $u \in W^{k+1,p}(U)$. Apply the difference quotient D^h_ℓ to the equation $Lu = f$: the function $u_h = D^h_\ell u$ satisfies

$$Lu_h = D^h_\ell f - (D^h_\ell a^{ij})\,\partial_{ij}u(\cdot + he_\ell) - (D^h_\ell b^i)\,\partial_i u(\cdot + he_\ell) - (D^h_\ell c)\,u(\cdot + he_\ell) =: f_h$$

an equation of the same elliptic form for u_h. The coefficient difference quotients $D^h_\ell a^{ij}$ are bounded in L^∞ uniformly in h because $a^{ij} \in C^{k-1,1} \subseteq C^{0,1}$ is Lipschitz, and $\partial_{ij}u \in W^{k-1,p} \subseteq L^p$; hence f_h is bounded in $W^{k-1,p}(U)$ uniformly in h. By the inductive hypothesis applied to u_h, one gets $u_h \in W^{k+1,p}$ with a bound uniform in h, so $D_\ell u = \lim_{h\to 0} u_h \in W^{k+1,p}(U)$. As ℓ ranges over all directions, every first derivative of u lies in $W^{k+1,p}$, that is $u \in W^{k+2,p}(U)$, and tracking constants gives the stated bound. This proves (2).

Step 3: Smooth data. If ∂U, the coefficients, and f are C^∞, then part (2) applies for every k, so $u \in W^{k+2,p}(U)$ for all $k \ge 0$. The Sobolev embedding theorem $W^{m,p}(U) \hookrightarrow C^{j,\gamma}(\overline{U})$ whenever $m - n/p > j$ ([2]) then places $u \in C^j(\overline{U})$ for every j, since $k+2-n/p \to \infty$ as $k \to \infty$. Therefore $u \in \bigcap_j C^j(\overline{U}) = C^\infty(\overline{U})$, completing the proof.

The base step is where the elliptic character does its work: the frozen-coefficient argument converts the ellipticity of L into the singular-integral bound for the frozen operator, and the continuity of the leading coefficients is exactly what lets the oscillation error be absorbed, producing the two missing derivatives in L^p from an H^1_0 solution. The higher steps, by contrast, do use coefficient difference quotients, which is why they must assume the coefficients are Lipschitz ($C^{k-1,1}$); they differentiate the equation and reuse the base step for the differentiated solution, so the whole tower stands on the interplay between the freezing method and the Calderón-Zygmund bound established in this chapter. The gain of two derivatives per pass, $f \in W^{k,p} \Rightarrow u \in W^{k+2,p}$, is the exact statement that L is a second-order operator: it costs two derivatives to apply, so it returns two to invert.

The smooth conclusion is the payoff a reader should carry away. An elliptic equation is *hypoelliptic* in the interior and, with smooth boundary and data, globally: wherever the data are smooth, so is the solution, with no loss. This is false for operators that are not elliptic. The heat operator $\partial_t - \Delta$ smooths in space but only forward in time, and the wave operator $\partial_{tt} - \Delta$ does not smooth at all, propagating singularities along characteristics; the absence of real characteristics is precisely what distinguishes the elliptic case and makes the bootstrap unconditional. The mechanism separating these behaviours, the wavefront set and the propagation of singularities, is the subject of [1].

One caveat belongs with the exponent. The Calderón-Zygmund bound underlying theorem 5.1.5, and hence every estimate in this chapter, holds only for $1 < p < \infty$. At the endpoints it fails: the second-derivative singular integral does not map L^1 to L^1 or L^∞ to L^∞. What survives are the substitute endpoint statements, $L^1 \to L^{1,\infty}$ (weak-L^1) at the low end and $L^\infty \to \mathrm{BMO}$ at the high end, so a bounded right-hand side gives a solution whose second derivatives lie in BMO but need not be bounded. This is why one cannot conclude $u \in C^2$ directly from $f \in L^\infty$: the correct L^∞-type regularity theory is the Schauder theory of chapter 4, which replaces L^∞ by a Hölder space and thereby restores a genuine sup-norm bound on the second derivatives.

Everything above concerns a single scalar equation. The same architecture extends to *systems*: a vector unknown $u = (u^1, \ldots, u^N)$ governed by a matrix of second-order operators, coupling the components. The prototype is the Lamé system of linear elasticity and the stationary Stokes system of fluid flow, where the components are the displacement or velocity fields. For such systems the pointwise ellipticity condition of definition 1.3.1 is too weak, and the right hypothesis is the *Legendre-Hadamard* or the stronger *Agmon-Douglis-Nirenberg* ellipticity condition on the principal symbol, which guarantees that the frozen constant-coefficient system has the same solvability the scalar half-space model enjoyed. Under it, the entire scalar theory of this chapter and the Schauder theory of chapter 4 carry over verbatim in their conclusions.

5.3.4 The Agmon-Douglis-Nirenberg Estimates for Systems Let $U \subseteq \mathbb{R}^n$ be bounded with $C^{1,1}$ boundary and let $\mathcal{L}$ be a second-order elliptic system in the sense of Agmon-Douglis-Nirenberg, with continuous leading coefficients and complementing boundary conditions of Dirichlet type. Then for $1 < p < \infty$ there is a constant C such that every $u \in W^{2,p}(U; \mathbb{R}^N)$ satisfying the boundary conditions obeys the vector-valued estimate

$$\|u\|_{W^{2,p}(U;\mathbb{R}^N)} \leq C \left(\|\mathcal{L}u\|_{L^p(U;\mathbb{R}^N)} + \|u\|_{L^p(U;\mathbb{R}^N)} \right)$$

and, under Hölder hypotheses on the data, the corresponding $C^{2,\alpha}$ Schauder estimate. The proof is the same freezing-plus-flattening argument, with the scalar Calderón-Zygmund bound replaced by its matrix-valued version, whose kernel is built from the fundamental solution of the frozen constant-coefficient system.

These vector-valued estimates are the direct generalization the scalar theory points to, and the boxed statement records them for reference; their proof adds no idea beyond the matrix bookkeeping and is left to the specialized literature. Two boundaries mark where this book stops. First, the systems above are still posed on flat domains $U \subseteq \mathbb{R}^n$. Elliptic operators on *manifolds and vector bundles*, where the operator is the Laplace-Beltrami operator, the Hodge Laplacian on differential forms, or a Dirac operator on a spinor bundle, need the language of manifolds and connections, and they are treated by the pseudodifferential-parametrix method rather than by frozen singular integrals; that is the province of [1], which builds the parametrix, recovers the Fredholm theory and the elliptic estimates in the bundle setting, and connects them to the index theorem. Second, the nonlinear elliptic theory, where the coefficients depend on u and its derivatives and the frozen-coefficient method must be married to a fixed-point or continuity argument, lies beyond the linear scope of this book; the Schauder and L^p estimates proved here are exactly the linear inputs that theory takes as given.

6

Spectral Theory of Elliptic Operators

The compact solution operator of chapter 2 does more than power the Fredholm alternative: it makes the Dirichlet Laplacian a self-adjoint operator with compact resolvent, and the spectral theorem then decomposes L^2 into its eigenfunctions. This chapter carries that decomposition through. It realizes $-\Delta$ with zero boundary values as an unbounded self-adjoint operator, reads off from the spectral theorem of the functional-analysis volume a complete orthonormal basis of smooth eigenfunctions with eigenvalues increasing to infinity, characterizes those eigenvalues variationally through the Rayleigh quotient and the Courant-Fischer min-max principle, and closes with Weyl's law, the leading-order count of eigenvalues below a threshold in terms of the volume of the domain.

6.1 The Dirichlet Laplacian and Its Spectrum

Every construction so far has treated $-\Delta$ as a map from one function space to another: a bounded operator $H_0^1(U) \to H^{-1}(U)$, or its inverse, the compact solution operator on $L^2(U)$. To speak of eigenvalues and an eigenbasis the operator must instead act *within* a single Hilbert space, from a domain inside $L^2(U)$ back to $L^2(U)$, and it must be self-adjoint there so the spectral theorem applies. This section sets $-\Delta$ up as such an operator, the Dirichlet Laplacian, and reads off its spectrum from the compactness already in hand.

The obstruction is that $-\Delta u$ need not lie in $L^2(U)$ for every $u \in H_0^1(U)$: differentiating twice can leave L^2. The remedy is to restrict the operator to the functions on which it does return to L^2, namely those with two L^2 derivatives and zero boundary trace. On that domain the operator is unbounded, in the precise sense that it is not defined on all of $L^2(U)$ and does not extend to a bounded map, so the theory of unbounded self-adjoint operators is the right frame.

Definition 6.1.1: The Dirichlet Laplacian

Let $U \subseteq \mathbb{R}^n$ be bounded and open with $C^{1,1}$ boundary. The *Dirichlet Laplacian* on U is the unbounded operator $A\colon \operatorname{dom}(A) \to L^2(U)$ given by $Au = -\Delta u$ on the domain

$$\operatorname{dom}(A) = H^2(U) \cap H_0^1(U)$$

the functions in $L^2(U)$ with two weak derivatives in $L^2(U)$ and vanishing boundary trace. Equivalently, A is the Friedrichs extension of the coercive symmetric form $a(u,v) = \int_U \nabla u \cdot \nabla v \, dx$ of chapter 1, densely defined on $H_0^1(U)$.

The two descriptions of A answer two different needs. The domain $H^2(U) \cap H_0^1(U)$ is concrete: it names exactly the functions for which $-\Delta u$ is an element of $L^2(U)$ and the boundary condition holds, and it is the domain on which one computes. The Friedrichs description is what *guarantees* self-adjointness. The form a is symmetric and coercive on $H_0^1(U)$, and the Friedrichs construction of the functional-analysis volume ([2]) manufactures from any such form a self-adjoint operator whose domain consists of those $u \in H_0^1(U)$ for which $v \mapsto a(u,v)$ is L^2-bounded; the $C^{1,1}$ boundary is exactly what lets the global elliptic regularity of theorem 5.3.1 identify that abstract domain with $H^2(U) \cap H_0^1(U)$, reconciling the two. That A is *positive* is immediate from the form: $(Au,u) = a(u,u) = \|\nabla u\|_{L^2}^2 > 0$ for $u \neq 0$ in $\operatorname{dom}(A)$, since a function in $H_0^1(U)$ with vanishing gradient vanishes.

Before extracting the spectrum abstractly, one case can be read off by hand, and it fixes the picture the general theory will reproduce.

Example 6.1.2: Eigenvalues of $-\Delta$ on an Interval

Take $n = 1$ and $U = (0, L)$. The eigenvalue problem for the Dirichlet Laplacian is the boundary-value problem

$$-\varphi'' = \lambda\varphi \quad \text{on } (0,L) \qquad \varphi(0) = \varphi(L) = 0$$

For $\lambda \le 0$ the only solution meeting both boundary conditions is $\varphi \equiv 0$: a solution of $-\varphi'' = \lambda\varphi$ with $\lambda < 0$ is a combination of real exponentials $e^{\pm\sqrt{-\lambda}\,x}$, which cannot vanish at two points unless it is trivial, and for $\lambda = 0$ the solutions are affine, again forced to zero by the endpoints. For $\lambda > 0$ the general solution is $\varphi(x) = c_1 \cos(\sqrt{\lambda}\,x) + c_2 \sin(\sqrt{\lambda}\,x)$; the condition $\varphi(0) = 0$ forces $c_1 = 0$, and then $\varphi(L) = c_2 \sin(\sqrt{\lambda}\,L) = 0$ with $c_2 \neq 0$ requires $\sqrt{\lambda}\,L = k\pi$ for a positive integer k. The eigenvalues and (normalized) eigenfunctions are therefore

$$\lambda_k = \left(\frac{k\pi}{L}\right)^2 \qquad \varphi_k(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{k\pi x}{L}\right) \qquad k = 1, 2, 3, \ldots$$

The eigenvalues are positive, simple, and increase to infinity like k^2, and the eigenfunctions are the classical Fourier sine basis of $L^2(0, L)$. This is the one-dimensional archetype of everything the general theorem asserts: a discrete, positive, unbounded spectrum with an orthonormal eigenbasis of smooth functions.

The interval computation was possible because separation of variables reduced the partial differential equation to an ordinary one. On a general domain no such formula is available, and the discreteness of the spectrum, the completeness of the eigenfunctions, and their smoothness must instead be extracted structurally. The structure is already present: the inverse of A (up to the shift by the identity that keeps it defined on all of L^2) is the compact solution operator of theorem 2.2.1, and the spectral theorem for compact self-adjoint operators does the rest.

The bridge is the resolvent. Rather than invert A itself, one inverts $A + I$, which is invertible on all of $L^2(U)$ because A is positive, and whose inverse is a compact self-adjoint operator to which the spectral theorem applies directly.

Theorem 6.1.3: Compact Resolvent and the Eigenbasis

Let $U \subseteq \mathbb{R}^n$ be bounded and open with $C^{1,1}$ boundary, and let A be the Dirichlet Laplacian of definition 6.1.1. Then $A + I$ is a bijection from $\operatorname{dom}(A)$ onto $L^2(U)$, and its inverse

$$(A + I)^{-1} \colon L^2(U) \to L^2(U)$$

is a compact, self-adjoint, positive operator, equal to the solution operator of theorem 2.2.1 for the shifted problem $-\Delta u + u = f$. Consequently A is self-adjoint and positive, and there is an orthonormal basis $\{\varphi_k\}_{k \geq 1}$ of $L^2(U)$ consisting of eigenfunctions of A,

$$A\varphi_k = \lambda_k \varphi_k$$

with real eigenvalues satisfying

$$0 < \lambda_1 \leq \lambda_2 \leq \cdots \to \infty$$

Each φ_k is smooth in the interior of U.

***Proof*:**
The argument identifies $(A + I)^{-1}$ with the compact solution operator, transports the spectral decomposition of that operator back to A, and finally invokes elliptic regularity for smoothness.

Step 1: $A + I$ is a bijection onto $L^2(U)$ with the solution operator as inverse. The equation $(A + I)u = f$ is $-\Delta u + u = f$ with $u \in H^2(U) \cap H^1_0(U)$. This is the Dirichlet problem for the operator $-\Delta + I$, which is of the divergence form of definition 1.3.1 with leading coefficient the identity matrix and zeroth-order coefficient $c = 1 \geq 0$, hence coercive. By the weak solvability of theorem 1.3.4, for every $f \in L^2(U)$ there is a unique weak solution $u \in H^1_0(U)$ of $-\Delta u + u = f$. Because U has $C^{1,1}$ boundary and the leading coefficients are constant (hence continuous), the global L^p regularity of theorem 5.3.1, taken at $p = 2$, applies: it places this weak solution in $H^2(U) = W^{2,2}(U)$ up to the boundary and identifies $-\Delta + I$ as a bijection of $H^2(U) \cap H^1_0(U)$ onto $L^2(U)$. The interior bootstrap of theorem 5.3.3 alone would give only $H^2_{\mathrm{loc}}(U)$, not the global membership the domain of A requires; the $C^{1,1}$ hypothesis is what carries the second derivatives to the boundary. Thus $u \in \operatorname{dom}(A)$ and $(A + I)u = f$ in $L^2(U)$, and uniqueness of the weak solution gives injectivity of $A + I$ on $\operatorname{dom}(A)$. So $A + I \colon \operatorname{dom}(A) \to L^2(U)$ is a bijection, and the map $f \mapsto u$ is exactly the solution operator $T = (A + I)^{-1}$ of the shifted problem. By theorem 2.2.1 applied to the coercive operator $-\Delta + I$, this T is a compact operator on $L^2(U)$.

Step 2: $T = (A + I)^{-1}$ is self-adjoint and positive. Let $f, h \in L^2(U)$, with weak solutions $u = Tf$ and $w = Th$, so that the shifted form $b(u, v) = \int_U \nabla u \cdot \nabla v + uv \, dx$ satisfies $b(u, v) = (f, v)$ and $b(w, v) = (h, v)$ for all $v \in H^1_0(U)$. Testing the first identity with $v = w$ and the second with $v = u$ gives

$$(Tf, h) = (u, h) = b(w, u) = b(u, w) = (f, w) = (f, Th)$$

where the middle equality is the symmetry $b(u, w) = b(w, u)$ of the form. Hence T is self-adjoint. It is positive because $(Tf, f) = (u, f) = b(u, u) = \|\nabla u\|_{L^2}^2 + \|u\|_{L^2}^2 > 0$ whenever $f \neq 0$, since $u = Tf = 0$ would force $f = (A + I)u = 0$.

Step 3: The spectral theorem for T. A compact self-adjoint operator on a Hilbert space admits an orthonormal basis of eigenvectors, with real eigenvalues accumulating only at 0 [2, Normal Compact, Then Diagonalizable, Spectrum of a Self-Adjoint Operator]. Applied to T on $L^2(U)$, this yields an orthonormal basis $\{\varphi_k\}_{k\geq 1}$ of $L^2(U)$ with $T\varphi_k = \mu_k\varphi_k$. Every eigenvalue μ_k is positive by Step 2, since $\mu_k\|\varphi_k\|_{L^2}^2 = (T\varphi_k, \varphi_k) > 0$, and 0 is not itself an eigenvalue because T is injective (it has the two-sided inverse $A + I$). Ordering the eigenvalues so that $\mu_1 \geq \mu_2 \geq \cdots > 0$, compactness forces $\mu_k \to 0$: an orthonormal sequence of eigenvectors with eigenvalues bounded below in modulus would have images $T\varphi_k$ with no convergent subsequence, contradicting compactness.

Step 4: Transporting the decomposition to A. Fix k. Since $\mu_k \neq 0$, the eigenrelation $T\varphi_k = \mu_k\varphi_k$ shows $\varphi_k = \mu_k^{-1}T\varphi_k$ lies in the range of T, which is $\mathrm{dom}(A)$; applying the bijection $A + I$ to $\varphi_k = T(\mu_k^{-1}\varphi_k)$ gives $(A + I)\varphi_k = \mu_k^{-1}\varphi_k$, that is,

$$A\varphi_k = \left(\frac{1}{\mu_k} - 1\right)\varphi_k$$

So φ_k is an eigenfunction of A with eigenvalue

$$\lambda_k = \frac{1}{\mu_k} - 1$$

equivalently $\mu_k = (\lambda_k + 1)^{-1}$. The map $\mu \mapsto \mu^{-1} - 1$ is decreasing on $(0, 1]$, so the ordering $\mu_1 \geq \mu_2 \geq \cdots$ becomes $\lambda_1 \leq \lambda_2 \leq \cdots$, and $\mu_k \to 0^+$ becomes $\lambda_k \to \infty$. Each λ_k is positive: positivity of A gives $\lambda_k\|\varphi_k\|_{L^2}^2 = (A\varphi_k, \varphi_k) > 0$. The same basis $\{\varphi_k\}$ that diagonalizes T therefore diagonalizes A, with the eigenvalues inverted and shifted. That $\{\varphi_k\}$ is a complete orthonormal system of eigenfunctions of A is exactly self-adjointness of A in the form the spectral theorem provides: on the domain $\mathrm{dom}(A) = \{u \in L^2(U) : \sum_k \lambda_k^2|(u, \varphi_k)|^2 < \infty\}$ one has $Au = \sum_k \lambda_k(u, \varphi_k)\varphi_k$, a real-diagonal operator, which is self-adjoint.

Step 5: Smoothness of the eigenfunctions. Each φ_k solves the elliptic equation $-\Delta\varphi_k = \lambda_k\varphi_k$ weakly, with right-hand side $\lambda_k\varphi_k \in L^2(U)$. The regularity bootstrap of theorem 5.3.3 applies: since $\varphi_k \in L^2(U)$ the equation places $\varphi_k \in H^2_{\mathrm{loc}}(U)$, then the right-hand side $\lambda_k\varphi_k$ inherits that regularity and the bootstrap gives $\varphi_k \in H^4_{\mathrm{loc}}(U)$, and iterating puts $\varphi_k \in H^m_{\mathrm{loc}}(U)$ for every m. By the Sobolev embedding [2, General Sobolev Embedding], φ_k is smooth in the interior of U, completing the proof.

The theorem is the payoff of the entire elliptic apparatus: coercivity gave the inverse, boundedness of U made it compact through Rellich, and self-adjointness came free from the symmetry of the Dirichlet form. The reduction to $(A + I)^{-1}$ is the mechanism worth remembering. An unbounded self-adjoint operator has no eigenbasis on its own terms, but its resolvent is a bounded, indeed compact, operator, and the spectral theorem for compact operators is elementary; shifting by I is only the device that makes the resolvent land on all of $L^2(U)$ and stay positive, so that inverting eigenvalues never divides by zero. The correspondence $\lambda_k = \mu_k^{-1} - 1$ turns the accumulation of the μ_k at 0, which is all a compact operator can offer, into the escape of the λ_k to infinity, which is what an unbounded operator must exhibit.

Two features of the spectrum deserve emphasis. First, the spectrum is *discrete*: a sequence of eigenvalues with no finite accumulation point and no continuous part, in contrast to the Laplacian on all of $\mathbb{R}^n$, whose spectrum is the whole half-line $[0, \infty)$. Boundedness of U is what makes the difference, through the compactness of the embedding $H_0^1(U) \hookrightarrow L^2(U)$; on an unbounded domain that embedding fails to be compact and the discrete spectrum dissolves into a continuum. Second, the eigenfunctions form a complete orthonormal basis, so every $f \in L^2(U)$ expands as $f = \sum_k (f, \varphi_k)\varphi_k$, the abstract analogue of the Fourier sine series that example 6.1.2 exhibited concretely. This expansion is what the eigenfunction method uses for boundary-value and evolution problems: writing an unknown against the basis $\{\varphi_k\}$ diagonalizes $-\Delta$, turning a partial differential equation into a decoupled family of scalar equations, one per eigenvalue.

6.2 The Rayleigh Quotient and the Min-Max Principle

The spectral decomposition of the previous section produces the eigenvalues $0 < \lambda_1 \leq \lambda_2 \leq \cdots \to \infty$ as an abstract list, extracted from the compact resolvent by the spectral theorem (theorem 6.1.3). That list carries no information about the domain U beyond what the resolvent already encodes, and in that form it is nearly useless for computation: to locate λ_k one would have to know the whole eigenbasis first. This section replaces the abstract list with a variational characterization. Each eigenvalue is the value of an optimization problem posed directly on the energy form, with no reference to the eigenfunctions, and that reformulation is what makes the eigenvalues estimable, comparable across domains, and monotone in the data.

The quantity being optimized is the ratio of the energy $a(u, u)$ to the mass $\|u\|_{L^2}^2$. For the Dirichlet Laplacian this is the ratio of the Dirichlet integral $\|\nabla u\|_{L^2}^2$ to the square of the L^2 norm, a scale-free measurement of how much a trial function oscillates relative to its size. The eigenfunctions are precisely the critical points of this ratio, and the eigenvalues are its critical values. Making that statement precise, and then turning it into the two-sided min-max formula that pins down λ_k without knowing $\varphi_1, \ldots, \varphi_{k-1}$, is the work of the section.

Definition 6.2.1: Rayleigh Quotient

Let $U \subseteq \mathbb{R}^n$ be bounded and open, and let a be the coercive symmetric bilinear form of the Dirichlet Laplacian on $H_0^1(U)$. The *Rayleigh quotient* of a function $u \in H_0^1(U) \setminus \{0\}$ is

$$R(u) = \frac{a(u, u)}{\|u\|_{L^2(U)}^2}$$

For $A = -\Delta$, where $a(u, u) = \int_U |\nabla u|^2 \, dx$, this is the ratio of the Dirichlet integral to the mass,

$$R(u) = \frac{\|\nabla u\|_{L^2(U)}^2}{\|u\|_{L^2(U)}^2}$$

The Rayleigh quotient is homogeneous of degree zero: $R(tu) = R(u)$ for every scalar $t \neq 0$, since both numerator and denominator scale by t^2. Its value therefore depends only on the direction of u in $H_0^1(U)$, not on its length, and optimizing R over $H_0^1(U) \setminus \{0\}$ is the same as optimizing $a(u, u)$ over the unit sphere $\{\|u\|_{L^2} = 1\}$. Coercivity of a makes R bounded below by a positive constant, so the infimum of R is a genuine positive number rather than zero. The name comes from Rayleigh's use of the ratio to estimate the fundamental tone of a vibrating membrane: a trial shape u that roughly matches the true fundamental mode gives a value $R(u)$ close to λ_1 from above, and the closer the shape, the sharper the estimate.

The connection to the eigenvalues is immediate once the quotient is evaluated on the eigenbasis. If $u = \sum_k c_k \varphi_k$ is the expansion of a trial function in the orthonormal eigenbasis $\{\varphi_k\}$ of theorem 6.1.3, then $\|u\|_{L^2}^2 = \sum_k c_k^2$, while the energy is $a(u,u) = \sum_k \lambda_k c_k^2$ because $a(\varphi_j, \varphi_k) = \lambda_k \langle \varphi_j, \varphi_k \rangle = \lambda_k \delta_{jk}$. The Rayleigh quotient is thus a weighted average of the eigenvalues,

$$R(u) = \frac{\sum_k \lambda_k c_k^2}{\sum_k c_k^2}$$

with weights c_k^2 proportional to the squared coordinates of u. This single identity drives every result in the section. The following example shows it on the one domain where the whole spectrum is available in closed form.

Example 6.2.2: The Rayleigh Quotient on an Interval

Let $U = (0, \pi) \subseteq \mathbb{R}$, so that $-\Delta = -d^2/dx^2$ on functions vanishing at the endpoints. The Dirichlet eigenfunctions are $\varphi_k(x) = \sqrt{2/\pi}\,\sin(kx)$ for $k = 1, 2, \ldots$, with eigenvalues $\lambda_k = k^2$; these solve $-\varphi_k'' = k^2 \varphi_k$ and vanish at 0 and π. Take the trial function $u(x) = x(\pi - x)$, a simple parabola vanishing at the endpoints, which lies in $H_0^1(0, \pi)$. Direct integration gives

$$\int_0^\pi u(x)^2\,dx = \frac{\pi^5}{30}, \qquad \int_0^\pi u'(x)^2\,dx = \int_0^\pi (\pi - 2x)^2\,dx = \frac{\pi^3}{3}$$

so that

$$R(u) = \frac{\pi^3/3}{\pi^5/30} = \frac{10}{\pi^2} \approx 1.0132$$

The true smallest eigenvalue is $\lambda_1 = 1$, and the trial value overestimates it by about 1.3 percent. The parabola is a crude stand-in for the true fundamental mode $\sin x$, yet the Rayleigh quotient already locates λ_1 to within a percent, and it does so from above, as every trial value must. Expanding u in the sine basis confirms the averaging identity: the odd coefficients dominate, with the $k = 1$ mode carrying the overwhelming share of the mass, which is why the weighted average $R(u) = \sum k^2 c_k^2 / \sum c_k^2$ sits just above $\lambda_1 = 1$.

The example exhibits the two features that the min-max principle will make exact. First, $R(u)$ never drops below λ_1: the weighted average of the eigenvalues is at least the smallest eigenvalue, with equality exactly when u is a multiple of φ_1. Second, restricting the trial function to a subspace where the low modes are excluded forces R upward, and by choosing the subspace correctly one can trap any λ_k between what a well-chosen k-dimensional space achieves and what every k-dimensional space is forced to exceed. Both statements are contained in the averaging identity $R(u) = \sum \lambda_k c_k^2 / \sum c_k^2$, and the theorem below extracts them.

The min-max characterization answers a concrete question. To compute λ_1 one minimizes R; but to reach λ_2 by minimization one must first restrict to the orthogonal complement of φ_1, which requires already knowing φ_1. The Courant-Fischer principle removes that dependence. It characterizes λ_k as a min-max over all k-dimensional subspaces at once, so that no eigenfunction need be known in advance, and the price of the freedom is only that one optimizes over subspaces rather than over vectors.

Theorem 6.2.3: Courant-Fischer Min-Max Principle

Let A be the Dirichlet Laplacian on the bounded open set $U \subseteq \mathbb{R}^n$, with eigenvalues $0 < \lambda_1 \leq \lambda_2 \leq \cdots \to \infty$ and orthonormal eigenbasis $\{\varphi_k\}$ of theorem 6.1.3. For each $k \geq 1$,

$$\lambda_k = \min_{\substack{V \subseteq H_0^1(U) \\ \dim V = k}} \max_{u \in V \setminus \{0\}} R(u)$$

where the minimum ranges over all k-dimensional subspaces V of $H_0^1(U)$, and dually

$$\lambda_k = \max_{\substack{W \subseteq H_0^1(U) \\ \operatorname{codim} W = k-1}} \min_{u \in W \setminus \{0\}} R(u)$$

where W ranges over the closed subspaces of $H_0^1(U)$ of codimension $k-1$. In particular $\lambda_1 = \min_{u \in H_0^1(U) \setminus \{0\}} R(u)$, and the minimum is attained at the ground state φ_1.

***Proof*:**
Write $H = H_0^1(U)$ throughout, and recall the averaging identity: for $u = \sum_j c_j \varphi_j \in H$ with $u \neq 0$,

$$R(u) = \frac{\sum_j \lambda_j c_j^2}{\sum_j c_j^2} \tag{6.1}$$

which holds because $\{\varphi_j\}$ is orthonormal in L^2 and $a(\varphi_i, \varphi_j) = \lambda_j \delta_{ij}$, so both the energy $a(u,u) = \sum_j \lambda_j c_j^2$ and the mass $\|u\|_{L^2}^2 = \sum_j c_j^2$ read off from the coordinates. The eigenfunction expansion converges in H, so this holds for every $u \in H$, not only for finite combinations. The proof establishes the min-max formula; the max-min formula follows by the same argument applied to a complementary family of subspaces, as noted at the end.

Step 1: The min-max value is at most λ_k. Test the outer minimum with the specific subspace

$$V_k = \operatorname{span}(\varphi_1, \ldots, \varphi_k)$$

which is k-dimensional because the φ_j are linearly independent. Any $u \in V_k$ has an expansion $u = \sum_{j=1}^k c_j \varphi_j$ supported on the first k modes, so by (6.1)

$$R(u) = \frac{\sum_{j=1}^k \lambda_j c_j^2}{\sum_{j=1}^k c_j^2} \leq \frac{\lambda_k \sum_{j=1}^k c_j^2}{\sum_{j=1}^k c_j^2} = \lambda_k$$

using $\lambda_j \leq \lambda_k$ for $j \leq k$. Thus $\max_{u \in V_k \setminus \{0\}} R(u) \leq \lambda_k$, and the maximum equals λ_k (attained at $u = \varphi_k$, where $R(\varphi_k) = \lambda_k$). Since the outer minimum ranges over all k-dimensional subspaces, it is bounded above by the value on this particular one:

$$\min_{\dim V = k} \max_{u \in V \setminus \{0\}} R(u) \leq \max_{u \in V_k \setminus \{0\}} R(u) = \lambda_k$$

Step 2: The min-max value is at least λ_k. Let V be any k-dimensional subspace of H. The claim is that V contains a nonzero vector on which R is at least λ_k; this forces $\max_{u \in V \setminus \{0\}} R(u) \geq \lambda_k$, and since V was arbitrary the outer minimum is at least λ_k. To find such a vector, intersect V with the tail space

$$W_k = \overline{\operatorname{span}}(\varphi_k, \varphi_{k+1}, \ldots) = \{u \in H : \langle u, \varphi_j \rangle_{L^2} = 0 \text{ for } j < k\}$$

the closed subspace of H on which the first $k-1$ coordinates vanish. Its L^2-orthogonal complement inside the span of the eigenbasis is $\operatorname{span}(\varphi_1, \dots, \varphi_{k-1})$, of dimension $k-1$. A dimension count now forces a nontrivial intersection: the linear map $H \to \mathbb{R}^{k-1}$ sending $u \mapsto (\langle u, \varphi_1 \rangle, \dots, \langle u, \varphi_{k-1} \rangle)$ has a kernel equal to W_k, and its restriction to the k-dimensional space V cannot be injective into the $(k-1)$-dimensional target. Hence there is a nonzero $u \in V \cap W_k$.

For this $u = \sum_{j \geq k} c_j \varphi_j$, the first $k-1$ coordinates vanish, so by (6.1)

$$R(u) = \frac{\sum_{j \geq k} \lambda_j c_j^2}{\sum_{j \geq k} c_j^2} \geq \frac{\lambda_k \sum_{j \geq k} c_j^2}{\sum_{j \geq k} c_j^2} = \lambda_k$$

using $\lambda_j \geq \lambda_k$ for $j \geq k$. Therefore $\max_{u \in V \setminus \{0\}} R(u) \geq R(u) \geq \lambda_k$. As V was an arbitrary k-dimensional subspace,

$$\min_{\dim V = k} \max_{u \in V \setminus \{0\}} R(u) \geq \lambda_k$$

Step 3: Conclusion and the ground state. Steps 1 and 2 give equality in the min-max formula. The minimizing subspace is $V_k = \operatorname{span}(\varphi_1, \dots, \varphi_k)$, on which the inner maximum λ_k is attained at φ_k. Specializing to $k = 1$, the subspaces are the lines $\mathbb{R}u$, the inner maximum over a line is just $R(u)$ by degree-zero homogeneity, and the formula reads $\lambda_1 = \min_{u \in H \setminus \{0\}} R(u)$. Testing with $u = \varphi_1$ gives $R(\varphi_1) = \lambda_1$, so the minimum is attained at the ground state.

Step 4: The dual max-min formula. Apply the same reasoning with the roles of the two eigenspace families exchanged. For a closed subspace W of codimension $k-1$, the map $u \mapsto (\langle u, \varphi_1 \rangle, \dots, \langle u, \varphi_k \rangle)$ restricted to $W \cap \operatorname{span}(\varphi_1, \dots, \varphi_k)$ shows W meets $\operatorname{span}(\varphi_1, \dots, \varphi_k)$ in a nonzero vector, on which $R \leq \lambda_k$ by the Step 1 estimate; hence $\min_{u \in W \setminus \{0\}} R(u) \leq \lambda_k$ for every such W. Testing with the specific choice $W = W_k$ (codimension $k-1$), every $u \in W_k$ has $R(u) \geq \lambda_k$ by the Step 2 estimate, so $\min_{u \in W_k \setminus \{0\}} R(u) = \lambda_k$, attained at φ_k. Taking the maximum over W gives the dual formula, completing the proof.

The two formulas capture λ_k from opposite sides. The min-max form builds the eigenvalue up from below: it searches for the k-dimensional subspace on which the worst-case Rayleigh quotient is as small as possible, and the optimal such space is spanned by the first k eigenfunctions. The max-min form corners the eigenvalue from above by imposing $k-1$ linear constraints and asking how small R can be made subject to them. Both dispense with the recursive obstruction that plain minimization runs into: neither requires knowledge of $\varphi_1, \dots, \varphi_{k-1}$, only optimization over subspaces of the right dimension. This is what makes the principle a tool rather than a tautology. A single well-chosen trial subspace furnishes an upper bound for λ_k, and a single family of constraints furnishes a lower bound, and neither bound presupposes solving the eigenvalue problem it is estimating.

The decisive structural payoff is that the characterization refers to the domain U only through the form a and the space $H_0^1(U)$, never through the eigenfunctions. Anything that enlarges $H_0^1(U)$ or shrinks the energy therefore moves every eigenvalue in a predictable direction. The two monotonicity principles below are the immediate consequences, and they are the reason the min-max formula, rather than the abstract eigenvalue list, is what the next section uses.

The first monotonicity compares two domains. Extending a function by zero embeds the Sobolev space of a smaller domain isometrically inside that of a larger one, and because the min-max formula sees only the form and the space, the smaller domain, having fewer trial functions to compete with, has the larger eigenvalues.

Corollary 6.2.4: Domain Monotonicity of Eigenvalues

Let $U \subseteq U'$ be bounded open sets in $\mathbb{R}^n$, and let $\lambda_k(U)$ and $\lambda_k(U')$ denote the Dirichlet eigenvalues of $-\Delta$ on U and U' respectively. Then

$$\lambda_k(U) \geq \lambda_k(U') \qquad \text{for every } k \geq 1$$

***Proof*:**
Extension by zero gives a natural inclusion $H_0^1(U) \hookrightarrow H_0^1(U')$: a function $u \in H_0^1(U)$, set equal to zero on $U' \setminus U$, lies in $H_0^1(U')$, because u is a limit in the H^1 norm of functions in $C_c^\infty(U)$, and each of these, extended by zero, lies in $C_c^\infty(U')$. The extension preserves both the Dirichlet integral and the L^2 norm, since the added region contributes nothing to either integral:

$$\int_{U'} |\nabla \tilde{u}|^2 \, dx = \int_U |\nabla u|^2 \, dx, \qquad \int_{U'} |\tilde{u}|^2 \, dx = \int_U |u|^2 \, dx$$

where $\tilde{u}$ is the zero-extension. Hence $R_{U'}(\tilde{u}) = R_U(u)$: the Rayleigh quotient of the extended function on U' equals that of the original function on U.

Now fix k and let $V \subseteq H_0^1(U)$ be any k-dimensional subspace. Its image $\tilde{V}$ under zero-extension is a k-dimensional subspace of $H_0^1(U')$, and by the equality of Rayleigh quotients

$$\max_{u \in \tilde{V} \setminus \{0\}} R_{U'}(u) = \max_{u \in V \setminus \{0\}} R_U(u)$$

Taking the minimum over all k-dimensional $V \subseteq H_0^1(U)$, the left side is bounded below by the minimum over *all* k-dimensional subspaces of $H_0^1(U')$, which is $\lambda_k(U')$ by theorem 6.2.3, because the extended subspaces $\tilde{V}$ form only part of the competition on U'. The right side is $\lambda_k(U)$. Thus $\lambda_k(U) \geq \lambda_k(U')$, as we sought to show.

Domain monotonicity says that confining a vibrating membrane raises its tones: a drum stretched over a smaller region vibrates at higher frequencies. The physical content matches the analytic content exactly, since the eigenvalues are the squared frequencies of the fundamental and overtone modes. Crucially, no regularity of the boundaries is needed, and U need not sit nicely inside U'; the argument uses only that zero-extension carries $H_0^1(U)$ into $H_0^1(U')$, which holds for arbitrary open sets. This is exactly the mechanism the Weyl asymptotics of the next section will exploit, where a general domain is bracketed between unions of small cubes and its eigenvalue count is squeezed between the counts for the two cube grids.

The second monotonicity fixes the domain and varies the operator. Passing from $-\Delta$ to a general divergence-form operator changes the energy form from the Dirichlet integral to $a(u,u) = \int_U (A\nabla u \cdot \nabla u + cu^2)\, dx$, and a pointwise comparison of the coefficients transfers directly to a comparison of the forms, hence of the min-max values.

Corollary 6.2.5: Monotonicity in the Coefficients

Let $Lu = -\nabla \cdot (A\nabla u) + cu$ and $\tilde{L}u = -\nabla \cdot (\tilde{A}\nabla u) + \tilde{c}u$ be two divergence-form elliptic operators on the same bounded open set U, both symmetric and coercive in the sense of definition 1.3.1, with associated forms a and $\tilde{a}$. Suppose the coefficients are pointwise ordered,

$$A(x)\xi \cdot \xi \le \tilde{A}(x)\xi \cdot \xi \quad \text{for all } \xi \in \mathbb{R}^n, \qquad c(x) \le \tilde{c}(x)$$

for almost every $x \in U$. Then the Dirichlet eigenvalues satisfy $\lambda_k(L) \le \lambda_k(\tilde{L})$ for every k.

***Proof*:**
The coefficient inequalities integrate to a pointwise-to-integral comparison of the two energy forms: for every $u \in H_0^1(U)$,

$$a(u,u) = \int_U \left(A\nabla u \cdot \nabla u + cu^2\right) dx \le \int_U \left(\tilde{A}\nabla u \cdot \nabla u + \tilde{c}u^2\right) dx = \tilde{a}(u,u)$$

Dividing by $\|u\|_{L^2}^2$ gives $R(u) \le \tilde{R}(u)$ for every $u \ne 0$, where R and $\tilde{R}$ are the Rayleigh quotients of L and $\tilde{L}$. The two problems share the space $H_0^1(U)$, so the pointwise inequality of Rayleigh quotients propagates through both optimizations: for any k-dimensional V,

$$\max_{u \in V \setminus \{0\}} R(u) \le \max_{u \in V \setminus \{0\}} \tilde{R}(u)$$

and taking the minimum over V preserves the inequality, so by theorem 6.2.3

$$\lambda_k(L) = \min_{\dim V = k} \max_{u \in V \setminus \{0\}} R(u) \le \min_{\dim V = k} \max_{u \in V \setminus \{0\}} \tilde{R}(u) = \lambda_k(\tilde{L})$$

which is the claim, completing the proof.

Monotonicity in the coefficients says that stiffening the operator raises its spectrum: increasing the conductivity matrix A or the zeroth-order potential c pushes every eigenvalue up. The two corollaries together illustrate the single organizing feature of the min-max principle, that λ_k is an increasing functional of the energy form and a decreasing functional of the admissible space. Every eigenvalue estimate in practice is an application of one of these two comparisons, replacing an intractable form or domain by a tractable one that bounds it on the correct side. The following example combines both with the closed-form spectrum of a rectangle to produce a two-sided estimate for a domain whose eigenvalues cannot be written down.

Example 6.2.6: Bracketing the Fundamental Tone of a Disk
Let $U \subseteq \mathbb{R}^2$ be the open disk of radius 1, whose Dirichlet eigenvalues are the squares of the zeros of Bessel functions and have no elementary closed form; the smallest is $\lambda_1(U) = j_{0,1}^2 \approx 5.783$, where $j_{0,1}$ is the first positive zero of the Bessel function J_0. The two monotonicity principles bracket $\lambda_1(U)$ using only rectangles, for which the spectrum is explicit. On the rectangle $(0, L_1) \times (0, L_2)$ the Dirichlet eigenvalues of $-\Delta$ are

$$\lambda_{m_1,m_2} = \pi^2\left(\frac{m_1^2}{L_1^2} + \frac{m_2^2}{L_2^2}\right), \qquad m_1, m_2 \ge 1$$

with fundamental value $\pi^2(L_1^{-2}+L_2^{-2})$, obtained by separation of variables from the one-dimensional modes $\sin(\pi m_i x_i/L_i)$.

The inscribed square Q_{in} of side $\sqrt{2}$ (its diagonal equal to the diameter 2) sits inside the disk, and the disk sits inside the circumscribed square Q_{out} of side 2. Domain monotonicity (corollary 6.2.4) applied to $Q_{\text{in}} \subseteq U \subseteq Q_{\text{out}}$ gives

$$\lambda_1(Q_{\text{out}}) \leq \lambda_1(U) \leq \lambda_1(Q_{\text{in}})$$

The circumscribed square has side 2, so $\lambda_1(Q_{\text{out}}) = \pi^2(\frac{1}{4}+\frac{1}{4}) = \pi^2/2 \approx 4.935$. The inscribed square has side $\sqrt{2}$, so $\lambda_1(Q_{\text{in}}) = \pi^2(\frac{1}{2}+\frac{1}{2}) = \pi^2 \approx 9.870$. Hence

$$4.935 \leq \lambda_1(U) \leq 9.870$$

and the true value 5.783 lies within the bracket. The estimate uses nothing about the disk beyond two rectangles that sandwich it, and it illustrates the general principle: a domain whose spectrum is inaccessible is trapped between domains whose spectra are known, and the trapping is exactly the content of domain monotonicity.

6.3 Weyl's Asymptotic Law

The min-max principle of section 6.2 locates each individual eigenvalue λ_k variationally, but it says nothing quantitative about how the whole sequence $0 < \lambda_1 \leq \lambda_2 \leq \cdots \to \infty$ spreads out. The question this section answers is the asymptotic one: as $\lambda \to \infty$, how many eigenvalues of the Dirichlet Laplacian lie below λ? The count is packaged in the *counting function*

$$N(\lambda) = \#\{k \mid \lambda_k \leq \lambda\}$$

and Weyl's law asserts that $N(\lambda)$ grows like a fixed constant times $\operatorname{vol}(U)\,\lambda^{n/2}$, with a constant depending only on the dimension n. The leading term sees only the volume of U: two domains of the same volume, however different in shape, have the same first-order eigenvalue asymptotics. The geometry of the boundary enters only in lower-order corrections, invisible to the leading term.

The heuristic behind the exponent is a phase-space count. An eigenfunction with eigenvalue near λ oscillates on the length scale $\lambda^{-1/2}$, so it occupies a phase-space cell of volume roughly $(2\pi)^n$ in the product of position space U (volume $\operatorname{vol}(U)$) and the frequency ball $|\xi|^2 \leq \lambda$ (volume $\omega_n\lambda^{n/2}$, where ω_n is the volume of the unit ball in $\mathbb{R}^n$). Dividing the available phase-space volume by the cell size gives

$$N(\lambda) \approx \frac{\operatorname{vol}(U)\cdot\omega_n\lambda^{n/2}}{(2\pi)^n}$$

which is exactly the assertion of Weyl's law. The proof below makes this count rigorous by carrying it out exactly on a cube, where separation of variables turns the eigenvalue problem into a lattice-point count, and then transferring the cube estimate to a general U by the domain monotonicity of theorem 6.2.3.

On a cube the eigenvalue problem is solved exactly, so it is worth recording as a worked case before the general theorem. The problem separates, and the count of eigenvalues below λ becomes the count of lattice points inside a Euclidean ball, whose leading asymptotics is the ball's volume.

Example 6.3.1: The Dirichlet Spectrum of a Box

Let $Q = \prod_{i=1}^{n}(0, L_i)$ be an open box with side lengths $L_1, \dots, L_n$. The Dirichlet eigenfunctions of $-\Delta$ on Q are the products of one-dimensional sines,

$$\varphi_m(x) = \prod_{i=1}^{n} \sin\Big(\frac{\pi m_i x_i}{L_i}\Big), \qquad m = (m_1, \dots, m_n) \in \mathbb{N}^n,\ m_i \geq 1$$

each vanishing on ∂Q and satisfying $-\Delta\varphi_m = \lambda_m \varphi_m$ with

$$\lambda_m = \pi^2 \sum_{i=1}^{n} \Big(\frac{m_i}{L_i}\Big)^2$$

Direct differentiation confirms the eigenvalue: applying $-\partial_{x_i}^2$ to the i-th factor multiplies it by $(\pi m_i/L_i)^2$, and summing over i gives λ_m. These products form an orthogonal basis of $L^2(Q)$ (the one-dimensional sines are a basis of $L^2(0, L_i)$ and L^2 of a product is the tensor product of the factors), so they exhaust the spectrum: every Dirichlet eigenvalue of $-\Delta$ on Q is one of the λ_m.

Counting the eigenvalues below λ is therefore counting the lattice points $m \in \mathbb{N}^n$ with $m_i \geq 1$ that satisfy $\pi^2 \sum_i (m_i/L_i)^2 \leq \lambda$, that is, the lattice points in the positive octant lying inside the ellipsoid

$$\sum_{i=1}^{n} \Big(\frac{m_i}{L_i}\Big)^2 \leq \frac{\lambda}{\pi^2}$$

Each such lattice point is the corner of a unit cube in the octant, and these cubes tile a region that fills out the octant portion of the ellipsoid as $\lambda \to \infty$. The full ellipsoid has volume $\omega_n \prod_i L_i\, (\lambda/\pi^2)^{n/2}$, of which the positive octant is the fraction 2^{-n}, so

$$N(\lambda) = \#\,\{m \in \mathbb{N}^n,\ m_i \geq 1 \mid \lambda_m \leq \lambda\} \sim \frac{1}{2^n}\,\omega_n \Big(\prod_{i=1}^{n} L_i\Big)\Big(\frac{\lambda}{\pi^2}\Big)^{n/2} = \frac{\omega_n}{(2\pi)^n}\,\mathrm{vol}(Q)\,\lambda^{n/2}$$

the last step using $\mathrm{vol}(Q) = \prod_i L_i$ and $2^n\pi^n = (2\pi)^n$. So for a box the counting function has exactly the Weyl asymptotics, with the volume appearing through the product of the side lengths and the dimensional constant $\omega_n/(2\pi)^n$ emerging from the volume of the frequency ellipsoid.

The box computation already contains the whole leading term, and it is the base case of the general theorem. Two ingredients carry it to a bounded U with negligible boundary. The first is that the same lattice-point count applies to any box, with only the side lengths changing. The second is a two-sided bracketing between an inner and an outer grid of cubes: a Dirichlet count on the inner cubes bounds $N(\lambda)$ below, and a Neumann count on the outer cubes bounds it above. Letting the boxes shrink, both bounds converge to the same volume integral, and the squeeze pins $N(\lambda)$ to the Weyl value; this is where the negligible boundary enters, ensuring the inner and outer grid volumes share the common limit $\mathrm{vol}(U)$.

Theorem 6.3.2: Weyl's Asymptotic Law

Let $U \subseteq \mathbb{R}^n$ be a bounded open set whose boundary is Lebesgue-negligible, $|\partial U| = 0$, and let $0 < \lambda_1 \le \lambda_2 \le \cdots \to \infty$ be the Dirichlet eigenvalues of $-\Delta$ on U (definition 6.1.1), listed with multiplicity. Then the counting function $N(\lambda) = \#\{k \mid \lambda_k \le \lambda\}$ satisfies

$$N(\lambda) \sim \frac{\omega_n}{(2\pi)^n}\operatorname{vol}(U)\,\lambda^{n/2} \qquad \text{as } \lambda \to \infty$$

where ω_n is the volume of the unit ball in $\mathbb{R}^n$, that is, $N(\lambda)\big/\big(\frac{\omega_n}{(2\pi)^n}\operatorname{vol}(U)\,\lambda^{n/2}\big) \to 1$.

***Proof Key ideas*:**
The full error analysis is compressed, but every step that produces the leading term is given. The argument brackets U between grids of small cubes and traps $N(\lambda)$ between a Dirichlet count on the inner cubes (a lower bound) and a Neumann count on the outer cubes (an upper bound), each computed by the box computation of example 6.3.1; the mesh limit then sends both counts to the volume integral. The two counts sit on opposite sides because a Neumann eigenvalue never exceeds the corresponding Dirichlet eigenvalue, so switching the outer cubes to Neumann boundary conditions can only raise the count.

Step 1: Bracketing U between grids of cubes. Fix a mesh size $h > 0$ and tile $\mathbb{R}^n$ by the closed cubes of the lattice $h\mathbb{Z}^n$, each of side h. Let

$$U_h^- = \operatorname{int}\bigcup\{Q \mid Q \text{ a grid cube with } \overline{Q} \subseteq U\}, \qquad U_h^+ = \operatorname{int}\bigcup\{Q \mid Q \text{ a grid cube with } \overline{Q} \cap U \neq \emptyset\}$$

the inner grid domain (the union of grid cubes contained in U) and the outer grid domain (the union of grid cubes meeting U). By construction $U_h^- \subseteq U \subseteq U_h^+$. Each $U_h^\pm$ is the interior of a finite union of closed cubes, hence a connected open set across the shared cube faces; it is *not* the disjoint union of the open cubes, since the interior grid walls have been reinstated. Write $G_h^\pm$ for the open set obtained by *deleting* those interior walls, a genuine disjoint union $G_h^\pm = \bigsqcup Q$ of the open cubes of side h. Then $G_h^\pm \subseteq U_h^\pm$, the two differ only by the grid walls (a finite union of hyperplane pieces, Lebesgue-null), so $\operatorname{vol}(G_h^\pm) = \operatorname{vol}(U_h^\pm)$. As $h \to 0$ the inner domain exhausts U and the outer domain shrinks to it, so

$$\operatorname{vol}(U_h^-) \to \operatorname{vol}(U) \qquad \text{and} \qquad \operatorname{vol}(U_h^+) \to \operatorname{vol}(U)$$

the discrepancy $\operatorname{vol}(U_h^+) - \operatorname{vol}(U_h^-)$ being the volume of the grid cubes that straddle ∂U, a boundary layer of thickness of order h whose volume tends to 0. (This is exactly where the hypothesis $|\partial U| = 0$ enters: the straddling cubes cover ∂U, so their total volume tends to $|\partial U| = 0$; without it the two grid volumes could tend to different limits and the squeeze would pin down no single constant. The boundary layer is the lower-order term that Weyl's leading asymptotics does not see.)

Step 2: Dirichlet-Neumann bracketing of the counting function. Write $N_U(\lambda)$ for the Dirichlet counting function of U. The lower and upper brackets use different boundary conditions, because pure Dirichlet-Dirichlet bracketing does not close the upper bound: $U \subseteq U_h^+$ forces $N_U \le N^{\mathrm{Dir}}_{U_h^+}$, but the disjoint-cube Dirichlet count carries *extra* interior walls that only lower the count, so it lands on the wrong side of $N^{\mathrm{Dir}}_{U_h^+}$ and cannot be chained to N_U. The Neumann count on the outer cubes fixes this.

For the lower bound, use the inner disjoint-cube domain $G_h^- \subseteq U$ with Dirichlet conditions. Domain monotonicity (theorem 6.2.3: extending an H_0^1 competitor by zero is admissible on the larger domain, so the min-max runs over a larger family and each eigenvalue can only decrease) gives $\lambda_k(U) \le \lambda_k(G_h^-)$, hence $N_U(\lambda) \ge N_{G_h^-}^{\mathrm{Dir}}(\lambda)$. And the Dirichlet spectrum of a disjoint union is the union of the spectra of the pieces (an H_0^1 eigenfunction is prescribed independently on each cube and extended by zero across the deleted walls), so $N_{G_h^-}^{\mathrm{Dir}}(\lambda) = M_h^- \, \nu_h^{\mathrm{D}}(\lambda)$, where ν_h^{D} is the single-cube Dirichlet count and M_h^- the number of inner cubes.

For the upper bound, put *Neumann* conditions on the outer cubes. On the fixed domain U_h^+ a Neumann eigenvalue never exceeds the corresponding Dirichlet one, $\mu_k^{\mathrm{Neu}}(U_h^+) \le \lambda_k^{\mathrm{Dir}}(U_h^+)$, because the Neumann Rayleigh quotient is minimized over the larger space $H^1(U_h^+)$ (no H_0^1 boundary constraint); and $U \subseteq U_h^+$ gives $\lambda_k^{\mathrm{Dir}}(U_h^+) \le \lambda_k(U)$ by the same domain monotonicity. Chaining, $\mu_k^{\mathrm{Neu}}(U_h^+) \le \lambda_k(U)$, so a Neumann count on U_h^+ has *more* entries below λ than N_U, that is $N_U(\lambda) \le N_{U_h^+}^{\mathrm{Neu}}(\lambda)$. The Neumann count on the connected U_h^+ is in turn bracketed above by the decoupled single-cube count, by a second application of the same monotonicity, now on the form domain. The Neumann form domain on U_h^+ is $H^1(U_h^+)$, whose members have matching traces across the interior grid walls (an H^1 function admits no jump across an interior hyperplane, since a jump would produce a surface-delta in its distributional derivative). Inserting Neumann walls along the interior faces enlarges the form domain from $H^1(U_h^+)$ to $\bigoplus_j H^1(Q_j)$, releasing exactly that trace-matching constraint; minimizing each Neumann Rayleigh quotient over the larger space can only *lower* the eigenvalues, hence *raise* the count, so $N_{U_h^+}^{\mathrm{Neu}}(\lambda) \le M_h^+ \, \nu_h^{\mathrm{N}}(\lambda)$ with ν_h^{N} the single-cube Neumann count and M_h^+ the number of outer cubes. (This is the Neumann counterpart of the Dirichlet wall insertion above, which *raised* the eigenvalues and lowered the count; inserting an interior wall changes the count either way, with the direction set by the boundary condition on the wall.) Both single-cube counts are placement-independent, so

$$M_h^- \, \nu_h^{\mathrm{D}}(\lambda) \le N_U(\lambda) \le M_h^+ \, \nu_h^{\mathrm{N}}(\lambda)$$

Step 3: The single-cube count. By example 6.3.1 applied to a cube of side h (all $L_i = h$),

$$\nu_h^{\mathrm{D}}(\lambda) \sim \frac{\omega_n}{(2\pi)^n} h^n \lambda^{n/2} \qquad \text{as } \lambda \to \infty$$

since a cube of side h has volume h^n. The Neumann count of the same cube has the same leading term,

$$\nu_h^{\mathrm{N}}(\lambda) \sim \frac{\omega_n}{(2\pi)^n} h^n \lambda^{n/2} \qquad \text{as } \lambda \to \infty$$

the Neumann eigenvalues on a box of side lengths L_i being $\pi^2 \sum_i (m_i/L_i)^2$ with $m_i \ge 0$ (cosines in place of sines), which is the same lattice-point count shifted to include the coordinate faces $m_i = 0$; the shift changes only the boundary-order correction, not the leading ball volume, so ν_h^{N} and ν_h^{D} agree to first order. Multiplying by the cube counts, and using that $M_h^{\pm} h^n = \mathrm{vol}(U_h^{\pm})$ is the total volume of the grid domain,

$$M_h^- \, \nu_h^{\mathrm{D}}(\lambda) \sim \frac{\omega_n}{(2\pi)^n} \mathrm{vol}(U_h^-) \, \lambda^{n/2}, \qquad M_h^+ \, \nu_h^{\mathrm{N}}(\lambda) \sim \frac{\omega_n}{(2\pi)^n} \mathrm{vol}(U_h^+) \, \lambda^{n/2}$$

as $\lambda \to \infty$ with h fixed.

Step 4: The mesh limit produces the volume. Combining Steps 2 and 3, for each fixed h,

$$\frac{\omega_n}{(2\pi)^n} \mathrm{vol}(U_h^-) \le \liminf_{\lambda \to \infty} \frac{N_U(\lambda)}{\lambda^{n/2}} \le \limsup_{\lambda \to \infty} \frac{N_U(\lambda)}{\lambda^{n/2}} \le \frac{\omega_n}{(2\pi)^n} \mathrm{vol}(U_h^+)$$

the outer inequalities from the single-cube asymptotics of Step 3 and the sandwiching of Step 2. This holds for every $h > 0$. Letting $h \to 0$ collapses the two ends onto the same value by Step 1, since $\mathrm{vol}(U_h^-)$ and $\mathrm{vol}(U_h^+)$ both tend to $\mathrm{vol}(U)$; the liminf and limsup are therefore equal to $\frac{\omega_n}{(2\pi)^n}\,\mathrm{vol}(U)$, that is,

$$\lim_{\lambda\to\infty} \frac{N_U(\lambda)}{\lambda^{n/2}} = \frac{\omega_n}{(2\pi)^n}\,\mathrm{vol}(U)$$

which is the asserted asymptotic, completing the proof. The passage $h \to 0$ turning the cube count into the volume integral is the crux: the volume $\mathrm{vol}(U)$ enters not through any single cube but as the limit of the total volume of the grid that brackets U.

Each grid cube contributes eigenvalues at the universal density $\omega_n/(2\pi)^n$ per unit volume per unit $\lambda^{n/2}$, and the domain contributes only its volume, assembled from the cubes; the shape of U affects only how the boundary cubes are apportioned, which is the discrepancy $\mathrm{vol}(U_h^+) - \mathrm{vol}(U_h^-)$ that vanishes in the mesh limit. The constant $\omega_n/(2\pi)^n$ is dimensional, not domain-dependent: it is the phase-space density of oscillation modes, the same for every domain in $\mathbb{R}^n$.

Solving the asymptotic for λ_k inverts the count into a growth law for the eigenvalues themselves. Setting $N(\lambda_k) \approx k$ and solving $\frac{\omega_n}{(2\pi)^n}\,\mathrm{vol}(U)\,\lambda_k^{n/2} \approx k$ gives

$$\lambda_k \sim \Big(\frac{(2\pi)^n}{\omega_n\,\mathrm{vol}(U)}\Big)^{2/n} k^{2/n} \qquad \text{as } k \to \infty$$

so the k-th eigenvalue grows like $k^{2/n}$, with a constant set by the volume. In one dimension this recovers $\lambda_k \sim (\pi k/L)^2$ for the interval $(0, L)$, matching the exact spectrum $\lambda_k = (\pi k/L)^2$; in two dimensions λ_k grows linearly in k, and the density of eigenvalues per unit λ is asymptotically constant, equal to $\mathrm{vol}(U)/(4\pi)$.

The same leading asymptotics can be read off a smoothed version of the counting function, the heat trace, which trades the sharp cutoff at λ for an exponential weight and thereby avoids the lattice-point error terms altogether. Weighting each eigenvalue by $e^{-\lambda_k t}$ and summing gives the trace of the heat semigroup e^{-tA},

$$Z(t) = \sum_{k=1}^{\infty} e^{-\lambda_k t}$$

and the reformulation of Weyl's law states that

$$Z(t) \sim \frac{1}{(4\pi t)^{n/2}}\,\mathrm{vol}(U) \qquad \text{as } t \to 0^+$$

Here $t \to 0^+$ plays the role that $\lambda \to \infty$ played before: small time probes high frequencies, and the leading singularity of the trace as $t \to 0^+$ records the same phase-space volume count. On the box Q the sum factors over the coordinates into a product of one-dimensional theta-like sums, each asymptotic to $L_i/\sqrt{4\pi t}$ as $t \to 0^+$, whose product is exactly $\mathrm{vol}(Q)/(4\pi t)^{n/2}$; the same cube-bracketing that proved the counting-function asymptotics transfers this to a general U. The two forms are equivalent through the Laplace transform, since $Z(t) = \int_0^\infty e^{-\lambda t}\,dN(\lambda)$, and a Tauberian theorem converts the $t \to 0^+$ asymptotics of Z back into the $\lambda \to \infty$ asymptotics of N; the heat-trace form is the one that generalizes cleanly to Riemannian manifolds, where the coefficient $\mathrm{vol}(U)$ is joined by lower-order heat coefficients carrying the curvature of the domain.

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